

PART I - DESIGN CRITERIA		PART IV - BUILDING EARTHWORK (CONTINUED)		PART VI - STRUCTURAL STEEL (CONTINUED)		PART VIII - STRUCTURAL TESTING AND INSPECTIONS		PART VIII - STRUCTURAL TESTING AND INSPECTIONS (CONTINUED)	
A. GENERAL BUILDING CODE		C. SOIL MATERIALS		7. WELDING PROCEDURE SPECIFICATION: ALL WELDING SHALL BE PERFORMED IN ACCORDANCE WITH A WELDING PROCEDURE SPECIFICATION (WPS) AS REQUIRED IN AWS D1.1 AND APPROVED BY THE TESTING AGENCY AND THE ENGINEER. THE WPS VARIABLES SHALL BE WITHIN THE PARAMETERS ESTABLISHED BY THE FILLER-METAL MANUFACTURER. THE CONTRACTOR SHALL ENGAGE THE SERVICES OF AN INDEPENDENT TESTING AGENCY TO PROVIDE THE QUALIFICATION TESTING REQUIRED BY AWS D 1.1, CHAPTER 4, PART 8 TO QUALIFY ANY NON-REGULATED WPS NEEDED FOR THE PROJECT. THE CONTRACTOR'S TESTING AGENCY SHALL PREPARE WELDING PROCEDURE QUALIFICATION RECORDS (WPQR) DOCUMENTING THE SUCCESSFUL QUALIFICATION OF EACH WELDING PROCEDURE SPECIFICATION.		A. GENERAL		7. NONDESTRUCTIVE EXAMINATION, IF REQUIRED, CONDUCTED IN ACCORDANCE WITH THE SPECIFIC REQUIREMENTS FOR THE ITEM BEING EXAMINED INCLUDING ULTRASONIC (UT) OR MAGNETIC PARTICLE (MT). NONDESTRUCTIVE INSPECTION PROCEDURES SHALL CONFORM TO AWS D1.1.	
1. THE CONSTRUCTION DOCUMENTS ARE BASED ON THE REQUIREMENTS OF THE 2021 VIRGINIA CONSTRUCTION CODE AND SUBSEQUENT REFERENCE STANDARDS.		1. GENERAL SOIL MATERIALS: SATISFACTORY SOIL MATERIALS ARE DEFINED AS THOSE COMPLYING WITH ASTM D2487 SOIL CLASSIFICATION GROUPS GW, GP, GM, SM, SW, AND SP.		1. NON-METALLIC NON-SHRINK GROUT: PREMIXED, FACTORY PACKAGED, NON-METALLIC, NON-SHRINK, NON-CORROSIVE, NON-STAINING, PRODUCT CONTAINING PORTLAND CEMENT, SILICA SANDS, SHRINKAGE COMPENSATING AGENTS, AND FLUIDITY IMPROVING COMPOUNDS. GROUT SHALL CONFORM TO ASTM C1107 AND HAVE A MINIMUM 28 DAY ASTM C109 COMPRESSIVE STRENGTH OF 8,000 PSI. ACCEPTABLE GROUT MATERIALS FOR GROUTING BASEPLATES ARE: a. SIKKA CORPORATION; MASTERFLOW 555 OR 4316 b. SIKKA CORPORATION; SIKAGROUT 212 OR 528 c. EUCILID CHEMICAL COMPANY; NS GROUT OR NS GROUT d. MAPEI CORPORATION; PLANIGROUT 755 OR 728		1. THE OWNER SHALL ENGAGE A TESTING AGENCY TO PERFORM FIELD TESTS AND INSPECTIONS IN ACCORDANCE WITH THE REFERENCED BUILDING CODE AND PREPARE TESTING INSPECTION REPORTS. a. RE-INSPECTION OF WORK SHALL BE AT CONTRACTOR'S EXPENSE. COSTS OF RE-INSPECTION SHALL BE PAID FOR BY THE OWNER. THESE COSTS SHALL BE REIMBURSED TO THE OWNER BY THE CONTRACTOR.		8. FILLET WELDS: MT TEST A MINIMUM OF 10% OF THE LENGTH OF EACH FILLET WELD EXCEEDING 5/16". THE FOLLOWING TESTING REGIMEN IS NOT REQUIRED BY SPECIAL INSPECTION. THOUGH RECOMMENDED, IT MAY BE DELETED WITHOUT VIOLATING CODE REQUIREMENTS IF SO DESIRED BY THE OWNER. PERIODIC MT TESTING OF REPRESENTATIVE FILLET WELDS 5/16" AND LESS BUT NEED NOT EXCEED 10% OF ALL SUCH WELDS, EXCEPT AS REQUIRED FOR HIGH REJECTION RATES AS INDICATED IN THE FOLLOWING PARAGRAPH. INCREASE MT TESTING RATE FOR WELDERS HAVING A HIGH REJECTION RATE AS REQUIRED TO ENSURE ACCEPTABLE WELDS.	
B. DEAD LOADS		D. SUB-BASE		F. CORROSION RESISTANCE		2. SPECIAL INSPECTIONS: THE TESTING AGENCY SHALL SERVE AS A SPECIAL INSPECTOR TO PROVIDE SPECIAL INSPECTION SERVICES, AS LEGALLY REQUIRED BY THE REFERENCED BUILDING CODE FOR THE ITEMS LISTED IN SECTION "SPECIAL INSPECTIONS."		9. COMPLETE JOINT PENETRATION (CJP) WELDS: ALL CJP WELDS EXCEEDING 5/16" THICKNESS SHALL BE 100% UT TESTED. THE TESTING AGENCY SHALL REVIEW THE CJP JOINTS TO DETERMINE WHERE GEOMETRY OR ACCESSIBILITY PRECLUDES THE USE OF STANDARD SCANNING PATTERNS. AT THESE LOCATIONS THE TESTING AGENCY SHALL DEVELOP AND SUBMIT FOR APPROVAL A WRITTEN TESTING PROCEDURE TO EXAMINE THESE AREAS. PERIODIC MT TESTING OF REPRESENTATIVE CJP WELDS 5/16" AND LESS NOT TO EXCEED 10% OF ALL SUCH WELDS, EXCEPT AS REQUIRED FOR HIGH REJECTION RATES AS INDICATED IN THE FOLLOWING PARAGRAPH.	
1. SEE LOAD PLANS ON S-004 AND S-005		1. INSTALL, TAMP, AND LEVEL NEW SAND OR AGGREGATE SUB-BASE TO (±2%) TO 85% RELATIVE DENSITY FOR COHESIONLESS MATERIAL PRIOR TO INSTALLING VAPOR BARRIER.		1. GALVANIZED FINISH: ALL STEEL SHAPES, PLATES, ANGLES, BOLTS, ANCHOR RODS, NUTS, WASHERS AND OTHER EXPOSED STEEL SHALL BE HOT DIP GALVANIZED PER ASTM A123.		1. CONTRACTOR RESPONSIBILITY: THE CONTRACTOR SHALL PROVIDE THE OWNER'S TESTING AGENCY WITH THE FOLLOWING: a. THE CONTRACTOR SHALL PROVIDE TO THE TESTING AGENCY CERTIFICATES AND REPRESENTATIVE SAMPLES OF MATERIALS PROPOSED FOR USE IN THE WORK IN QUANTITIES SUFFICIENT FOR ACCURATE TESTING AS SPECIFIED b. DOCUMENTATION OF WELDERS CERTIFICATION TO PERFORM THE INDICATED WORK AND THE APPLICABLE WPS DOCUMENTATION AS INDICATED ABOVE. c. PROPER FACILITIES, INCLUDING SCAFFOLDING, TEMPORARY WORK PLATFORMS, HOISTING FACILITIES, CASUAL LABOR, ETC, FOR INSPECTION OF WORK IN THE MILLS, SHOP, AND FIELD.		10. MARK FOR REMEDIATION ANY AREA NOT MEETING CONTRACT DOCUMENT REQUIREMENTS. CORRECTION OF REJECTED WELDS SHALL BE MADE IN ACCORDANCE WITH AWS D1.1.	
C. LIVE LOADS		E. BELOW GRADE VAPOR BARRIER		3. ZINC COATING: THE ZINC COATING FOR STEEL SHAPES AND PLATES SHALL CONFORM TO ASTM A123. WEIGHT OF ZINC COATING PER SQUARE FT OF SURFACE FOR 1/8" AND 3/16" THICK STEELS SHALL AVERAGE NOT LESS THAN 2.0 OZ WITH NO INDIVIDUAL THICKNESS LESS THAN 1.8 OZ. THE COATING WEIGHT SHALL AVERAGE NOT LESS THAN 2.3 OZ WITH NO INDIVIDUAL THICKNESS LESS THAN 2.0 OZ FOR 1/4" THICK AND HEAVIER STEEL.		2. OWNER'S TESTING AGENCY RESPONSIBILITY: THE TESTING AGENCY SHALL SAMPLE AND TEST MATERIALS AS THEY ARE BEING INSTALLED IN ACCORDANCE WITH THE SPECIFIED ACCEPTANCE CRITERIA. THE TESTING AGENCY WILL REPORT AND INTERPRET THE TEST RESULTS. THE TESTING AGENCY SHALL MONITOR AND REPORT ON THE INSTALLATION OF CONSTRUCTION WORK AND SHALL PERFORM TESTS OF THE COMPLETED CONSTRUCTION AS REQUIRED TO INDICATE CONTRACTOR'S COMPLIANCE WITH THE VARIOUS PROJECT REQUIREMENTS. a. THE TESTING AGENCY SHALL MEET THE BASIC REQUIREMENTS OF ASTM E520 AND SHALL SUBMIT TO THE OWNER (ARCHITECT), AND ENGINEER EVIDENCE OF CURRENT ACCREDITATION FROM THE AMERICAN ASSOCIATION FOR LABORATORY ACCREDITATION, THE ANSITO ACCREDITATION PROGRAM OR THE "NIST" NATIONAL VOLUNTARY LABORATORY ACCREDITATION PROGRAM. b. THE TESTING AGENCY SHALL BE AN APPROVED AGENCY BY THE BUILDING OFFICIAL OF THE CITY WHEREIN THE PROJECT IS LOCATED TO PERFORM SPECIAL INSPECTIONS AND OTHER TESTS AND INSPECTIONS AS OUTLINED IN THE APPLICABLE BUILDING CODE. c. THE INSPECTION BY THE TESTING AGENCY OF THE WELDER'S WORK SHALL BE IN SEQUENCE, TIMELY, AND PERFORMED IN SUCH A MANNER SO THAT CORRECTIONS CAN BE MADE WITHOUT DELAYING THE PROGRESS OF THE WORK. INSPECTIONS OF THE WELDER'S WORK SHALL BE PERFORMED BY CERTIFIED WELDING INSPECTORS IN ACCORDANCE WITH AWS D1.1 SECTION 6.1.4 FOR VISUAL WELD INSPECTION AND AWS D1.1, SECTION 6.1.4.6 FOR NONDESTRUCTIVE EXAMINATION. QUALIFIED TECHNICIANS WITH A MINIMUM OF TWO YEARS' EXPERIENCE IN STRUCTURAL STEEL TESTING AND INSPECTION. d. INSPECTIONS OF STRUCTURAL STEEL AND BOLTED CONNECTIONS SHALL BE PERFORMED BY QUALIFIED TECHNICIANS WITH A MINIMUM OF TWO YEAR'S EXPERIENCE IN STRUCTURAL STEEL TESTING AND INSPECTION. e. THE TESTING AGENCY SHALL PROVIDE REPORTS OF ALL TESTING AND INSPECTIONS. ALL REPORTS SHALL INDICATE TYPES AND LOCATIONS OF ALL DEFECTS FOUND DURING INSPECTION. THE MEASURING PERSONNEL AND PERFORMED TO CORRECT SUCH DEFECTS, STATEMENTS OF FINAL APPROVAL OF ALL WELDING OF FIELD CONNECTIONS, AND OTHER FABRICATION AND ERECTION DATA PERTINENT TO THE SAFE AND PROPER WELDING OF FIELD CONNECTIONS.		11. RE-EXAMINE ALL REWORKED AREAS AND INTERPRET, RECORD, AND REPORT THE RESULTS OF EXAMINATIONS OF REWORKED WELDS.	
1. RAISED ACCESS FLOOR PANELS, STRINGERS, AND PEDESTALS: SHALL BE DESIGNED AS A DEFERRED SUBMITTAL BY THE MANUFACTURER'S ENGINEER TO SUPPORT THE WORST-CASE OF THE FOLLOWING LOAD CONDITIONS: a. A UNIFORM LIVE LOAD OF 250 PSF IN GENERAL ACCESS FLOOR AREAS, OR b. SERVER RACKS WEIGHING UP TO 4,000 LBS EACH WITH FOOTPRINT OF 24" X 48", INSTALLED SIDE-BY-SIDE IN CONTINUOUS ROWS AS SHOWN IN THE ARCHITECTURAL DRAWINGS.		1. MATERIALS a. PLASTIC SHEET VAPOR BARRIER AND RELATED MATERIALS SHALL ALL BE FROM THE SAME MANUFACTURER. b. PLASTIC SHEET VAPOR BARRIER: 15 MIL THICK PLASTIC SHEET CONFORMING TO ASTM E745 CLASS A. ASTM F1249 WATER VAPOR PERMEANCE SHALL BE LESS THAN 0.01 US PERMS AFTER MANDATORY CONDITIONING. ACCEPTABLE SYSTEMS ARE: 1) STEGO INDUSTRIES, LLC; STEGO WRAP VAPOR BARRIER (15 MIL) 2) EPRO; ECOSHIELD-E (15 MIL) 3) RAVEN INDUSTRIES; VAPORBLOCK V615 4) W.R. MEADOWS; PERMIMATON MLL 5) FORTIFIBER BUILDING SYSTEMS GROUP; MOISTOP ULTRA 15 c. SEAM TAPE FOR PLASTIC VAPOR RETARDERS: HIGH-DENSITY POLYETHYLENE TAPE WITH PRESSURE SENSITIVE ADHESIVE HAVING A MINIMUM WIDTH OF 0.75" HAVING A MAXIMUM WATER VAPOR PERMEANCE RATE OF 0.3 PERMS. ACCEPTABLE SYSTEMS ARE: 1) STEGO INDUSTRIES, LLC; STEGO TAPE 2) EPRO; ECOTAPE 3) RAVEN INDUSTRIES; VAPORBOND TAPE 4) W.R. MEADOWS; PERMIMATON TAPE 5) FORTIFIBER BUILDING SYSTEMS GROUP; MOISTOP TAPE		2. SURFACE PREPARATION: PREPARE THE SURFACE OF ALL STRUCTURAL STEEL SPECIFIED TO BE HOT DIP GALVANIZED IN ACCORDANCE WITH THE SURFACE PREPARATION AS SPECIFIED BY THE STEEL STRUCTURES PAINTING COUNCIL (SSPC), VOLUME 2: a. REMOVE ALL GREASE, OIL, GRIME AND FOREIGN CONTAMINANTS BY THOROUGH CLEANING WITH AN ALKALINE OR ORGANIC SOLVENT FOLLOWED BY THOROUGH RINSING IN COLD WATER. b. SURFACE PREPARATION FOR HOT DIP GALVANIZING SHALL BE BLAST CLEANED TO "WHITE METAL" ACCORDING TO (SSPC-SP5). c. DIP IN A FLUX SOLUTION OF ZINC AMMONIA CHLORIDE FOLLOWED BY DRYING AT ROOM TEMPERATURE.		12. VERIFY THAT QUALITY OF WELDS MEET THE REQUIREMENTS OF AWS D1.1.			
3. PEDESTALS SUPPORTING THE RAISED FLOOR PANELS AND STRINGERS SHALL BE ARRANGED ON A 24"x24" GRID. THE MAXIMUM LOAD IMPOSED ON THE STRUCTURAL SLAB FROM EACH PEDESTAL, INCLUDING FLOOR SYSTEM SELF WEIGHT, SHALL BE LIMITED TO 2,750 LBS (ASD).		2. INSTALLATION a. INSTALL NEW VAPOR RETARDER IN ACCORDANCE WITH ASTM E1643 AND MANUFACTURER'S INSTRUCTIONS. b. DO NOT PUNCTURE VAPOR BARRIER WITH REINFORCEMENT. c. OVERLAP NEW VAPOR BARRIER WITH EXISTING VAPOR BARRIER AT LEAST 4" AND TAPE ALL SEAM.		4. ALL FIELD WELDS AND PENETRATIONS IN ZINC COATING SHALL BE PAINTED IN THE FIELD WITH ZRC GALVALITE REPAIR PAINT OR ENGINEER APPROVED EQUAL.		13. THE COSTS OF REWORKING ALL DEFECTIVE WELDS AND THE COSTS OF RETESTING BY THE TESTING AGENCY PROVIDING SERVICES FOR THE OWNER SHALL BE BORNE BY THE CONTRACTOR.			
D. WIND LOADS		3. REPAIRS a. REPAIR DAMAGED AREAS BY CUTTING PATCHES OF VAPOR BARRIER, OVERLAPPING DAMAGED AREA 6" MINIMUM AND TAPING ALL FOUR SIDES WITH SEAM TAPE.		5. PAINTED FINISH: ALL STEEL SHAPES, PLATES, ANGLES, BOLTS, ANCHOR RODS, NUTS, WASHERS AND OTHER EXPOSED STEEL SHALL BE SHOP PAINTED AND FIELD TOUCHED-UP WHERE REQUIRED USING THE FOLLOWING SURFACE PREPARATION AND PAINTS: a. SURFACE PREPARATION: COMMERCIAL BLAST (SSPC-SP6) b. SHOP PRIMER: GLID-GUARD ALKALOY METAL PRIMER 4570 AS MANUFACTURED BY GLIDDEN, 2.0 MILS MINIMUM DRY FILM OR ENGINEER APPROVED EQUAL. c. FINISH COAT: GLID-GUARD ALKALOY INDUSTRIAL ENAMEL NO. 4550 SERIES AS MANUFACTURED BY GLIDDEN (COLOR AS SELECTED BY ARCHITECT) 2.0 MILS MINIMUM DRY THICKNESS OR ENGINEER APPROVED EQUAL.		F. INSPECTION OF POST-INSTALLED ANCHOR, BOLTS, AND REINFORCING BARS INSTALLED IN HARDENED CONCRETE: THE TESTING AGENCY OR MANUFACTURE SHALL MAKE THE FOLLOWING INSPECTIONS OF POST-INSTALLED ANCHOR INSTALLATIONS IN ACCORDANCE WITH THE PUBLISHED, CURRENTLY VALID, EVALUATION SERVICE REPORT (ESR) FOR EACH ANCHOR PRODUCT. INSPECTIONS SHALL BE PERFORMED AT THE FREQUENCY REQUIRED BY THE ESR.			
1. WIND PRESSURES ARE BASED ON THE PROVISIONS OF THE AMERICAN SOCIETY OF CIVIL ENGINEERS, MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22 AND THE FOLLOWING CRITERIA: a. BASIC DESIGN WIND SPEED (V): 119 MPH (3 SECOND GUST) b. WIND EXPOSURE CATEGORY: B c. INTERNAL PRESSURE COEFFICIENT (GCP1): +0.18/-0.18		2. SEISMIC LOADS a. PLASTIC SHEET VAPOR BARRIER AND RELATED MATERIALS SHALL ALL BE FROM THE SAME MANUFACTURER. b. PLASTIC SHEET VAPOR BARRIER: 15 MIL THICK PLASTIC SHEET CONFORMING TO ASTM E745 CLASS A. ASTM F1249 WATER VAPOR PERMEANCE SHALL BE LESS THAN 0.01 US PERMS AFTER MANDATORY CONDITIONING. ACCEPTABLE SYSTEMS ARE: 1) STEGO INDUSTRIES, LLC; STEGO WRAP VAPOR BARRIER (15 MIL) 2) EPRO; ECOSHIELD-E (15 MIL) 3) RAVEN INDUSTRIES; VAPORBLOCK V615 4) W.R. MEADOWS; PERMIMATON MLL 5) FORTIFIBER BUILDING SYSTEMS GROUP; MOISTOP ULTRA 15 c. SEAM TAPE FOR PLASTIC VAPOR RETARDERS: HIGH-DENSITY POLYETHYLENE TAPE WITH PRESSURE SENSITIVE ADHESIVE HAVING A MINIMUM WIDTH OF 0.75" HAVING A MAXIMUM WATER VAPOR PERMEANCE RATE OF 0.3 PERMS. ACCEPTABLE SYSTEMS ARE: 1) STEGO INDUSTRIES, LLC; STEGO TAPE 2) EPRO; ECOTAPE 3) RAVEN INDUSTRIES; VAPORBOND TAPE 4) W.R. MEADOWS; PERMIMATON TAPE 5) FORTIFIBER BUILDING SYSTEMS GROUP; MOISTOP TAPE		6. SUBSTRATE CLEANING OF HOT-DIPPED GALVANIZED SURFACES TO RECEIVE PAINT 1. GALVANIZED SURFACES: ALL GALVANIZED SURFACES SHALL BE CLEANED BY ACID WASHING WITH 10% PHOSPHORIC ACID SOLUTION. REMOVE THE ACID FROM THE GALVANIZED SURFACE WITH WATER AND ALLOW THE SURFACE TO DRY. PROPERLY ETCHED SURFACES WITH PHOSPHORIC ACID APPEAR BLACK. AREAS CONTAMINATED WITH CHROMIUM OR OTHER CHEMICALS WILL NOT APPEAR BLACK AND SHALL BE RE-ETCHED UNTIL THE SURFACE APPEARS BLACK. WASHING WITH TSP (TRI SODIUM PHOSPHATE) SOLUTION MAY BE USED IN LIEU OF PHOSPHORIC ACID WASHING. PRIME THE ETCHED SURFACE WITH A FLAT EPOXY PRIMER COMPATIBLE WITH APPLIED CEMENTITIOUS FIREPROOFING AFTER THE SURFACE HAS DRIED.		3. INSPECTION OF REINFORCEMENT STEEL PLACEMENT: THE TESTING AGENCY SHALL MAKE THE FOLLOWING INSPECTIONS OF REINFORCEMENT STEEL PLACEMENT: 1. INSPECTION FREQUENCY: PROVIDE PERIODIC INSPECTION OF REINFORCING STEEL PLACEMENT. INSPECT 100% OF REINFORCEMENT BEFORE EACH CONCRETE PLACEMENT TO VERIFY THE INFORMATION NOTED BELOW. a. REINFORCEMENT HAS CORRECT SIZE, SPACING, AND PROPER COVER. b. SUFFICIENT SPACING BETWEEN REINFORCEMENT FOR CONCRETE PLACEMENT. c. REINFORCEMENT IS PROPERLY SUPPORTED AND BRACED TO FORMWORK TO PREVENT MOVEMENT DURING CONCRETE PLACEMENT. d. DOWEL REINFORCEMENT IS OF PROPER SIZE, AT PROPER SPACING, AND HAS PROPER LAP LENGTH AND EMBEDMENT LENGTH. e. PROPER HOOKS ARE PROVIDED AT BAR ENDS AS DETAILED. f. WELDED WIRE REINFORCEMENT IS COMPOSED OF FLAT SHEETS, HAS PROPER WIRE GAGE AND SPACING, IS PROPERLY SUPPORTED, AND IS PROPERLY LAPPED WITH A LENGTH OF ONE SQUARE PLUS TWO IN.. g. PROPER CONSTRUCTION/CONTROL/EXPANSION JOINT SPACING AND REINFORCEMENT. h. REINFORCEMENT AROUND EMBEDDED ITEMS IS PLACED PER THE DRAWINGS. i. PROPER INSTALLATION OF FLAT SLAB SHEAR HEAD REINFORCEMENT. j. MECHANICAL TENSION SPLICES: THE TESTING AGENCY SHALL PROVIDE 100% VISUAL INSPECTION OF MECHANICAL TENSION SPLICES ON THE PROJECT AND CONSULT WITH THE MANUFACTURER REGARDING RECOMMENDATIONS FOR INSTALLATION. INSPECTION SHALL VERIFY COMPLIANCE WITH THE REQUIREMENTS IN THE GENERAL NOTES AND CONFORMANCE WITH THE MANUFACTURER'S RECOMMENDATIONS FOR INSTALLATION AFTER CONSULTING WITH THE MANUFACTURER, WHO IS TO BE PRESENT FOR THE FIRST INSTALLATION OF THE SPLICE ON THE PROJECT. k. WELDED REINFORCEMENT HAS BEEN DONE PER AWS REQUIREMENTS.			
E. SNOW LOADS		1. SNOW LOADS ARE BASED ON THE AMERICAN SOCIETY OF CIVIL ENGINEERS, MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES, ASCE 7-22 AND THE FOLLOWING CRITERIA: a. GROUND SNOW LOAD (Pg): 90 PSF b. FLAT-ROOF SNOW LOAD (PF): 56.7 PSF c. SNOW EXPOSURE FACTOR (Ce): 0.9 d. THERMAL FACTOR (CT): 1		7. INSPECTION OF STRUCTURAL STEEL AND BOLTING: THE TESTING AGENCY SHALL MAKE THE FOLLOWING INSPECTIONS OF STRUCTURAL STEEL AND BOLTING: 1. INSPECTION FREQUENCY: PROVIDE PERIODIC INSPECTION OF STRUCTURAL STEEL AND BOLTING. VERIFY INITIAL INSTALLATION OF STRUCTURAL STEEL AND BOLTING IN ACCORDANCE WITH THE REQUIREMENTS STATED BELOW. PERIODICALLY INSPECT STRUCTURAL STEEL AND BOLTING AFTER THE INITIAL VERIFICATION. PROVIDE FINAL INSPECTION AFTER ALL STRUCTURAL STEEL AND BOLTING HAS BEEN INSTALLED. 2. VERIFY THAT THE FABRICATOR'S FABRICATION AND QUALITY CONTROL PROCEDURES PROVIDE A SOUND BASIS FOR INSPECTION CONTROL OF WORKMANSHIP AND OF THE ABILITY TO CONFORM TO CONSTRUCTION DOCUMENTS AND INDUSTRY STANDARDS. REVIEW THE PROCEDURES FOR COMPLETENESS AND ADEQUACY RELATIVE TO CODE REQUIREMENTS FOR THE FABRICATOR'S FINISHED PRODUCT. 3. CHECK STRUCTURAL STEEL AS RECEIVED IN THE FIELD FOR POSSIBLE SHIPPING DAMAGE, WORKMANSHIP, AND IDENTIFICATION MARKING TO CONFORM TO AISC 360 FOR STRUCTURAL STEEL AND SPECIFIED ASTM STANDARDS FOR OTHER STEEL. 4. VERIFY THAT SURVEYS ARE OCCURRING AS SPECIFIED TO CHECK PLUMBNESS AND FRAME ALIGNMENT AS ERECTION PROGRESSES. REVIEW THE SUBMITTED SURVEY REPORT. 5. PERIODICALLY INSPECT THE STEEL FRAME FOR SUCH ITEMS AS BRACING AND STIFFENING DETAILS, MEMBER LOCATIONS, AND JOINT DETAILS AT EACH CONNECTION FOR COMPLIANCE WITH APPROVED CONSTRUCTION DOCUMENTS. 6. INSPECT 100% OF THE COLUMN COMPRESSION AND BASE JOINTS FOR VERIFICATION THAT GAPS IN CONTACT BEARING DO NOT EXCEED 1/16 IN. (1.5MM). GAPS GREATER THAN 1/16 (1.5MM) IN, BUT LESS THAN 1/4 IN. (6MM) SHALL BE REPORTED TO THE OWNER AND ENGINEER FOR ASSESSMENT. ALL GAPS GREATER THAN 1/4 IN. (6MM) SHALL BE SHIMMED. 7. ENDEAVOR TO GUARD THE OWNER AGAINST THE CONTRACTOR CUTTING, GRINDING, BEAMING, OR MAKING ANY OTHER FIELD MODIFICATION TO STRUCTURAL STEEL WITHOUT THE PRIOR APPROVAL OF THE ENGINEER. REPORT ANY NOTED UNAUTHORIZED MODIFICATIONS TO THE OWNER AND ENGINEER.		1. INSPECTION FREQUENCY: PROVIDE PERIODIC INSPECTION OF REINFORCING STEEL PLACEMENT. INSPECT 100% OF REINFORCEMENT BEFORE EACH CONCRETE PLACEMENT TO VERIFY THE INFORMATION NOTED BELOW. a. REINFORCEMENT HAS CORRECT SIZE, SPACING, AND PROPER COVER. b. SUFFICIENT SPACING BETWEEN REINFORCEMENT FOR CONCRETE PLACEMENT. c. REINFORCEMENT IS PROPERLY SUPPORTED AND BRACED TO FORMWORK TO PREVENT MOVEMENT DURING CONCRETE PLACEMENT. d. DOWEL REINFORCEMENT IS OF PROPER SIZE, AT PROPER SPACING, AND HAS PROPER LAP LENGTH AND EMBEDMENT LENGTH. e. PROPER HOOKS ARE PROVIDED AT BAR ENDS AS DETAILED. f. WELDED WIRE REINFORCEMENT IS COMPOSED OF FLAT SHEETS, HAS PROPER WIRE GAGE AND SPACING, IS PROPERLY SUPPORTED, AND IS PROPERLY LAPPED WITH A LENGTH OF ONE SQUARE PLUS TWO IN.. g. PROPER CONSTRUCTION/CONTROL/EXPANSION JOINT SPACING AND REINFORCEMENT. h. REINFORCEMENT AROUND EMBEDDED ITEMS IS PLACED PER THE DRAWINGS. i. PROPER INSTALLATION OF FLAT SLAB SHEAR HEAD REINFORCEMENT. j. MECHANICAL TENSION SPLICES: THE TESTING AGENCY SHALL PROVIDE 100% VISUAL INSPECTION OF MECHANICAL TENSION SPLICES ON THE PROJECT AND CONSULT WITH THE MANUFACTURER REGARDING RECOMMENDATIONS FOR INSTALLATION. INSPECTION SHALL VERIFY COMPLIANCE WITH THE REQUIREMENTS IN THE GENERAL NOTES AND CONFORMANCE WITH THE MANUFACTURER'S RECOMMENDATIONS FOR INSTALLATION AFTER CONSULTING WITH THE MANUFACTURER, WHO IS TO BE PRESENT FOR THE FIRST INSTALLATION OF THE SPLICE ON THE PROJECT. k. WELDED REINFORCEMENT HAS BEEN DONE PER AWS REQUIREMENTS.			
F. SEISMIC LOADS		1. THE STRUCTURAL ALTERATIONS HAVE BEEN DESIGNED TO WITHSTAND THE SEISMIC FORCES SPECIFIED IN SECTION 1619.0 OF THE INTERNATIONAL BUILDING CODE, USING THE FOLLOWING DESIGN DATA: DESIGNED SPECTRAL RESPONSE SS = 0. xxx DESIGNED SPECTRAL RESPONSE SD1 = 0. xxx DESIGN SHORT PERIOD RESPONSE: SDS = 0. xxx DESIGN ONE SECOND PERIOD RESPONSE: SD1 = 0. xxx SITE CLASS C SEISMIC IMPORTANCE FACTOR: I = 1.25 SEISMIC DESIGN CATEGORY: B SEISMIC RESPONSE COEFFICIENT: CSD1 = 0.01 SEISMIC BASE SHEAR: V= X.XX KIPS (NORTH SOUTH) V= X.XX KIPS (EAST WEST) BASIC SEISMIC-FORCE-RESISTING SYSTEM: INTERMEDIATE PRECAST SHEAR WALLS STEEL SYSTEMS NOT SPECIFICALLY DETAILED FOR SEISMIC RESISTANCE: OMEGA=3.0 DEFLECTION AMPLIFICATION FACTOR: CDS1 = 0. xxx ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE		2. INSPECTION OF STRUCTURAL STEEL AND BOLTING: THE TESTING AGENCY SHALL MAKE THE FOLLOWING INSPECTIONS OF STRUCTURAL STEEL AND BOLTING: 1. INSPECTION FREQUENCY: PROVIDE PERIODIC INSPECTION OF STRUCTURAL STEEL AND BOLTING. VERIFY INITIAL INSTALLATION OF STRUCTURAL STEEL AND BOLTING IN ACCORDANCE WITH THE REQUIREMENTS STATED BELOW. PERIODICALLY INSPECT STRUCTURAL STEEL AND BOLTING AFTER THE INITIAL VERIFICATION. PROVIDE FINAL INSPECTION AFTER ALL STRUCTURAL STEEL AND BOLTING HAS BEEN INSTALLED. 2. VERIFY THAT THE FABRICATOR'S FABRICATION AND QUALITY CONTROL PROCEDURES PROVIDE A SOUND BASIS FOR INSPECTION CONTROL OF WORKMANSHIP AND OF THE ABILITY TO CONFORM TO CONSTRUCTION DOCUMENTS AND INDUSTRY STANDARDS. REVIEW THE PROCEDURES FOR COMPLETENESS AND ADEQUACY RELATIVE TO CODE REQUIREMENTS FOR THE FABRICATOR'S FINISHED PRODUCT. 3. CHECK STRUCTURAL STEEL AS RECEIVED IN THE FIELD FOR POSSIBLE SHIPPING DAMAGE, WORKMANSHIP, AND IDENTIFICATION MARKING TO CONFORM TO AISC 360 FOR STRUCTURAL STEEL AND SPECIFIED ASTM STANDARDS FOR OTHER STEEL. 4. VERIFY THAT SURVEYS ARE OCCURRING AS SPECIFIED TO CHECK PLUMBNESS AND FRAME ALIGNMENT AS ERECTION PROGRESSES. REVIEW THE SUBMITTED SURVEY REPORT. 5. PERIODICALLY INSPECT THE STEEL FRAME FOR SUCH ITEMS AS BRACING AND STIFFENING DETAILS, MEMBER LOCATIONS, AND JOINT DETAILS AT EACH CONNECTION FOR COMPLIANCE WITH APPROVED CONSTRUCTION DOCUMENTS. 6. INSPECT 100% OF THE COLUMN COMPRESSION AND BASE JOINTS FOR VERIFICATION THAT GAPS IN CONTACT BEARING DO NOT EXCEED 1/16 IN. (1.5MM). GAPS GREATER THAN 1/16 (1.5MM) IN, BUT LESS THAN 1/4 IN. (6MM) SHALL BE REPORTED TO THE OWNER AND ENGINEER FOR ASSESSMENT. ALL GAPS GREATER THAN 1/4 IN. (6MM) SHALL BE SHIMMED. 7. ENDEAVOR TO GUARD THE OWNER AGAINST THE CONTRACTOR CUTTING, GRINDING, BEAMING, OR MAKING ANY OTHER FIELD MODIFICATION TO STRUCTURAL STEEL WITHOUT THE PRIOR APPROVAL OF THE ENGINEER. REPORT ANY NOTED UNAUTHORIZED MODIFICATIONS TO THE OWNER AND ENGINEER.		2. GENERAL ANCHORS INSPECTION REQUIREMENTS: VERIFY THAT THE ANCHOR IS INSTALLED IN ACCORDANCE WITH MANUFACTURER'S PRINTED INSTALLATION INSTRUCTIONS, ESR REPORT SPECIAL INSPECTION REQUIREMENTS, AS WELL AS THE FOLLOWING DESIGN REQUIREMENTS: a. ANCHOR MANUFACTURER AND PRODUCT, INCLUDING MATERIAL, IS IN ACCORDANCE WITH DESIGN DRAWINGS OR APPROVED SUBSTITUTION. b. ANCHOR DIAMETER, LENGTH, AND INSTALLED EMBEDMENT DEPTH. c. DRILL BIT TYPE AND DIAMETER. d. ANCHOR LOCATION, EDGE DISTANCE, AND SPACING. e. HOLE DIAMETER, DEPTH, CLEANING PROCEDURE, AND CLEANLINESS. f. ANCHOR MAXIMUM TIGHTENING TORQUE. g. CONCRETE TYPE, CONCRETE STRENGTH, AND CONCRETE THICKNESS ARE IN ACCORDANCE WITH DESIGN DRAWINGS.			
PART II - NON-DESTRUCTIVE EVALUATION		PART V - CONCRETE		PART VII - SPECIAL INSPECTIONS		PART IX - SUBMITTALS			
A. ITEMS EMBEDDED IN CONCRETE STRUCTURES		A. CONCRETE REPAIR MATERIALS		A. STATEMENT OF SPECIAL INSPECTIONS		A. SUBMITTAL LIST AND SCHEDULE			
1. ITEMS EMBEDDED IN CONCRETE STRUCTURES SHALL NOT BE DAMAGED DURING REPAIR WORK OR INSTALLATION OF NEW MEMBERS REQUIRING POST-INSTALLED ANCHORS. EMBEDDED ITEMS MAY INCLUDE MILD REINFORCEMENT, PRESTRESSING REINFORCEMENT, DOWELS, EMBEDDED CONNECTIONS, ELECTRICAL CONDUITS, PLUMBING, ETC.		1. ALL CONCRETE SHALL CONFORM TO THE REQUIREMENTS AS SPECIFIED IN SPECIFICATION SECTION "CONCRETE REPAIR MATERIALS."		1. SPECIAL INSPECTION IS REQUIRED FOR THE ITEMS LISTED BELOW. REFER TO GENERAL NOTES PART "STRUCTURAL TESTING AND INSPECTIONS" FOR TYPE AND EXTENT OF EACH SPECIAL INSPECTION AND TEST. GENERAL NOTES PART "STRUCTURAL TESTING AND INSPECTIONS" ALSO INDICATES WHETHER CONTINUOUS OR PERIODIC INSPECTION IS REQUIRED FOR THE ITEMS LISTED ABOVE ADDITIONAL INFORMATION.		1. THE CONTRACTOR SHALL PREPARE A DETAILED LIST AND SCHEDULE OF ALL SUBMITTAL ITEMS TO BE SENT TO THE STRUCTURAL ENGINEER PRIOR TO THE START OF CONSTRUCTION. THIS LIST SHALL BE UPDATED AND REVISED AND KEPT CURRENT AS THE JOB PROGRESSES. THE SUBMITTAL LIST SHALL BE ORGANIZED AS SHOWN BELOW: a. SHOP DRAWINGS b. DESIGN CALCULATIONS c. PRODUCT TECHNICAL DATA d. PRODUCT MSDS e. CERTIFICATES, REPORTS, AND OTHER LITERATURE			
2. ITEMS EMBEDDED IN CONCRETE SHALL BE LOCATED BY NON-DESTRUCTIVE EVALUATION PRIOR TO PERFORMING ANY WORK. CONTRACTOR SHALL MARK ON THE STRUCTURE THE LOCATION OF EMBEDDED ITEMS AND PROVIDE A REPORT TO THE ENGINEER.		B. HORIZONTAL CONSTRUCTION JOINTS IN CONCRETE POURS		1. REFER TO GENERAL NOTES PART "STRUCTURAL TESTING AND INSPECTIONS" FOR ADDITIONAL INFORMATION.		B. SUBMITTALS TO BE PROVIDED TO STRUCTURAL ENGINEER			
3. CONTRACTOR SHALL NOT START FABRICATION OF NEW MEMBERS UNTIL ITEMS EMBEDDED IN CONCRETE HAVE BEEN LOCATED. CONTACT ENGINEER IF EXISTING EMBEDDED ITEMS INTERFERE WITH LOCATION OF POST-INSTALLED ANCHORS SPECIFIED IN DRAWINGS.		1. THERE SHALL BE NO HORIZONTAL CONSTRUCTION JOINTS IN ANY CONCRETE POURS UNLESS SHOWN ON THE DRAWINGS. THE ENGINEER SHALL APPROVE ALL DEVIATIONS OR ADDITIONAL JOINTS IN WRITING.		1. SOILS: a. PREPARED EARTH FILL b. COMPACTION, DENSITY, AND WATER CONTENT c. FORMWORK AND BRACING		1. PRODUCT SUBMITTALS: IN ADDITION TO THE SUBMITTALS REQUIRED BY THE PROJECT SPECIFICATIONS, THE FOLLOWING SUBMITTALS SHALL BE PROVIDED: a. SELECT FILL MATERIALS. b. EPOXY-MODIFIED CEMENTITIOUS COATING FOR EMBEDDED REINFORCEMENT. c. CONCRETE REPAIR MATERIALS. d. FIREPROOFING SYSTEM FOR STRUCTURAL STEEL.			
4. ENGINEER MAY REQUIRE A PARTICULAR NON-DESTRUCTIVE EVALUATION METHOD FOR THE LOCATION OF EMBEDDED ITEMS.		C. REINFORCING STEEL		2. STEEL CONSTRUCTION: a. ALL FIELD WELDING b. HIGH-STRENGTH BOLTING c. STRUCTURAL STEEL AND BOLTING d. WELDING OF STRUCTURAL STEEL		2. DEFERRED SUBMITTALS: a. THE FOLLOWING ITEMS ARE CONSIDERED DEFERRED SUBMITTALS BY THE REGISTERED DESIGN PROFESSIONAL IN RESPONSIBLE CHARGE: 1) FORMWORK, SHORING, RESHORING, AND BACKSHORING (S&S, REC) 2) DISTRIBUTED SYSTEM (PIPING AND CONDUIT) ANCHORAGE AND BRACING 3) UNISTRUT FRAMING 4) STRUCTURAL CEILING GRIDS 5) COLD FORMED STEEL FRAMING 6) STRUCTURAL STEEL CONNECTIONS 7) SOIL IMPROVEMENTS 8) SLIDE AND THERMAL BEARING PADS 9) PRE-MANUFACTURED COMPONENTS, INCLUDING ATTACHMENT TO PRIMARY STRUCTURE 10) CONTRACTOR SHALL OBTAIN DELEGATED DESIGN SUBMITTAL FOR INDUSTRIAL RACKING AND ANCHORAGE, TO BE INSTALLED BY CONTRACTOR. ANY DEVIATIONS FROM THE DESIGN SHALL NOT BE PERMITTED WITHOUT APPROVAL FROM THE INDUSTRIAL RACKING DESIGN ENGINEER AND OWNER 11) RAISED ACCESS FLOOR SYSTEM 12) EQUIPMENT ANCHORAGE TO SUPPORTS, SUPPORT CURBS, HANGERS. 13) PRE-FABRICATED METAL STAIRS AND PLATFORMS, ARCHITECTURAL METALS, ACCESS LADDER AND FALL ARREST SYSTEM			
PART III - FOUNDATION		E. SPLICES AND HOOKS IN REINFORCING STEEL		3. CONCRETE CONSTRUCTION: a. POST-INSTALLED ANCHORS b. REINFORCING STEEL PLACING c. FORMWORK d. CONCRETE MATERIALS AND PLACEMENT		3. SUBMITTAL REQUIREMENTS: a. ALL SHOP DRAWINGS MUST BE REVIEWED AND ELECTRONICALLY STAMPED BY THE CONTRACTOR PRIOR TO SUBMITTAL b. CONTRACTOR SHALL PROVIDE THE SUBMITTAL IN ELECTRONIC PORTABLE DOCUMENT FORMAT (PDF) PER THE SPECIFICATIONS. c. THE OMISSION FROM THE SHOP DRAWINGS OF ANY MATERIALS REQUIRED BY THE CONTRACT DOCUMENTS TO BE FURNISHED SHALL NOT RELIEVE THE CONTRACTOR OF THE RESPONSIBILITY OF FURNISHING AND INSTALLING SUCH MATERIALS, REGARDLESS OF WHETHER THE SHOP DRAWINGS HAVE BEEN REVIEWED AND APPROVED.			
A. GEOTECHNICAL REPORT		1. REFER TO DRAWING DETAILS FOR SPLICE AND HOOK REQUIREMENTS OF REINFORCING STEEL BEING SUPPLEMENTED.		4. PASSIVE FIRE PROTECTION SYSTEMS a. SPRAYED FIRE-RESISTANT MATERIALS b. MASTIC AND INTUMESCENT FIRE-RESISTANT COATINGS c. FIRE BARRIER JOINTS AND PENETRATIONS		4. CHECK STRUCTURAL STEEL AS RECEIVED IN THE FIELD FOR POSSIBLE SHIPPING DAMAGE, WORKMANSHIP, AND IDENTIFICATION MARKING TO CONFORM TO AISC 360 FOR STRUCTURAL STEEL AND SPECIFIED ASTM STANDARDS FOR OTHER STEEL.			
1. FOUNDATION DESIGN IS BASED ON THE GEOTECHNICAL INVESTIGATION REPORT AS FOLLOWS 480 SPRINGPARK PLACE - GENERATOR ADDITION PREPARED BY EGS MID-ATLANTIC, LLC DATED JANUARY 31, 2025. THE STRUCTURAL FOUNDATION DESIGN IS PENDING AN ADDENDUM TO THE GEOTECHNICAL REPORT CONTAINING REQUIRED UPDATED FOUNDATION RECOMMENDATIONS.		PART VI - STRUCTURAL STEEL		A. MATERIAL		5. VERIFY THAT SURVEYS ARE OCCURRING AS SPECIFIED TO CHECK PLUMBNESS AND FRAME ALIGNMENT AS ERECTION PROGRESSES. REVIEW THE SUBMITTED SURVEY REPORT.			
2. THE GEOTECHNICAL REPORT IS AVAILABLE TO THE GENERAL CONTRACTOR UPON REQUEST TO THE OWNER. THE INFORMATION INCLUDED THEREIN MAY BE USED BY THE GENERAL CONTRACTOR FOR HIS GENERAL INFORMATION ONLY. THE ARCHITECT AND ENGINEER WILL NOT BE RESPONSIBLE FOR THE ACCURACY OR APPLICABILITY OF SUCH DATA THEREIN.		1. ASTM SPECIFICATION AND GRADE: CLEARLY MARK THE GRADE OF STEEL ON EACH PIECE, WITH A DISTINGUISHING MARK VISIBLE FROM FLOOR SURFACES, FOR THE PURPOSE OF FIELD INSPECTION OF PROPER GRADE OF STEEL. UNLESS NOTED OTHERWISE ON THE DRAWINGS, STRUCTURAL STEEL SHALL BE AS FOLLOWS: a. W-SHAPES: ASTM A992 OR ASTM A572, GRADE 50 b. OTHER STEEL: ANY OTHER STEEL NOT INDICATED OTHERWISE SHALL CONFORM TO ASTM A992 OR ASTM A572, GRADE 50, EXCEPT PLATES AND ANGLES THAT SHALL BE ASTM A36.		1. ALL BOLTS, 3/4" DIAMETER AND GREATER, FOR STRUCTURAL CONNECTIONS, SHALL CONFORM TO ASTM F3125 GRADE A325 TYPE 1 PLAIN, UNLESS INDICATED OTHERWISE ON THE DRAWINGS. THE NUTS, BOLTS, AND WASHERS SHALL BE SHIPPED TOGETHER AS AN ASSEMBLY, UNLESS INDICATED OTHERWISE ON THE DRAWINGS.		6. INSPECT 100% OF THE COLUMN COMPRESSION AND BASE JOINTS FOR VERIFICATION THAT GAPS IN CONTACT BEARING DO NOT EXCEED 1/16 IN. (1.5MM). GAPS GREATER THAN 1/16 (1.5MM) IN, BUT LESS THAN 1/4 IN. (6MM) SHALL BE REPORTED TO THE OWNER AND ENGINEER FOR ASSESSMENT. ALL GAPS GREATER THAN 1/4 IN. (6MM) SHALL BE SHIMMED.			

[illegible]

VSW

STEEL ANNOTATION STYLES

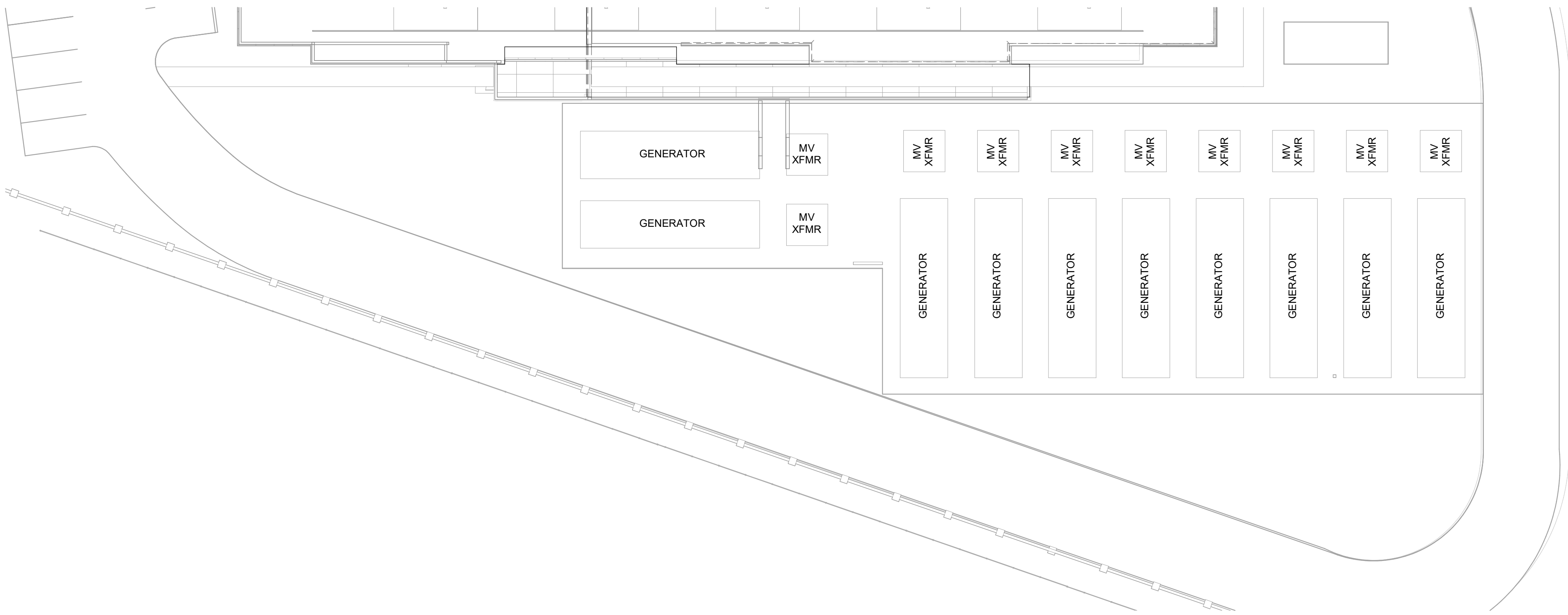


S-002

GROUND LEVEL MEP WEIGHT	
DESCRIPTION	OPERATIONAL WEIGHT
MV XFMR	22,170 LBS
GENERATOR	209,500 LBS

NOTE:

1. GENERATOR WEIGHT INCLUDES THE 3,000 KW GENERATOR AND THE BASE FUEL TANK AND FUEL LOCATED BENEATH IT.



10 GROUND LEVEL- SITE LOAD DIAGRAM
3/64" = 1'-0"

LEVEL 01 UPPER LEVEL PLAN			
DESCRIPTION	SUPER IMPOSED DEAD LOAD	LIVE LOAD	HATCH PATTERN
DATA HALL	65 PSF	350 PSF	

NOTES:

1.  INDICATES OPENING IN FLOOR.
2. ALL LOADS ARE UNFACTORED SERVICE LOADS.

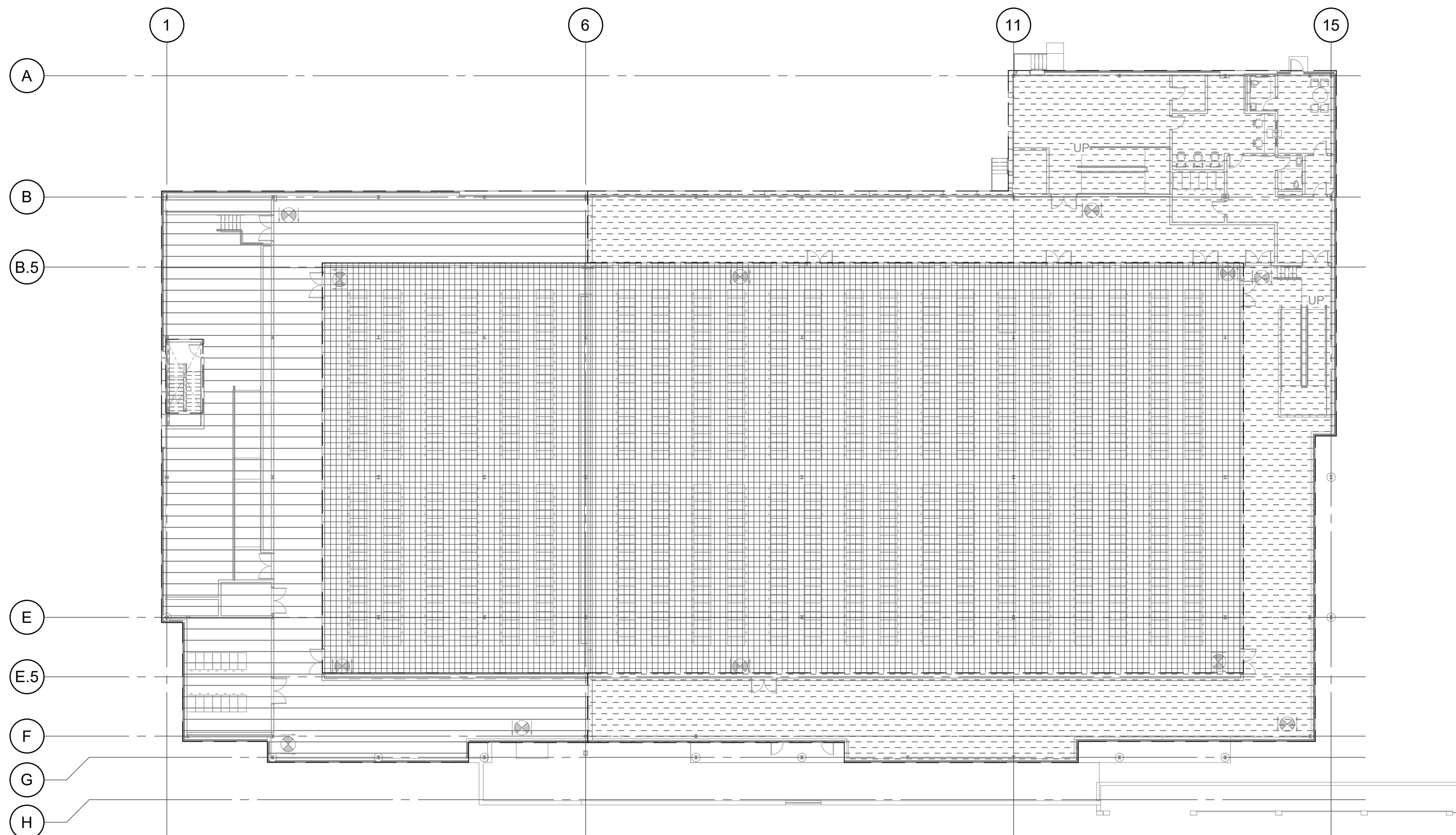


15 LEVEL 01 - LOAD DIAGRAM
3/64" = 1'-0"

LEVEL 02 UPPER LEVEL PLAN			
DESCRIPTION	SUPER IMPOSED DEAD LOAD	LIVE LOAD	HATCH PATTERN
DATA HALL	65 PSF	350 PSF	
GALLERY-OVER THE GRADE	-	250 PSF	
GALLERY-ELEVATED	65 PSF	200 PSF	

NOTES:

1.  INDICATES OPENING IN FLOOR.
2. DATA HALL DEAD LOAD INCLUDES MISCELLANEOUS RAISED FLOOR, HUNG GRID SUPPORT SYSTEM, COLLATERAL LOADS, OWNER-FURNISHED EQUIPMENT, AND STRUCTURAL STRENGTHENING SYSTEMS.
3. DATA HALL LIVE LOAD INCLUDES THE WEIGHT OF RACKS.
4. ALL LOADS ARE UNFACTORED SERVICE LOADS.



20 LEVEL 02 - LOAD DIAGRAM
3/64" = 1'-0"

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S-004 - LOADING DIAGRAMS / DESIGN CRITERIA

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Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026

VSW JOB NO. 250202 SCALE 3/64" = 1'-0"
LOADING DIAGRAMS / DESIGN CRITERIA

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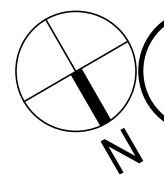
S-004

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LEVEL 03 ROOF PLAN		
DESCRIPTION	DEAD LOAD	HATCH PATTERN
SUB-FRAMING	30 PSF	

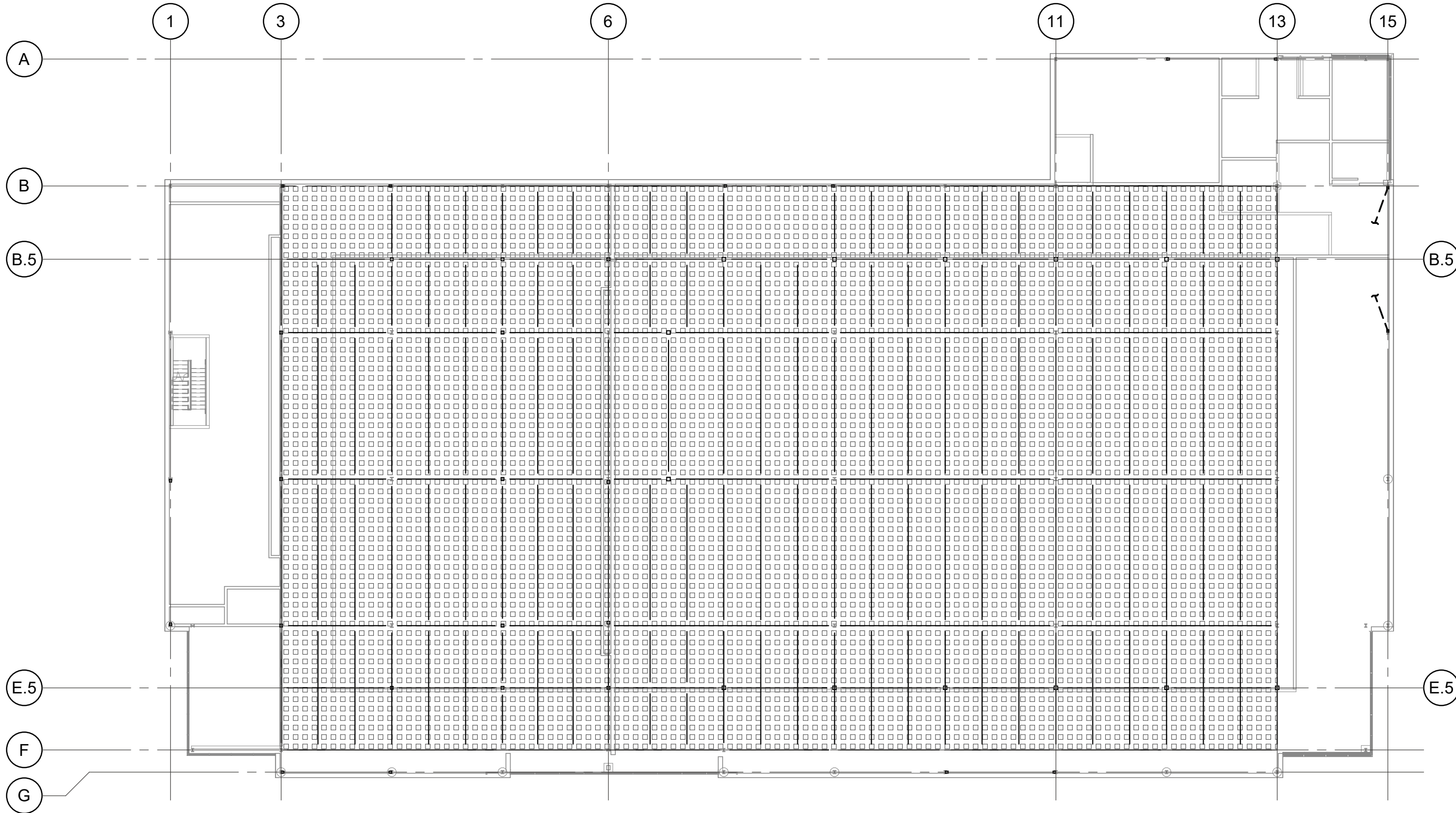
NOTES:

- ALL LOADS ARE UNFACTORED SERVICE LOADS.
- SUB-FRAMING LOAD APPLIES ON THE COLUMNS BELOW IT.



10 LEVEL 03 - SUB-FRAMING LOAD DIAGRAM

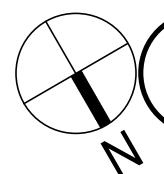
3/64" = 1'-0"



LEVEL 03 ROOF PLAN				
DESCRIPTION	DEAD LOAD	LIVE LOAD	FLAT ROOF SNOW LOAD	HATCH PATTERN
ROOF BELOW DUNNAGE	20 PSF	30 PSF	-	
ROOF - NO DUNNAGE ABOVE	20 PSF	30 PSF	56.7 PSF	

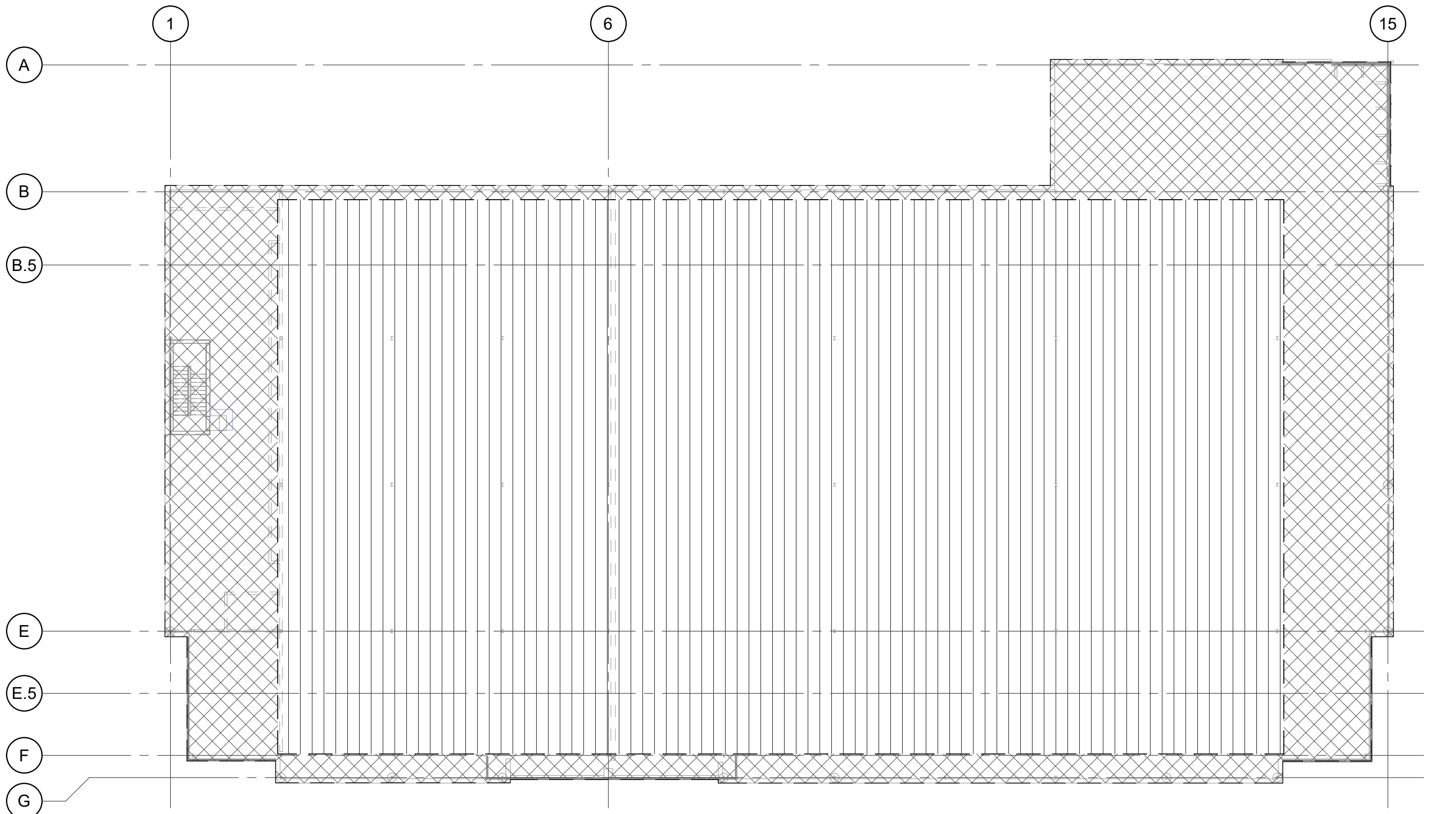
NOTES:

- ALL LOADS ARE UNFACTORED SERVICE LOADS.



15 LEVEL 03 - ROOF LOAD DIAGRAM

3/64" = 1'-0"

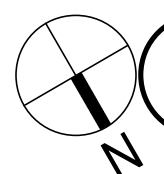


LEVEL 04 MEP WEIGHT ON DUNNAGE	
DESCRIPTION	OPERATIONAL WEIGHT
CHILLER	42,600 LBS
TES	65,200 LBS

NOTES:

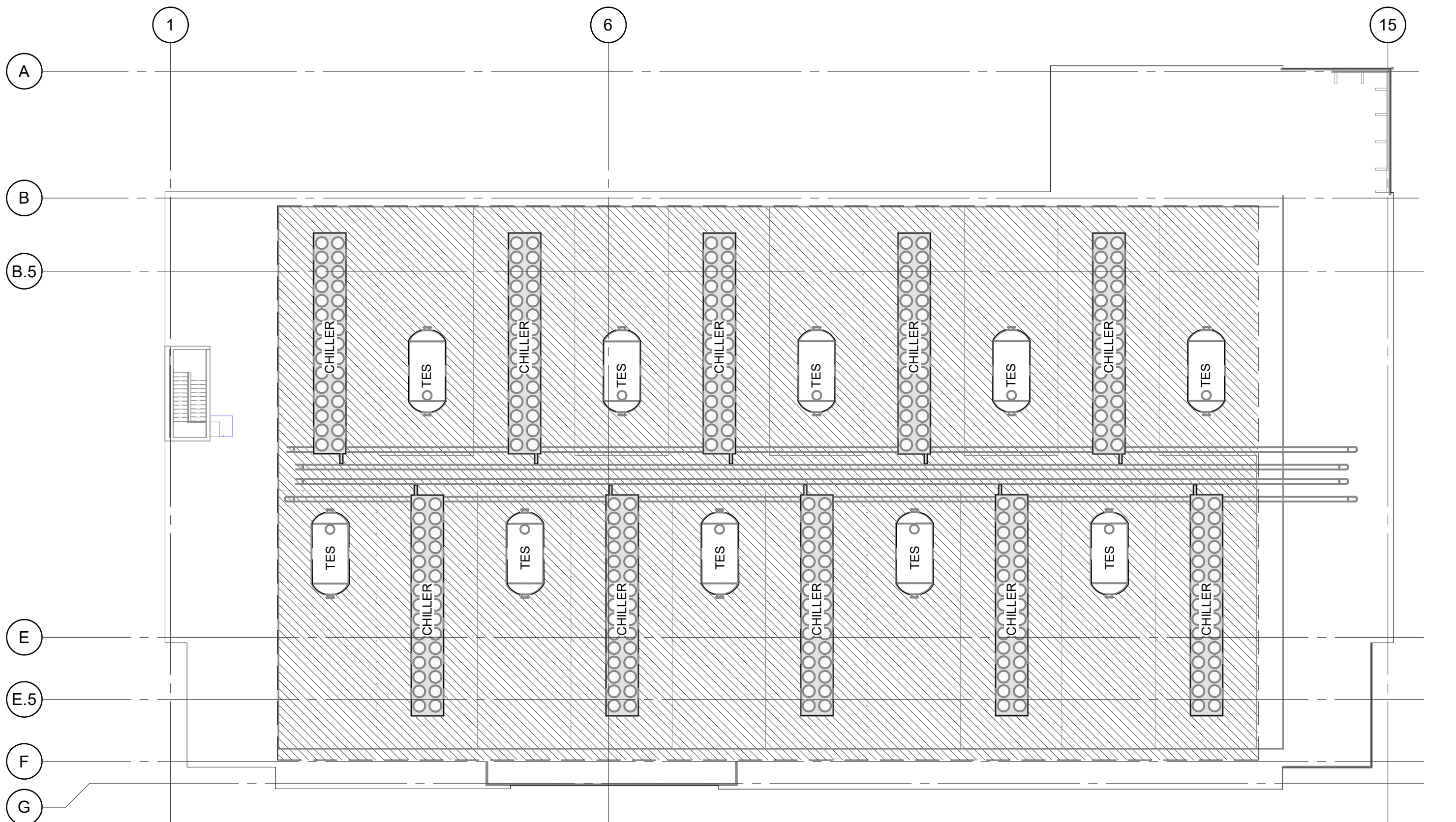
- ICE LOAD IS APPLIED AS A UNIFORM LINE LOAD (PLF) ALONG THE LENGTH OF EXPOSED STEEL DUNNAGE MEMBERS.
- ALL LOADS ARE UNFACTORED SERVICE LOADS.

LEVEL 04 DUNNAGE PLAN				
DESCRIPTION	SUPER IMPOSED DEAD LOAD	LIVE LOAD	FLAT ROOF SNOW LOAD	HATCH PATTERN
DUNNAGE	10 PSF	40 PSF	56.7 PSF	



20 LEVEL 04 - DUNNAGE LOAD DIAGRAM

3/64" = 1'-0"



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S-005 - LOADING DIAGRAMS / DESIGN CRITERIA

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Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

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04/06/2026

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EXISTING COLUMN SCHEDULE	
MARK	COLUMN SIZE
C-1	W8x24
C-2	W8x24
C-3	W8x31
C-4	W8x24
C-5	W8x28
C-6	W8x35
C-7	W8x28
C-8	W8x40
C-9	W8x58
C-10	W8x31
C-11	W8x48

RI-X OR RE-X
STRUCTURAL STEEL
COLUMN REINFORCING
DETAIL, SEE SCHEDULES ON
SHEET S-800

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04/06/26	DD SUBMISSION	

SECOND FLOOR FRAMING PLAN
DRAWING

S-101

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UPPER LEVEL PLAN NOTES:

1. CONTRACTOR SHALL FIELD VERIFY ALL EXISTING DIMENSIONS, ELEVATIONS, AND CONDITIONS SHOWN ON THE DRAWINGS. EXISTING INFORMATION IS BASED ON ORIGINAL STRUCTURAL DRAWINGS BY "CAD CONCEPTS INC." DATED OCTOBER 1, 1997. NOTIFY THE ARCHITECT AND STRUCTURAL ENGINEER OF ANY DISCREPANCIES.

2. TOP OF STEEL DUNNAGE ELEVATION = " " - UNLESS NOTED OTHERWISE.

3. CONTRACTOR TO NOTE THAT HEAVY WELDING OF SEVERAL MISCELLANEOUS PLATES ARE REQUIRED AT ALL COLUMN TO BEAM INTERFACES TO LOAD THROUGH 2ND AND ROOF LEVEL (NOT CURRENTLY SHOWN), TO BE DESIGNED AND DETAILED IN FUTURE SUBMISSIONS.

4. CONTRACTOR TO NOTE: MAJORITY OF THE EXISTING SLAB ON GROUND WILL NEED TO BE DEMOLISHED AND REPLACED TO MATCH EXISTING IN ORDER TO INSTALL NEW FOOTINGS AND FOOTING ENLIGHTENMENTS. EXISTING SLAB ON GROUND IS A STRUCTURAL MEMBER THAT BRACES THE EXISTING PERIMETER RETAINING WALLS.

5. CONTRACTOR TO PROVIDE TEMPORARY LATERAL BRACING OF EXISTING PERIMETER RETAINING WALLS AS REQUIRED TO ENSURE STRUCTURAL STABILITY. DEMO OF SLAB ON GROUND CAN ONLY BE PERFORMED IN LIMITED SECTIONS AND STRIPS AND NOT ALL AT ONCE. SEQUENCING REQUIREMENTS AND LIMITATIONS OF SLAB ON GROUND TO BE PROVIDED IN FUTURE SUBMISSIONS.

6. BF-X ON PLAN, DENOTES (N) FULL-HEIGHT STEEL BRACED FRAME.

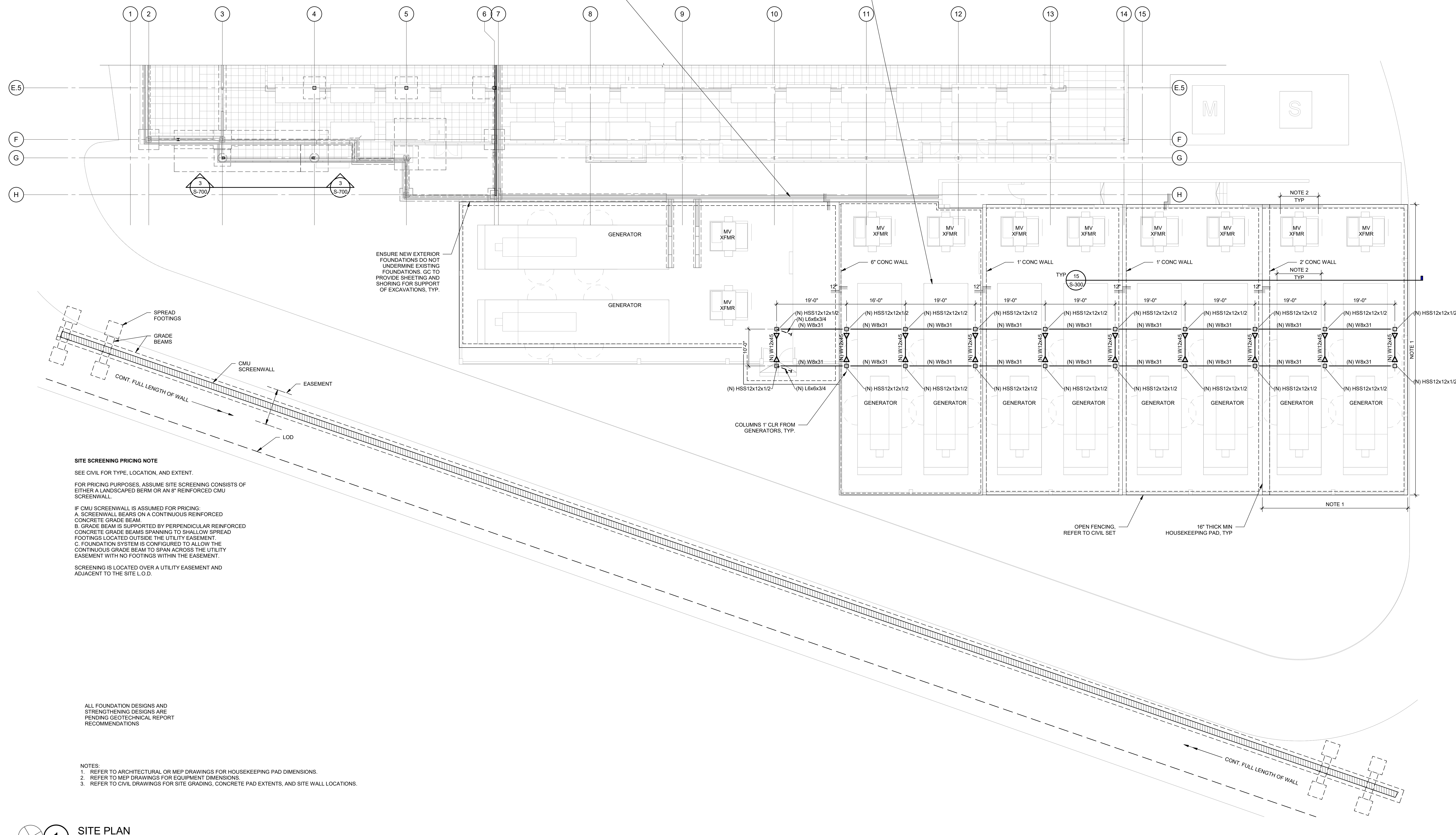


Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026VSW JOB NO. SCALE
250202 3/32" = 1'-0"SITE PLAN
DRAWING

S-102

EXISTING BUILDING FOUNDATION
PROTECTION - PRICING NOTECIVIL GRADING IS PRELIMINARY. NEW
CONSTRUCTION IS LOCATED ADJACENT TO
EXISTING BUILDING FOUNDATIONS.FOR PRICING PURPOSES, CONTRACTOR SHALL
ACCOUNT FOR MEANS AND METHODS
REQUIRED TO PREVENT UNDERMINING OR
LOSS OF SUPPORT OF EXISTING FOUNDATIONS,
INCLUDING TEMPORARY AND/OR PERMANENT
SUPPORT SYSTEMS AS NECESSARY.FINAL GRADING AND FOUNDATION DETAILS
WILL BE CONFIRMED IN FINAL DESIGN.OVERHEAD ELECTRICAL SUPPORT PILLARS -
PRICING NOTECONDUIT ROUTING, ELEVATION, AND
SUPPORT REQUIREMENTS ARE TBD. REFER
TO ELECTRICAL DRAWINGS.FOR PRICING PURPOSES, ASSUME OVERHEAD
CONDUITS ARE SUPPORTED BY (2) HSS 8x8
STEEL TUBE COLUMNS, APPROXIMATELY 15
FT TALL, WITH W8 STEEL BEAMS
APPROXIMATELY 6 FT LONG. THE POSTS ARE
SUPPORTED ON 24" DIAMETER REINFORCED
CONCRETE DRILLED PIERS OR ON THE
GENERATOR PAD.ASSUME (4) SUPPORT POSTS PER
GENERATOR LINEUP FOR PRICING ONLY.

SITE SCREENING PRICING NOTE

SEE CIVIL FOR TYPE, LOCATION, AND EXTENT.

FOR PRICING PURPOSES, ASSUME SITE SCREENING CONSISTS OF
EITHER A LANDSCAPED BERM OR AN 8" REINFORCED CMU
SCREENWALL.IF CMU SCREENWALL IS ASSUMED FOR PRICING:
A. SCREENWALL BEARS ON A CONTINUOUS REINFORCED
CONCRETE GRADE BEAM.
B. GRADE BEAM IS SUPPORTED BY PERPENDICULAR REINFORCED
CONCRETE GRADE BEAMS SPANNING TO SHALLOW SPREAD
FOOTINGS LOCATED OUTSIDE THE UTILITY EASEMENT.
C. FOUNDATION SYSTEM IS CONFIGURED TO ALLOW THE
CONTINUOUS GRADE BEAM TO SPAN ACROSS THE UTILITY
EASEMENT WITH NO FOOTINGS WITHIN THE EASEMENT.SCREENING IS LOCATED OVER A UTILITY EASEMENT AND
ADJACENT TO THE SITE L.O.D.ALL FOUNDATION DESIGNS AND
STRENGTHENING DESIGNS ARE
PENDING GEOTECHNICAL REPORT
RECOMMENDATIONSNOTES:
1. REFER TO ARCHITECTURAL OR MEP DRAWINGS FOR HOUSEKEEPING PAD DIMENSIONS.
2. REFER TO MEP DRAWINGS FOR EQUIPMENT DIMENSIONS.
3. REFER TO CIVIL DRAWINGS FOR SITE GRADING, CONCRETE PAD EXTENTS, AND SITE WALL LOCATIONS.

Revisions

02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

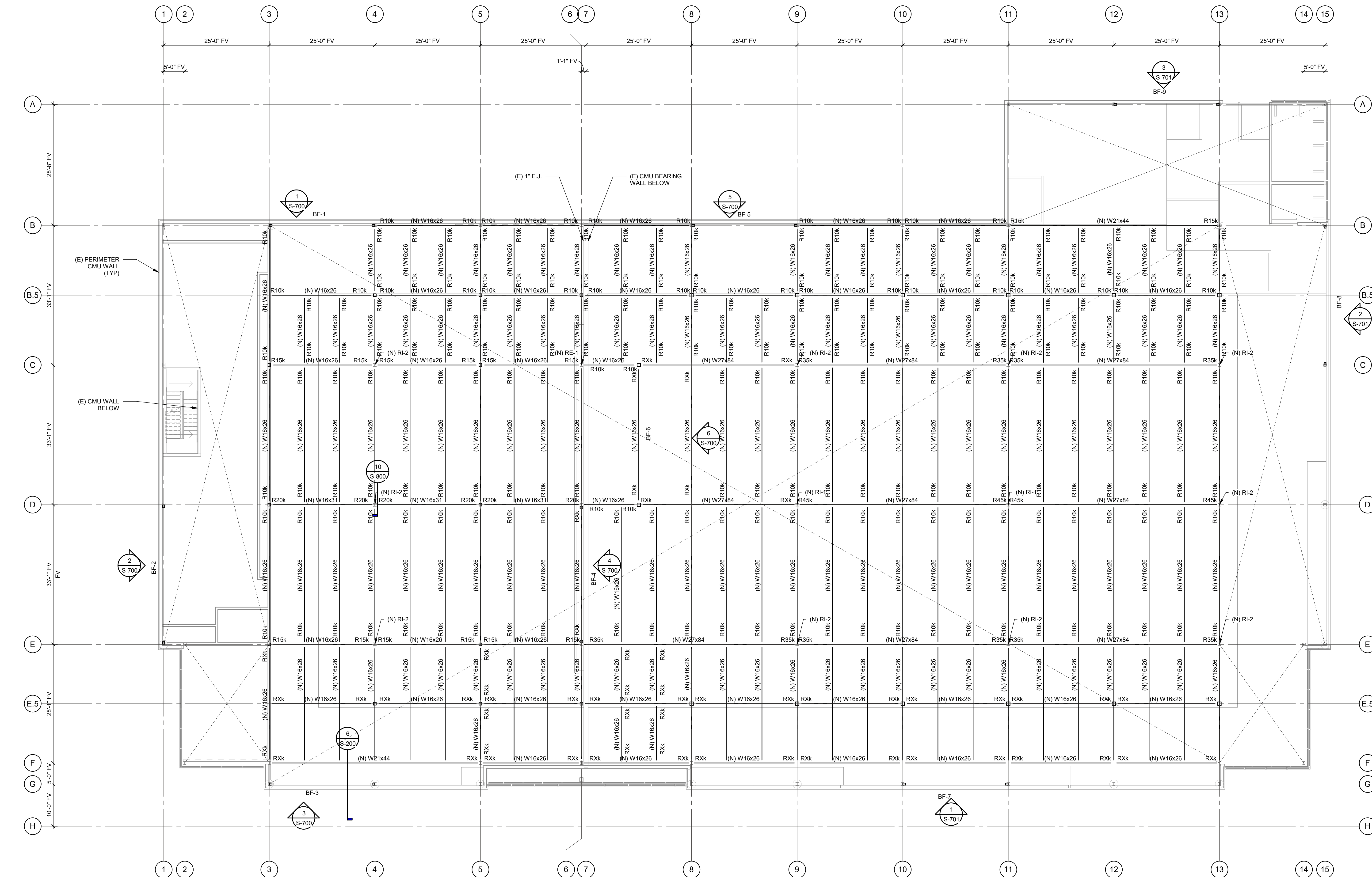
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04/06/2026VSW JOB NO. 250202 SCALE
As indicated

SUB-FRAMING PLAN

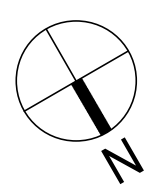
DRAWING

S-103

PLAN SYMBOLS:

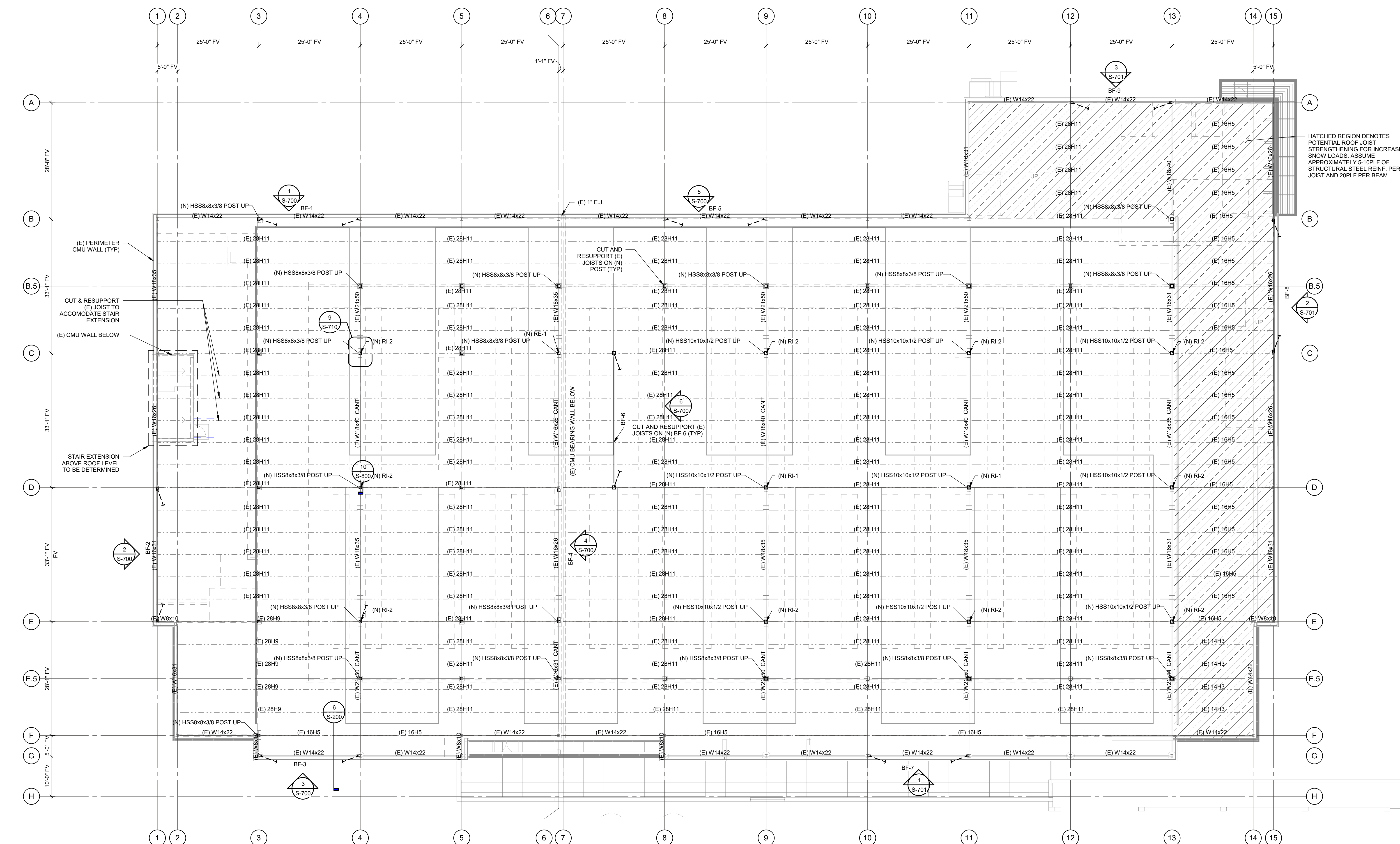
R1-X OR RE-X
STRUCTURAL STEEL
COLUMN REINFORCING
DETAIL, SEE SCHEDULES ON
SHEET S-800

- NOTES:
- STRUCTURAL STEEL IS INTENDED FOR INTERIOR, NON-OCCUPIABLE COLLATERAL SUPPORT FRAMING ONLY.
 - MAINTAIN 15'-6\"/>



1 SUB-FRAMING PLAN

3/32\"/>



1. ROOF PLAN NOTES:

1. THE EXISTING ROOF STRUCTURE CONSISTS OF STEEL DECK SPANNING BETWEEN STEEL JOISTS, AND SUPPORTED BY THE STEEL BEAMS/GIRDERS AND STEEL COLUMNS.

2. CONTRACTOR SHALL VERIFY ALL EXISTING DIMENSIONS AND ELEVATIONS SHOWN ON THE DRAWINGS. EXISTING DIMENSIONS AND ELEVATIONS SHOWN ARE BASED ON THE ORIGINAL STRUCTURAL DRAWINGS BY "CAD CONCEPTS INC." DATED OCTOBER 1, 1997.

3. THE EXISTING VENT AND VENT MECHANISM UNIT, SIZE, LOCATION, AND DIMENSIONS ARE SHOWN ON THE DRAWING. INFORMATION WITH MEP ENGINEER, "TECHNISTADJUST," IS REQUIRED TO VERIFY THE EXISTING MECHANISM UNIT IS THE FIELD TO AVOID INTERFERING WITH EXISTING JOIST MEMBERS.

4. AS TO THE EXISTING STRUCTURE, NO DRAWINGS FOR ADDITIONAL INFORMATION REQUIRED NOT SHOWN.

5. AT ALL EXISTING COLUMN TO BEAM CONNECTIONS AT THE ROOF LEVEL A TYPICAL DIAGONAL KICKER BRACE EXISTS (NOT CURRENTLY SHOWN ON PLAN). ASSUME BRACES TO REMAIN IN CURRENT LOCATIONS UNLESS NOTED OTHERWISE IN THE FUTURE SUBMISSIONS.

6. AT ALL EXISTING GIRDER TO GIRDER LEVELS, PROVIDE 2" X 4" BRACING AT THIRD POINTS WITH STEEL ANGLE MEMBERS EXTENDING FROM BOTTOM CHORD OF JOISTS TO BOTTOM FLANGE OF GIRDER.

7. CONTRACTOR TO NOTE THAT HEAVY WELDING OF SEVERAL MISCELLANEOUS PLATES ARE REQUIRED AT ALL COLUMN TO BEAM CONNECTIONS TO LOAD THROUGH 2ND AND ROOF LEVEL (NOT CURRENTLY SHOWN), TO BE DESIGNED AND DETAILED IN FUTURE SUBMISSIONS.

8. CONTRACTOR TO NOTE SOME EXISTING OPENINGS TO BE INFILLED AND SOME NEW ROOF DECK OPENINGS ARE REQUIRED (NOT CURRENTLY SHOWN). THE SHOWING OF THE OPENINGS TO BE DESIGNED AND DETAILED IN FUTURE SUBMISSIONS.

9. STEEL COLUMNS EXTENDING THROUGH ROOF SHALL INCLUDE A STRUCTURAL THERMAL BREAK BETWEEN INTERIOR AND EXTERIOR STEEL PLATES, LOCATED WITHIN THE ROOF INSULATION ZONE. INCLUDE ALLOWANCE FOR LOAD-TRANSFER-CAPABLE THERMAL BREAK MATERIAL.

ROOF DIAPHRAGM – PRICING NOTES:
GENERAL
NOTES BELOW ARE INTENDED FOR CONTRACTOR PRICING PURPOSES ONLY AND REPRESENT CONCEPTUAL SCOPE FOR ROOF DIAPHRAGM STRENGTHENING AT THIS STAGE. FINAL MEMBER SIZES, CONNECTION TYPES, FASTENERS, WELDS, AND INSTALLATION DETAILS SHALL BE DETERMINED DURING SUBSEQUENT DESIGN PHASES.

CASE A – ROOFING NOT REPLACED
PROVIDE DITCHING PER ARCHITECTURAL DRAWINGS AT ALL COLUMNS AND ROOF PENETRATIONS.
2. REINFORCE THE EXISTING ROOF DIAPHRAGM BY PROVIDING NEW STEEL ANGLE MEMBERS WELDED TO THE UNDERSIDE OF THE EXISTING STEEL ROOF DECK.
3. PROVIDE ADDITIONAL STEEL ANGLE MEMBERS AROUND THE ENTIRE ROOF PERIMETER, WELDED AT THE SUBFRAMING LEVEL, FOR DIAPHRAGM LOAD TRANSFER.

PRICING ASSUMPTIONS:
1. ASSUME ADDITIONAL STEEL EQUAL TO 5 PSF OVER THE ENTIRE ROOF AREA ASSOCIATED WITH THE DIAPHRAGM STRENGTHENING.

CASE B – ROOFING REPLACED (TO TOP OF STEEL ROOF DECK)
ROOFING IS REMOVED DOWN TO THE TOP OF THE EXISTING STEEL ROOF DECK.
1. ADDING SIDE-LAP AND END-LAP FASTENERS THROUGHOUT THE EXISTING ROOF DECK.
2. PROVIDING ADDITIONAL STEEL ANGLE MEMBERS AROUND THE ENTIRE ROOF PERIMETER, WELDED AT THE ROOF LEVEL, TO SUPPORT DIAPHRAGM LOAD TRANSFER.

CLARIFYING NOTE
THESE CASES REPRESENT ALTERNATIVE CONCEPTUAL APPROACHES FOR ROOF DIAPHRAGM STRENGTHENING. ONLY ONE CASE WOULD BE IMPLEMENTED IN CONSTRUCTION.



ROOF FRAMING PLAN
 3/32" = 1'-0"

ROOF FRAMING PLAN
DRAWING
S-104

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Revisions

NO.	DATE	DESCRIPTION
02/03/26	DESIGN DEVELOPMENT /PRICING	
1 04/06/26	DD SUBMISSION	

ISSUED:
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04/06/2026VSW JOB NO. SCALE
250202 As indicated

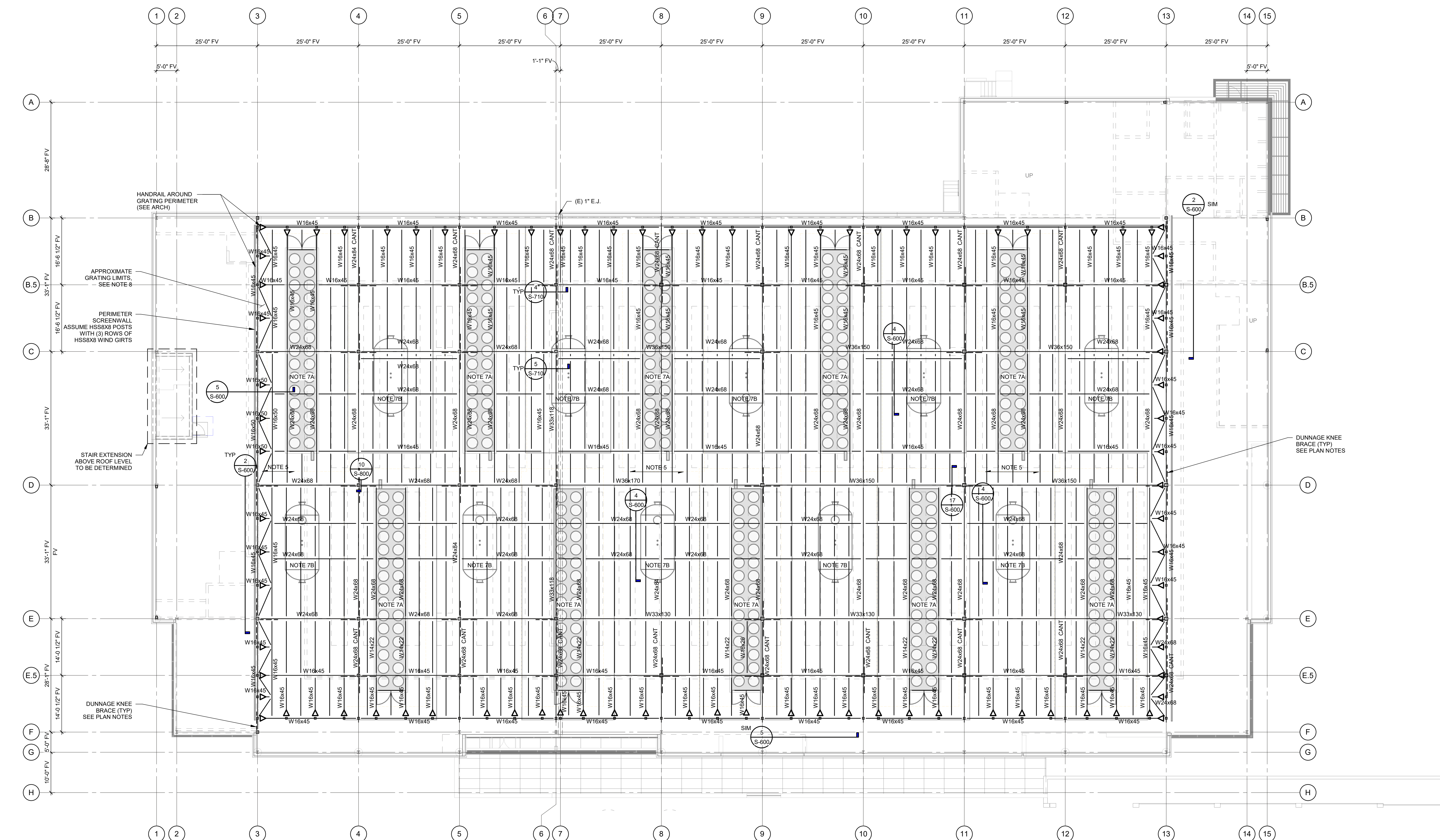
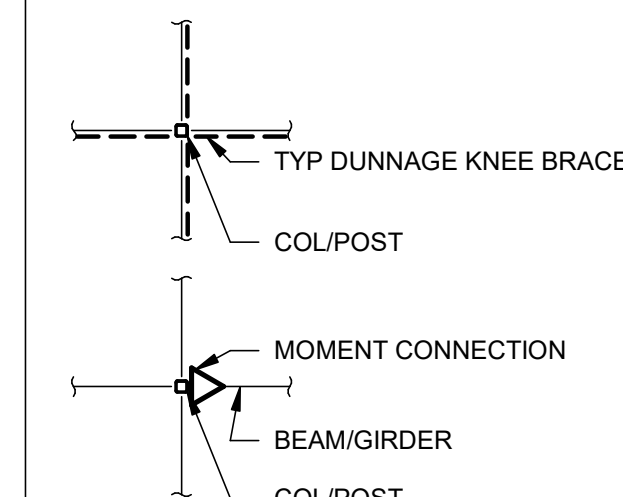
DUNNAGE FRAMING PLAN

DRAWING

S-105

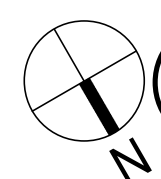
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PLAN SYMBOLS:



ROOF DUNNAGE PLAN NOTES:

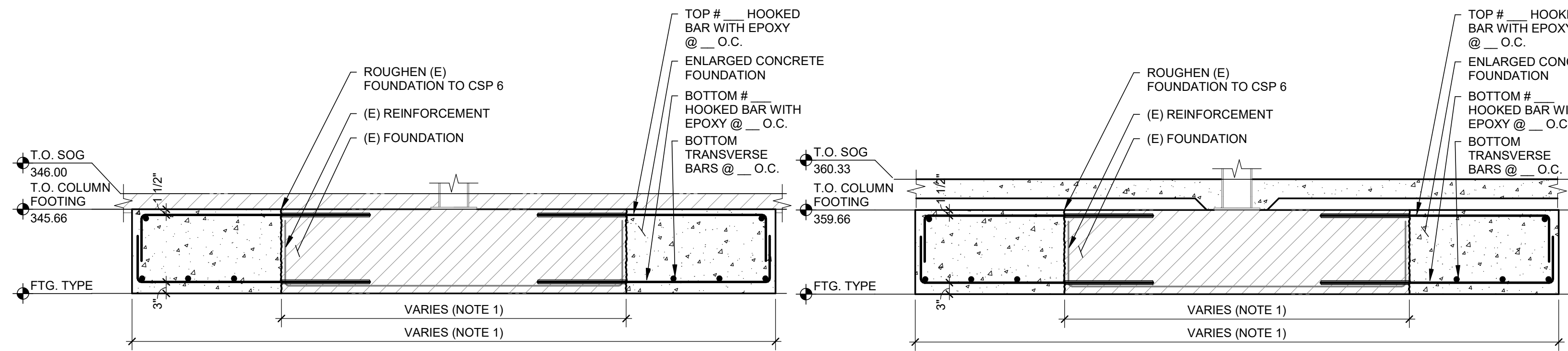
- CONTRACTOR SHALL FIELD VERIFY ALL EXISTING DIMENSIONS, ELEVATIONS, AND CONDITIONS SHOWN ON THE DRAWINGS. EXISTING INFORMATION IS BASED ON ORIGINAL STRUCTURAL DRAWINGS BY "CAD CONCEPTS INC." DATED OCTOBER 1, 1997. NOTIFY THE ARCHITECT AND STRUCTURAL ENGINEER OF ANY DISCREPANCIES.
- TOP OF STEEL DUNNAGE ELEVATION = " " UNLESS NOTED OTHERWISE.
- ALL NEW STEEL INCLUDING CONNECTIONS AND ATTACHMENTS TO BE HOT-DIP GALVANIZED. REPAIR GALVANIZATION AT SITE. WELDED AREAS AND OTHER AREAS DAMAGED DURING CONSTRUCTION AS PER STEEL SPECIFICATIONS.
- ALL STEEL DUNNAGE POST TO BE PLACED CONCENTRICALLY ON TOP OF EXISTING STEEL COLUMNS OF BUILDING, UNLESS NOTED OTHERWISE.
- PROVIDE 19-W4 SERRATED WELDED STEEL BAR GRATING WITH 1-1/4" x 3/16" SPACING AS INDICATED. MAXIMUM SUPPORT SPACING SHALL NOT EXCEED 4'-0" O.C. PROVIDE BAND BAR AT ALL PANEL PERIMETERS, EXPOSED EDGES. ATTACH GRATING TO STRUCTURAL STEEL IN ACCORDANCE WITH MANUFACTURER'S RECOMMENDATIONS.
- DRAWINGS INDICATE THE APPROXIMATE LOCATIONS OF MECHANICAL UNITS. REFER TO SHEET S-710 FOR STEEL DUNNAGE FRAMING DETAILS. ACTUAL LOCATIONS, SIZES, AND CONNECTIONS OF MECHANICAL UNITS SHALL BE COORDINATED WITH THE MECHANICAL DRAWINGS.
- MAXIMUM OPERATING WEIGHTS FOR MECHANICAL UNITS ARE SHOWN ON PLAN AND SHALL NOT BE EXCEEDED. IF MECHANICAL UNITS EXCEED THE MAXIMUM OPERATING WEIGHTS SHOWN, NOTIFY THE ENGINEER OF RECORD.
A. CHILLER UNIT: MAXIMUM OPERATING WEIGHT = 42,600 LBS PER UNIT.
B. TES TANK: MAXIMUM OPERATING WEIGHT = 65,200 LBS PER UNIT.
- FULL GRATING COVERAGE WITHIN THE STEEL DUNNAGE AREA IS REQUIRED UP TO THE INTERIOR FACE OF THE SCREENWALL. ADDITIONAL STEEL FRAMING AND/OR MEMBER RECONFIGURATION MAY BE NECESSARY AS DESIGN PROGRESSES. CONTRACTOR TO CARRY CONTINGENCY FOR ASSOCIATED STRUCTURAL MODIFICATIONS.
- ROOF DUNNAGE ACCESS STAIR SHALL BE A DELEGATED DESIGN BY A LICENSED PROFESSIONAL ENGINEER IN THE PROJECT JURISDICTION. CONTRACTOR TO PROVIDE STAIR SYSTEM, CONNECTIONS, AND SUPPORTING MEMBERS WITH SHOP DRAWINGS AND CALCULATIONS FOR REVIEW.
- ALL INFILL BEAMS NOT SPECIFICALLY LABELED ON PLAN SHALL BE W10x22. REFER TO FRAMING PLAN FOR MEMBER ORIENTATION.
- ALL KNEE BRACES SHOWN ON PLAN SHALL BE HSS4x4x1/4. INSTALL KNEE BRACES AFTER FINAL INSTALLATION OF ALL TES AND CHILLER EQUIPMENT.



1

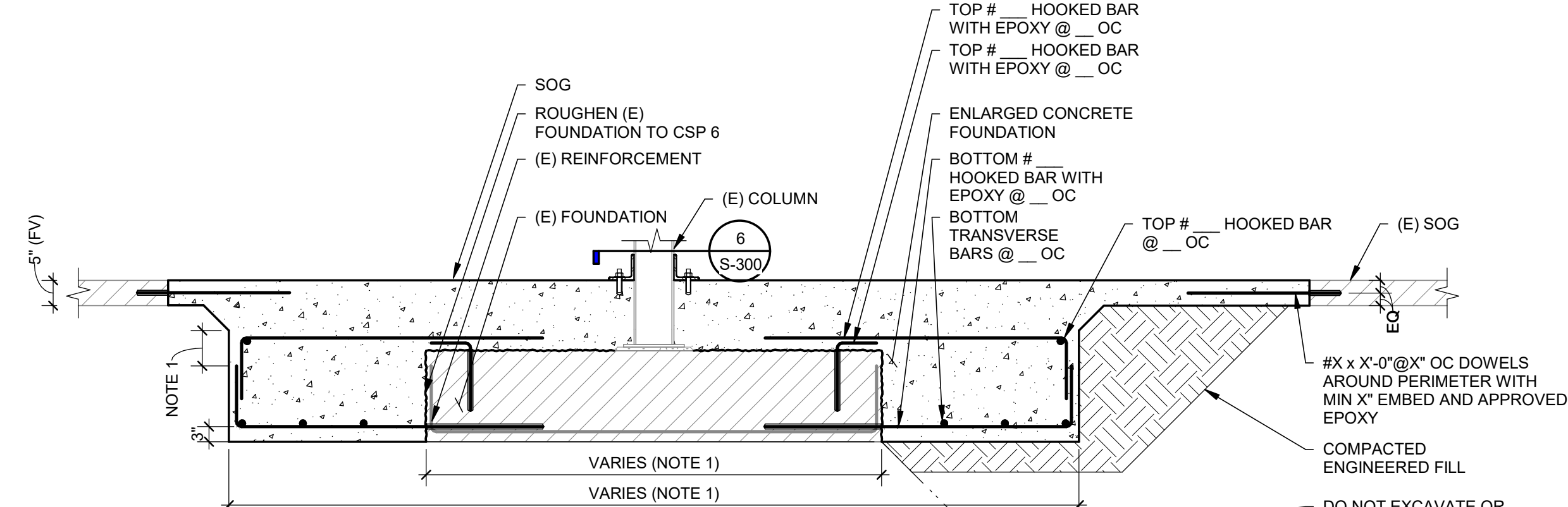
DUNNAGE FRAMING PLAN

3/32" = 1'-0"

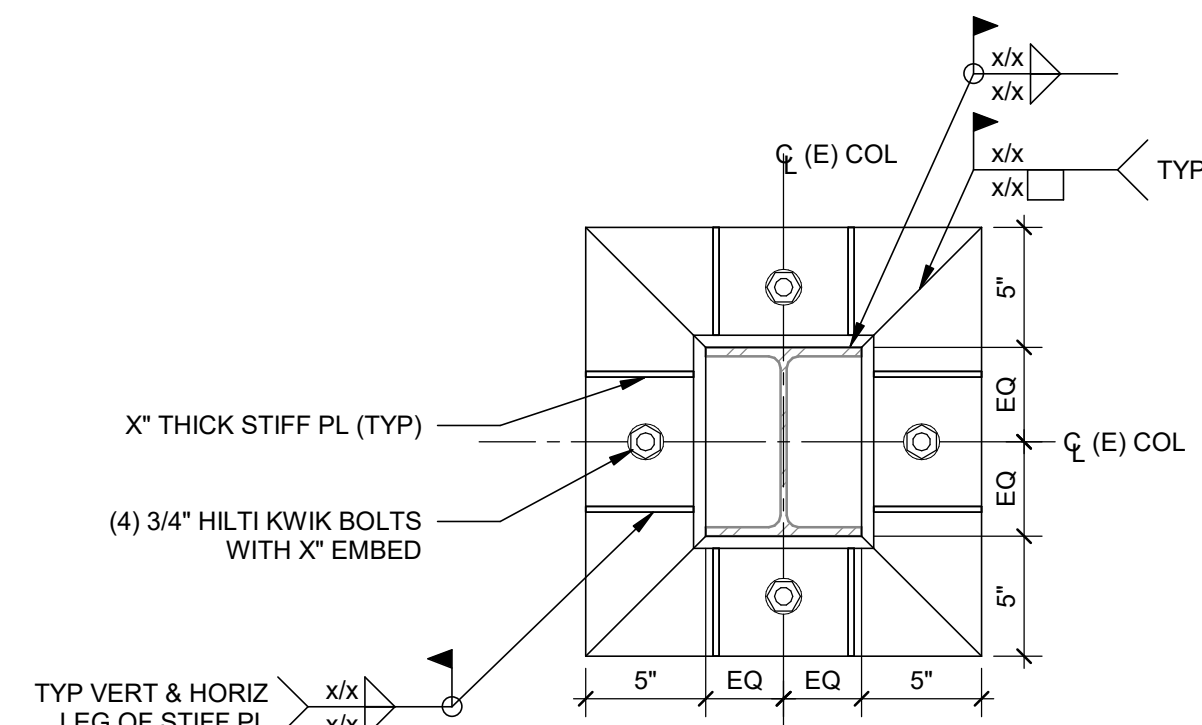


4 TYPICAL DETAIL AT G1-G7
1/2" = 1'-0"

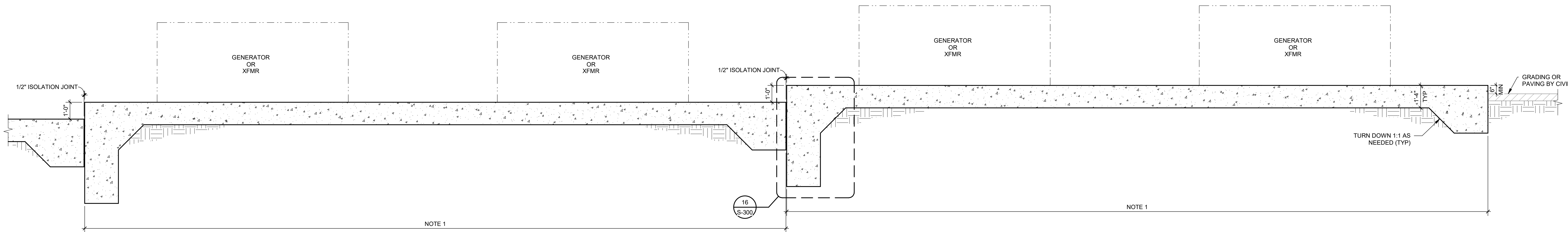
3 TYPICAL DETAIL AT G7-G15
1/2" = 1'-0"



2 TYPICAL FOOTING ENLARGEMENT
1/2" = 1'-0"

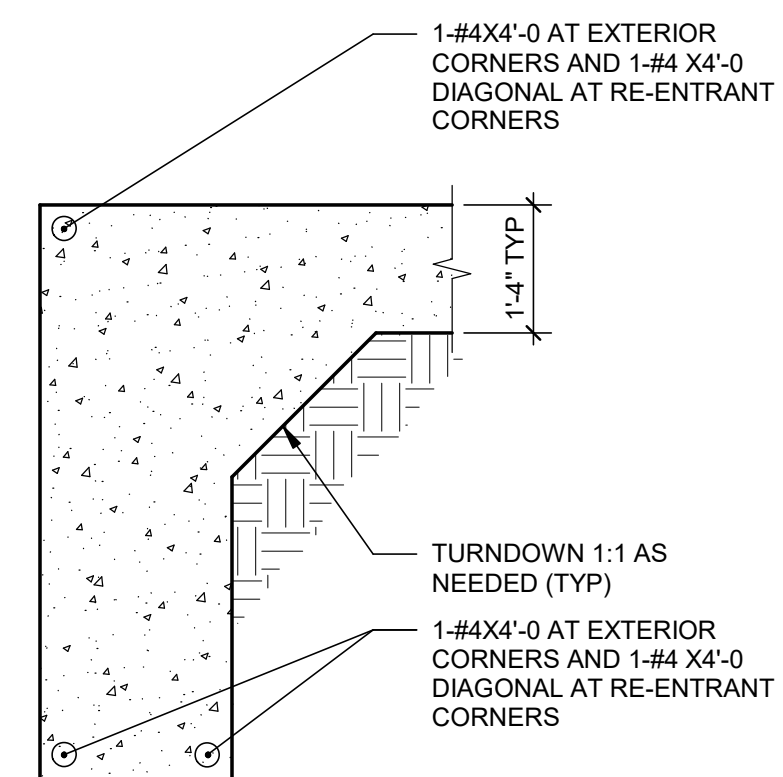


6 STEEL COLLAR AT EXISTING COLUMN
1 1/2" = 1'-0"



NOTES:
1. REFER TO ARCHITECTURAL AND MEP DRAWINGS FOR THE EXTENT OF HOUSEKEEPING PAD.

15 TYPICAL HOUSEKEEPING PAD
3/8" = 1'-0"



16 TYPICAL HOUSEKEEPING PAD REINFORCEMENT
1/2" = 1'-0"

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S-300 - TYPICAL DETAILS

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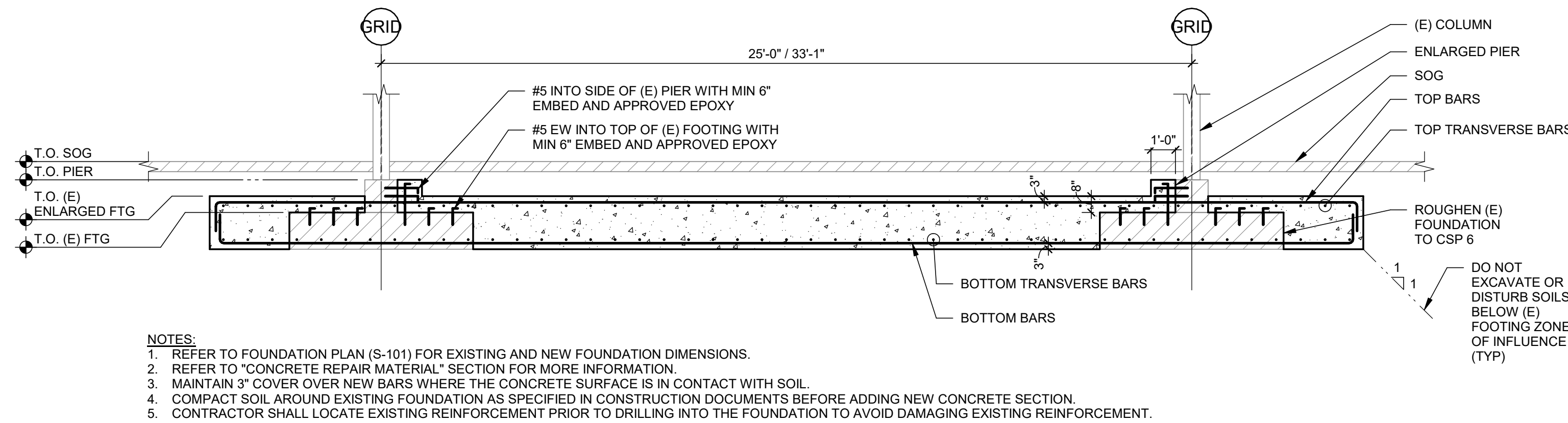
Revisions
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DEVELOPMENT
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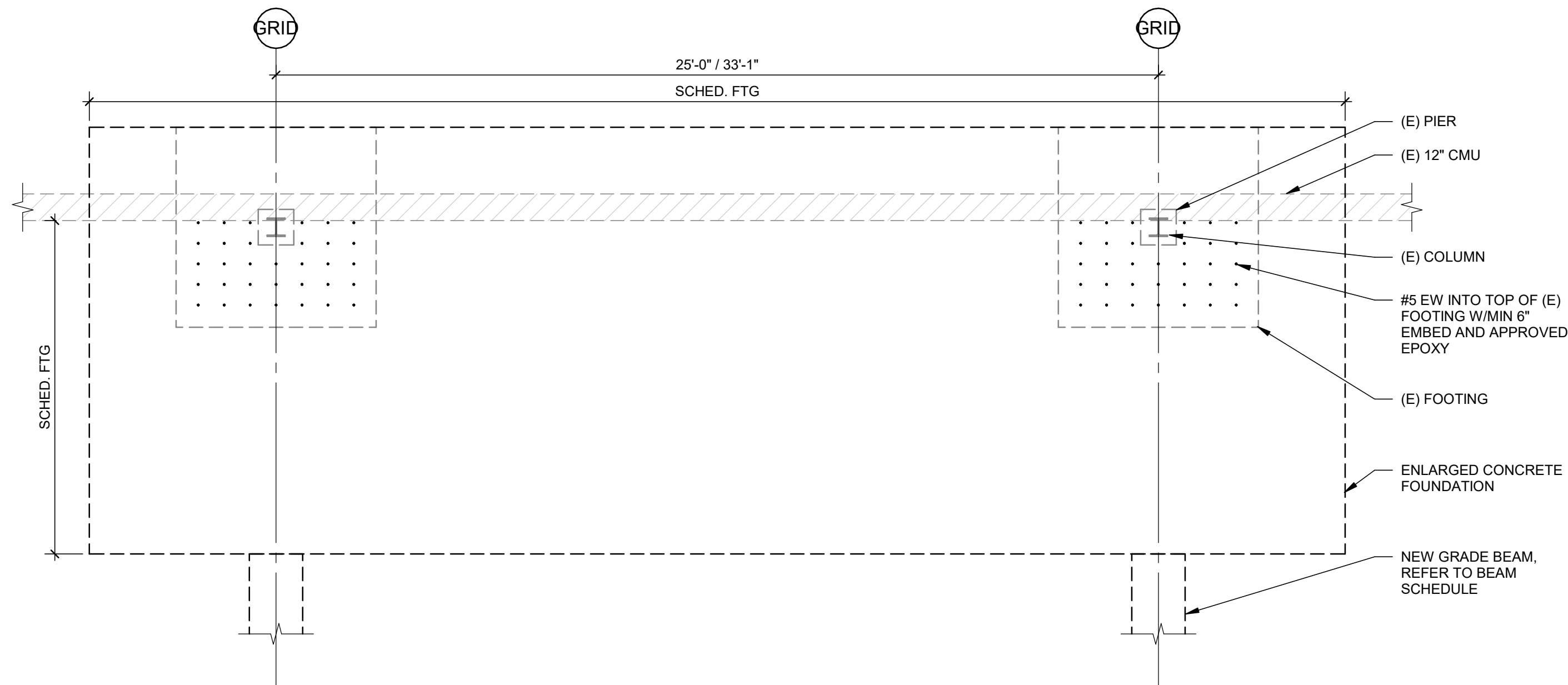
VSW JOB NO. SCALE
250202 As indicated

TYPICAL DETAILS
DRAWING
S-300

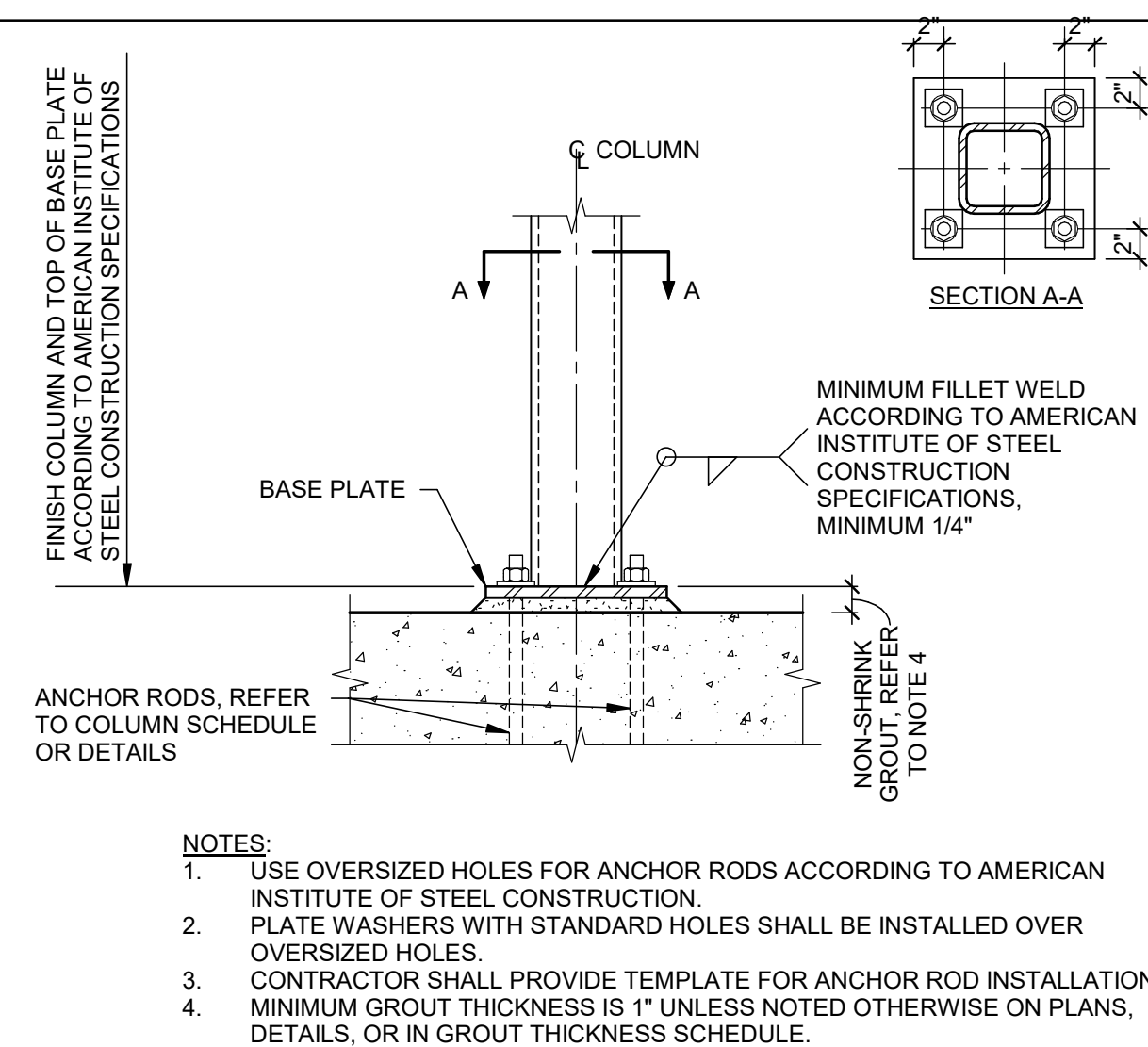
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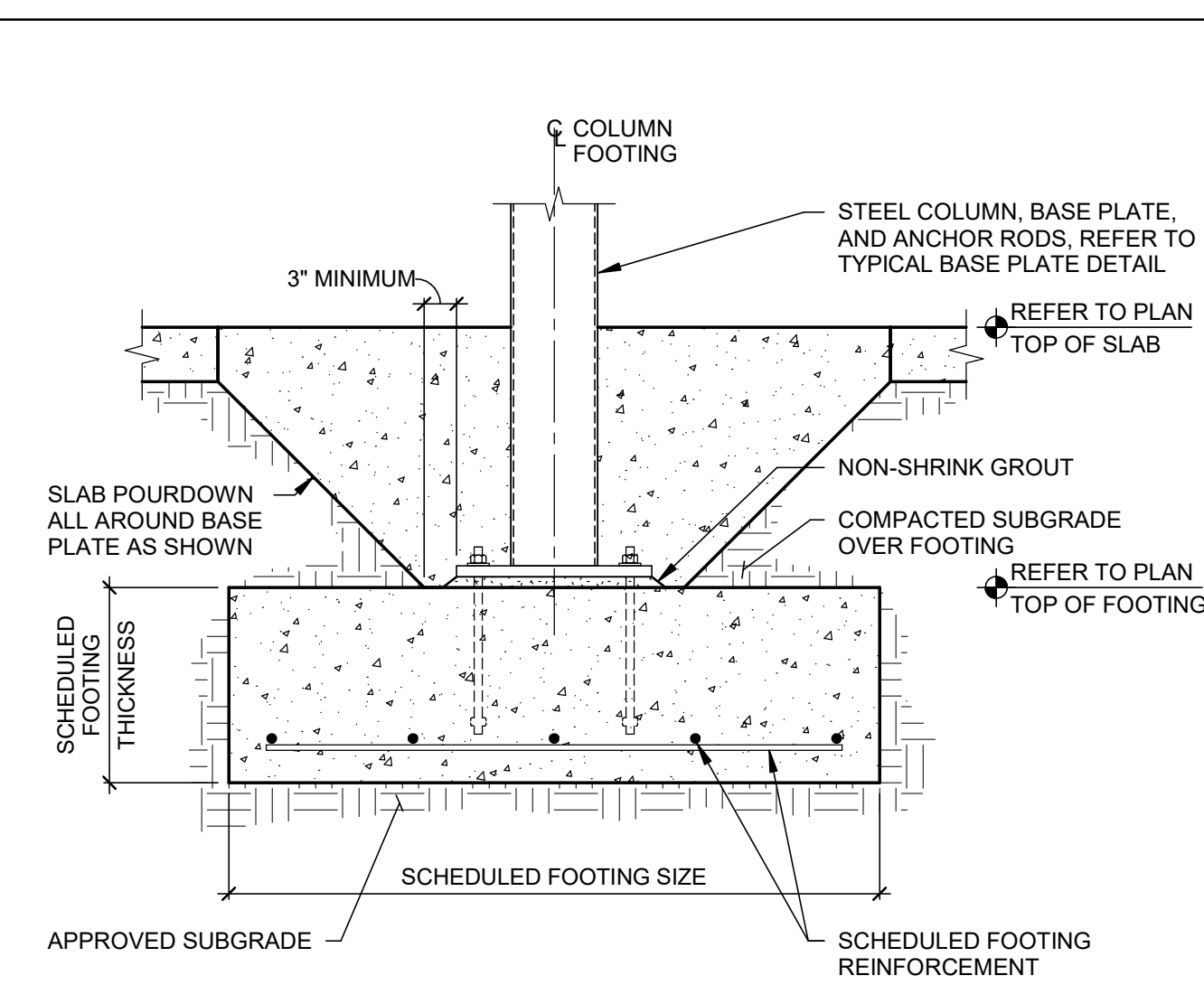
4 TYPICAL FOOTING ENLARGEMENT FOR BRACED FRAME
1/4" = 1'-0"



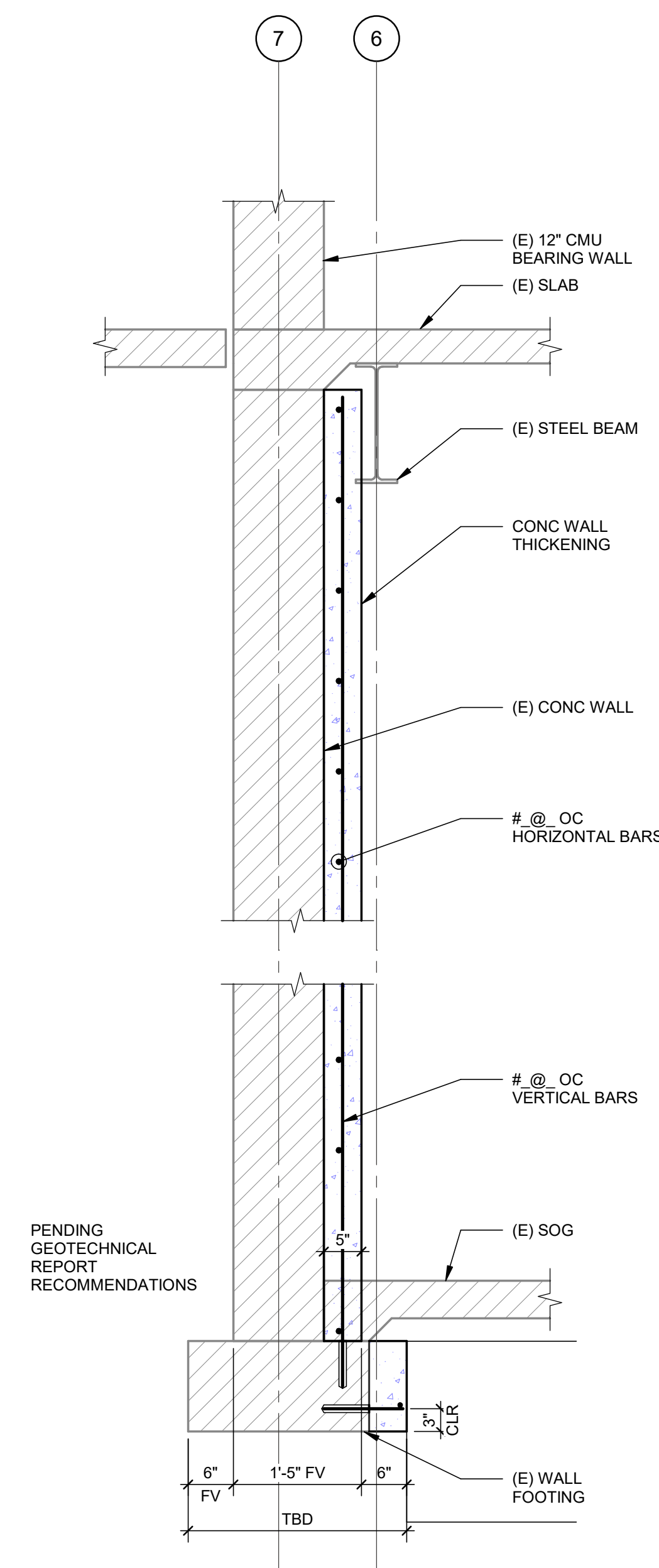
9 TYPICAL FOOTING ENLARGEMENT AT BRACED FRAME - PLAN
1/4" = 1'-0"



2 TYPICAL RECTANGULAR/SQUARE HSS COLUMN BASE CONNECTION
NO SCALE



1 TYPICAL STEEL COLUMN ON SPREAD FOOTING
NO SCALE



16 CONCRETE BASEMENT WALL STRENGTHENING DETAIL
3/4" = 1'-0"

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S-400 - FOUNDATION DETAILS AND SECTIONS

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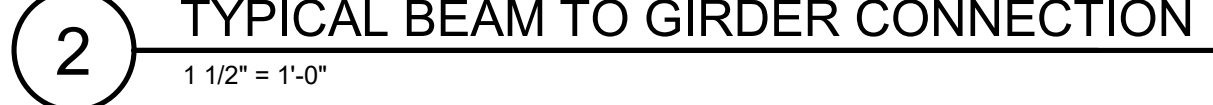
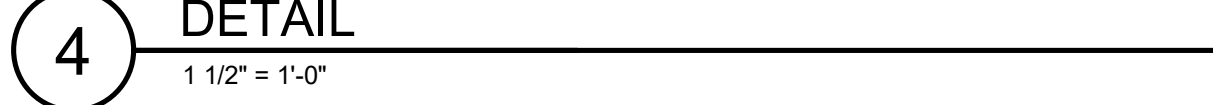
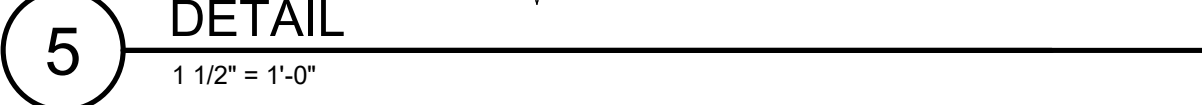
VSW JOB NO. 250202 SCALE As indicated

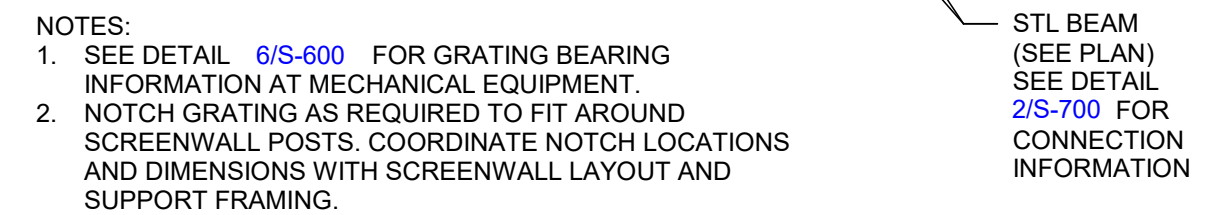
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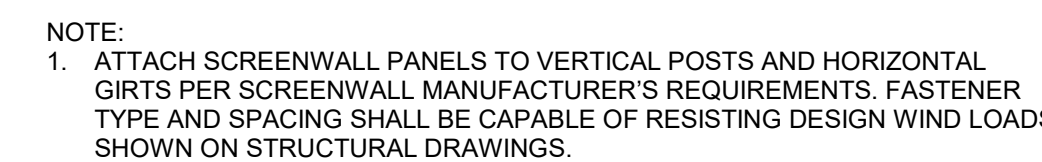
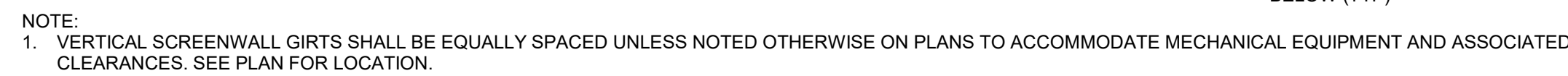
DRAWING

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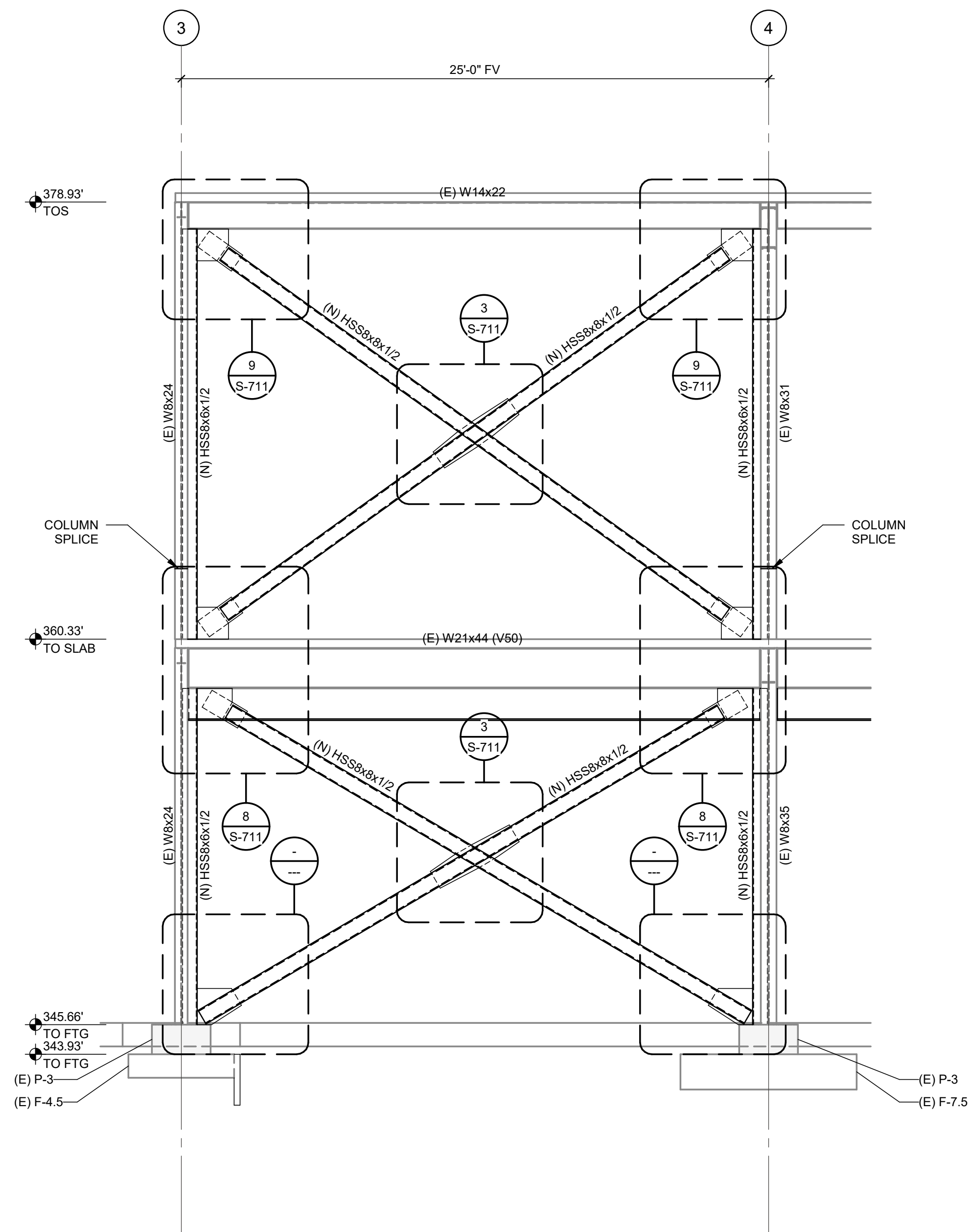
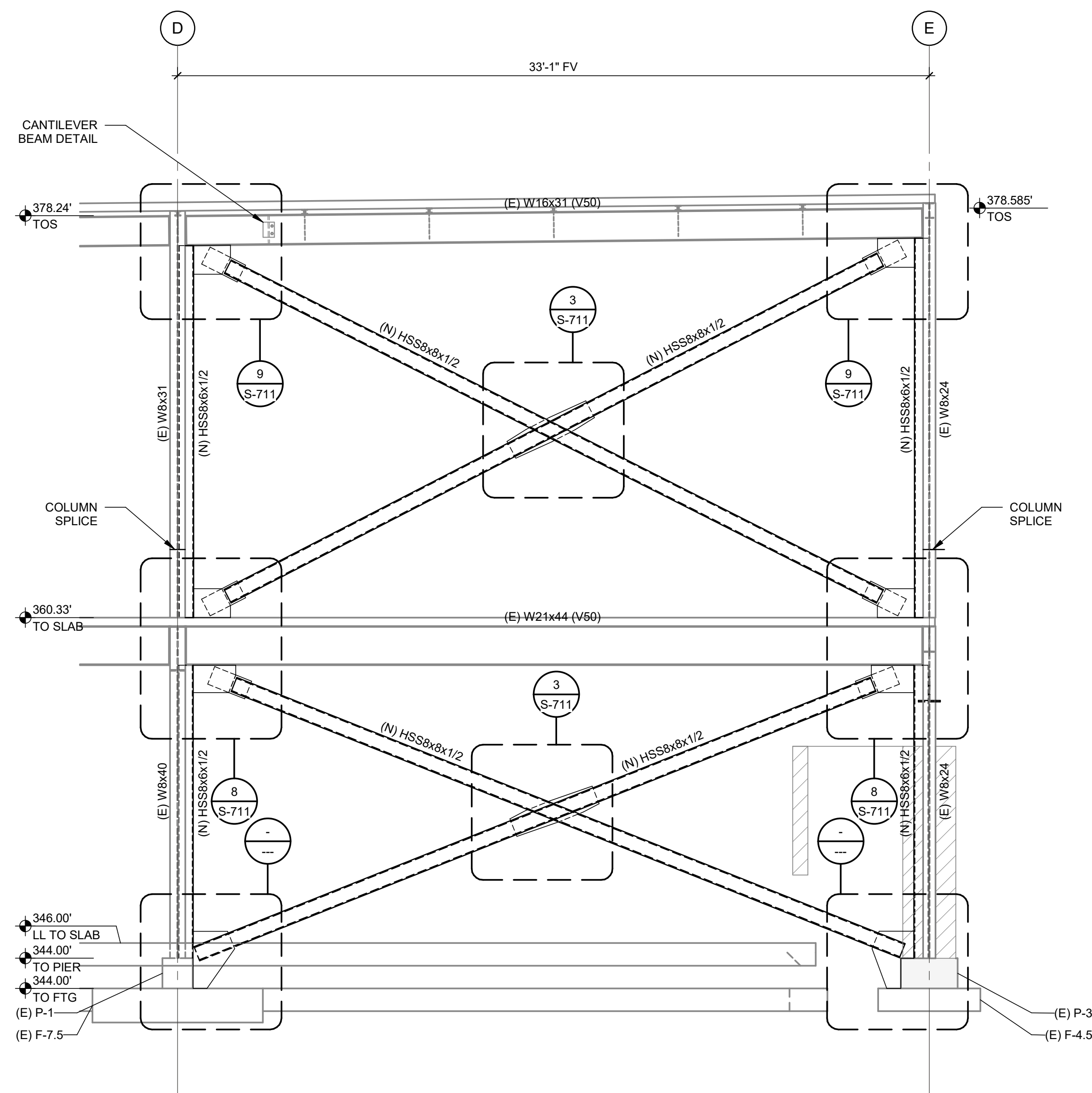
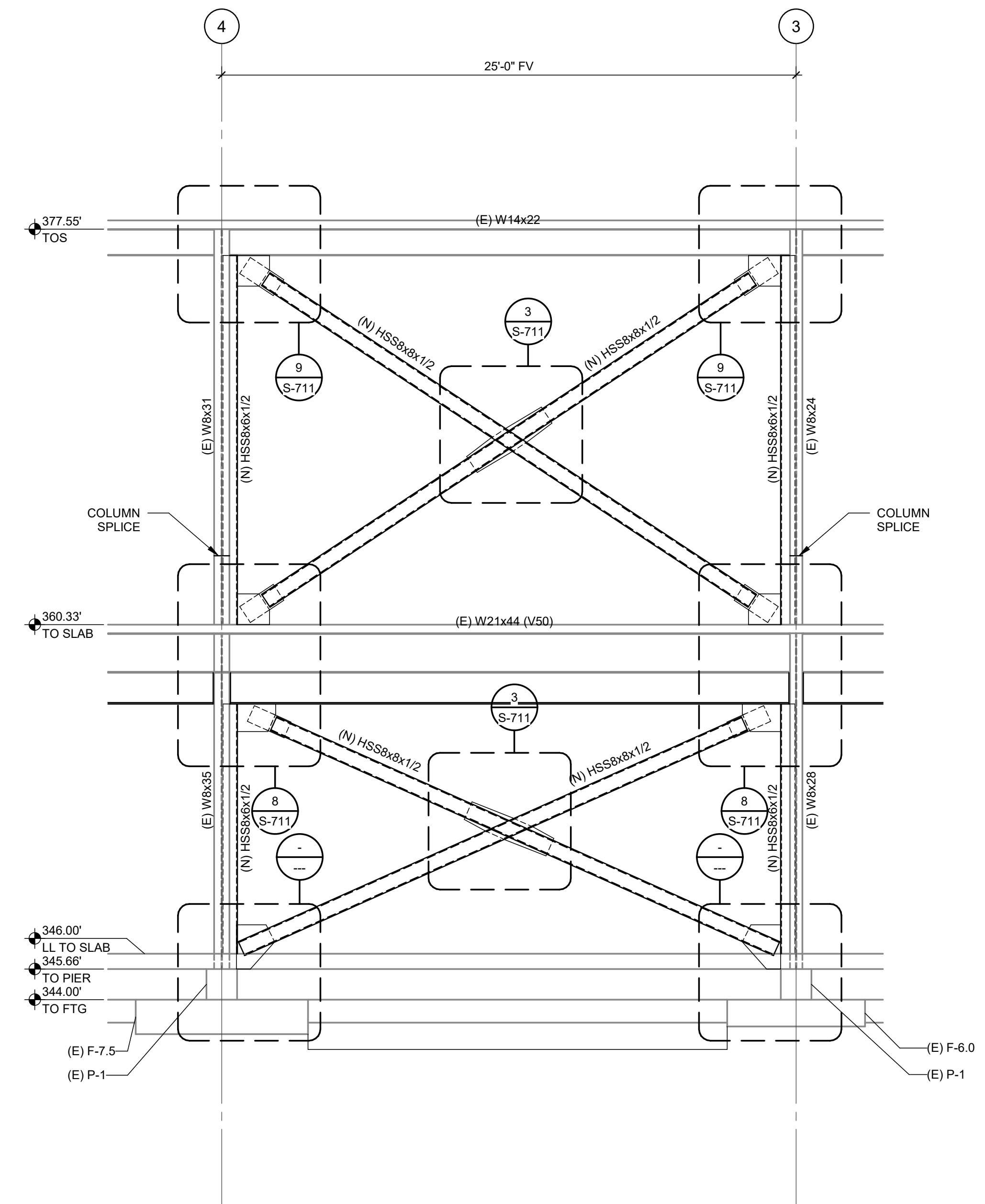
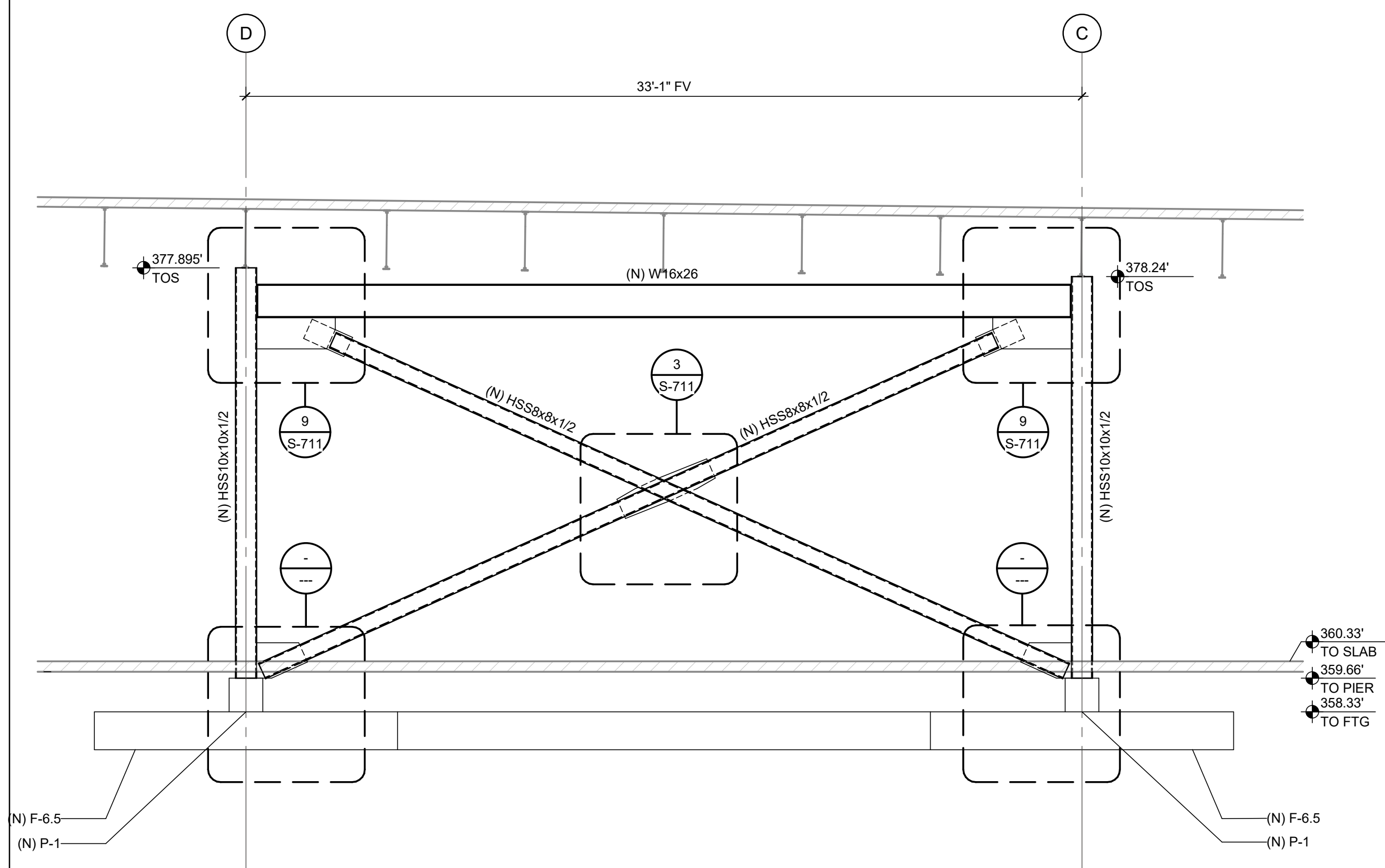
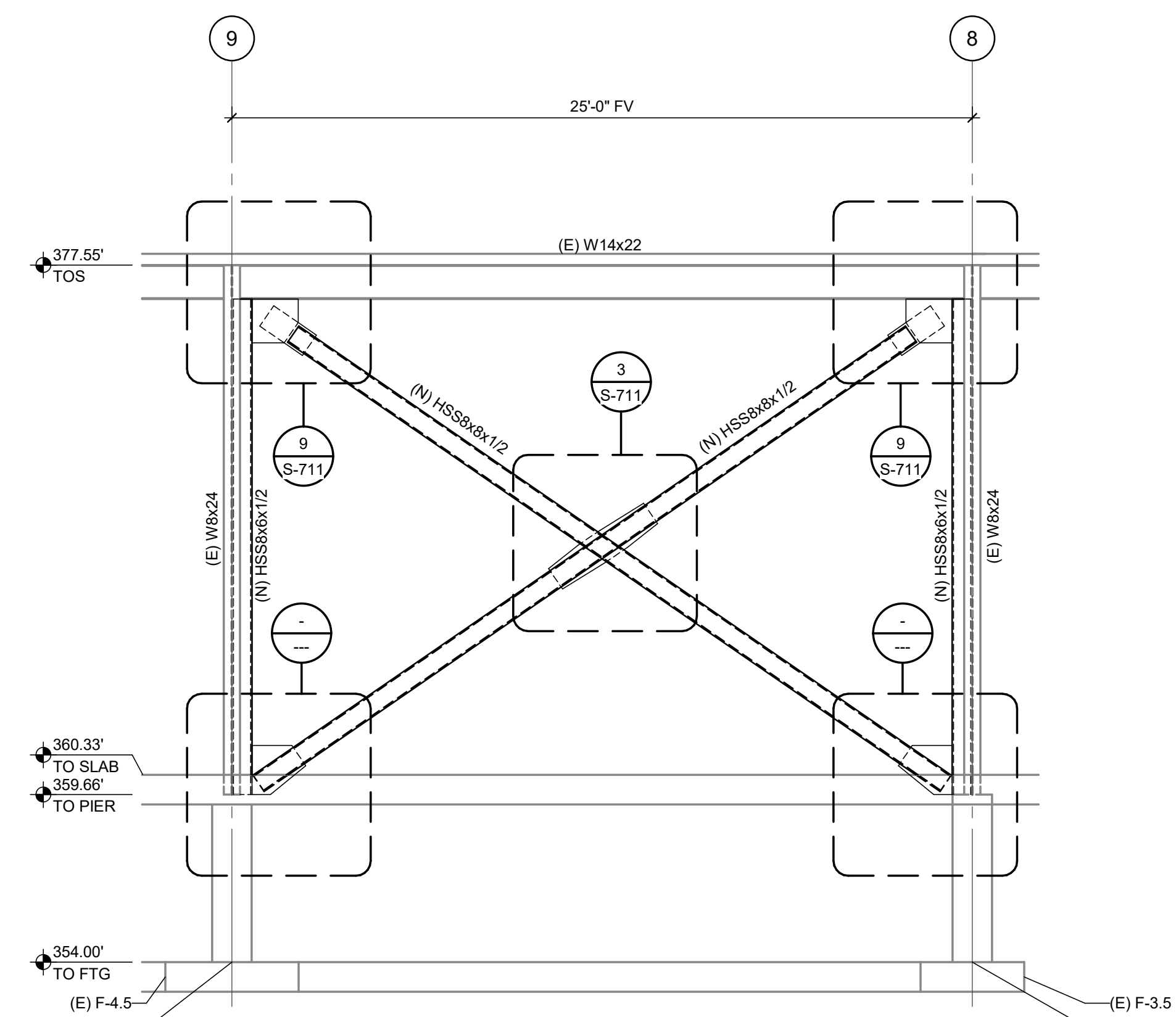
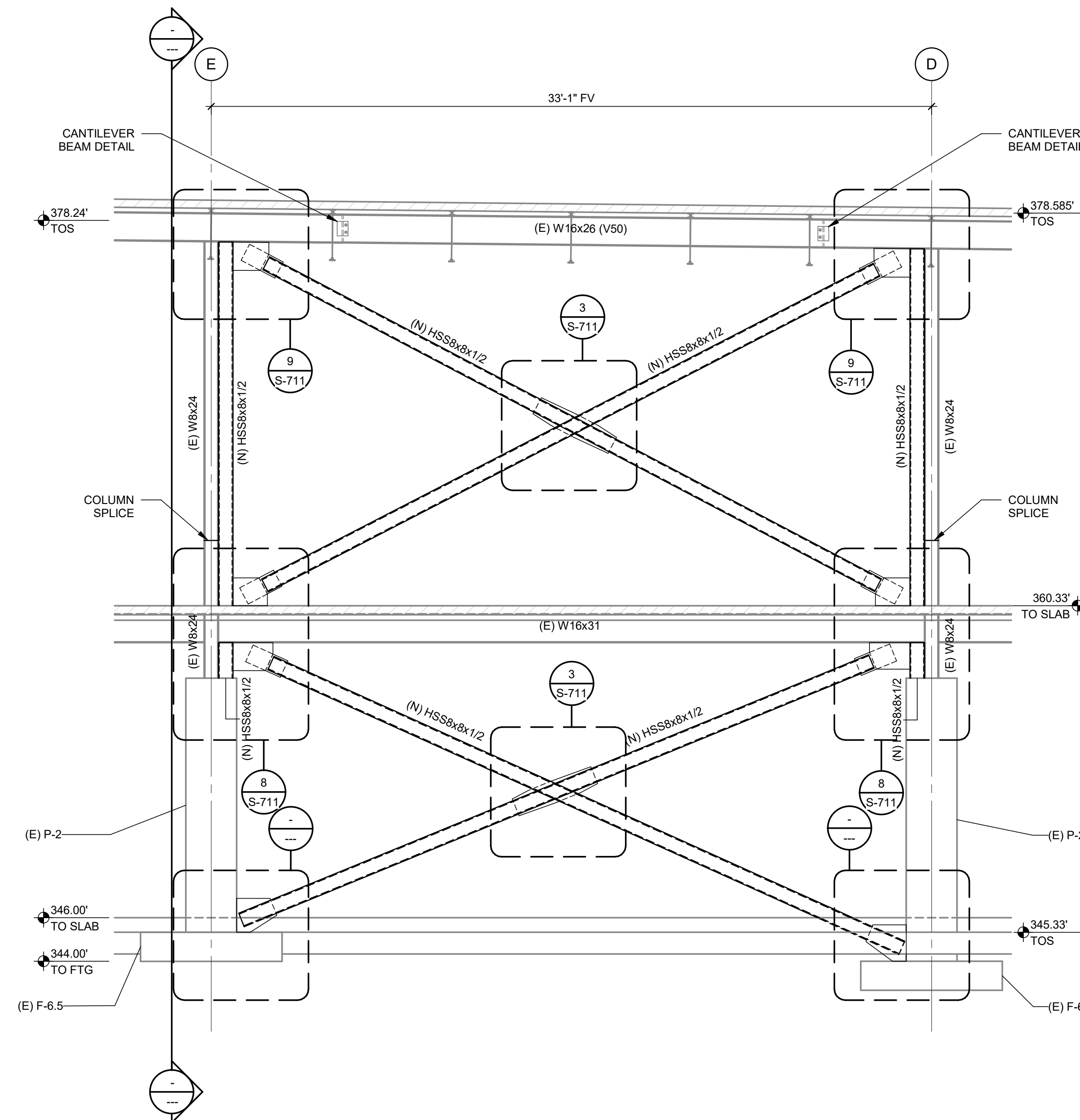


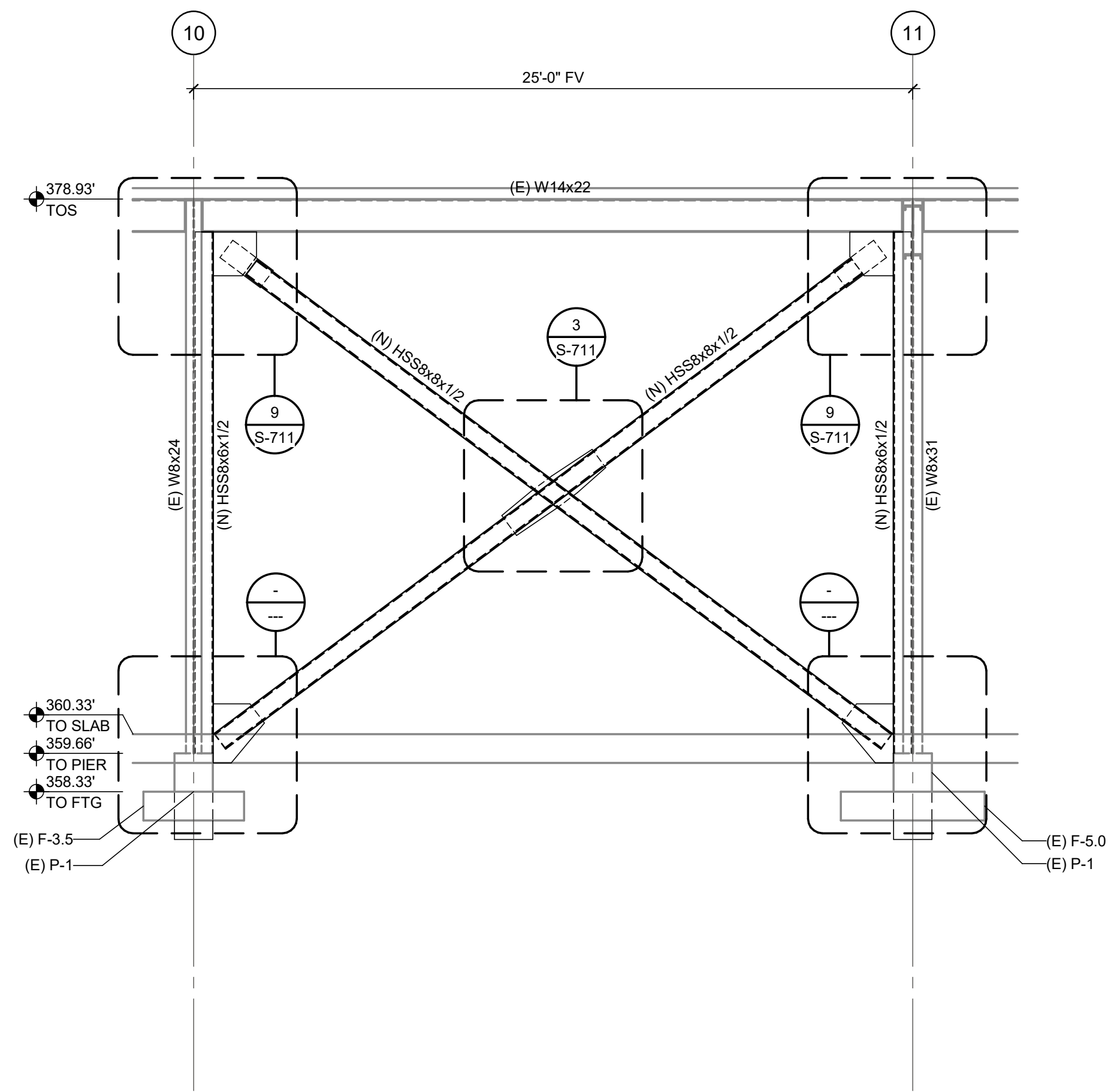
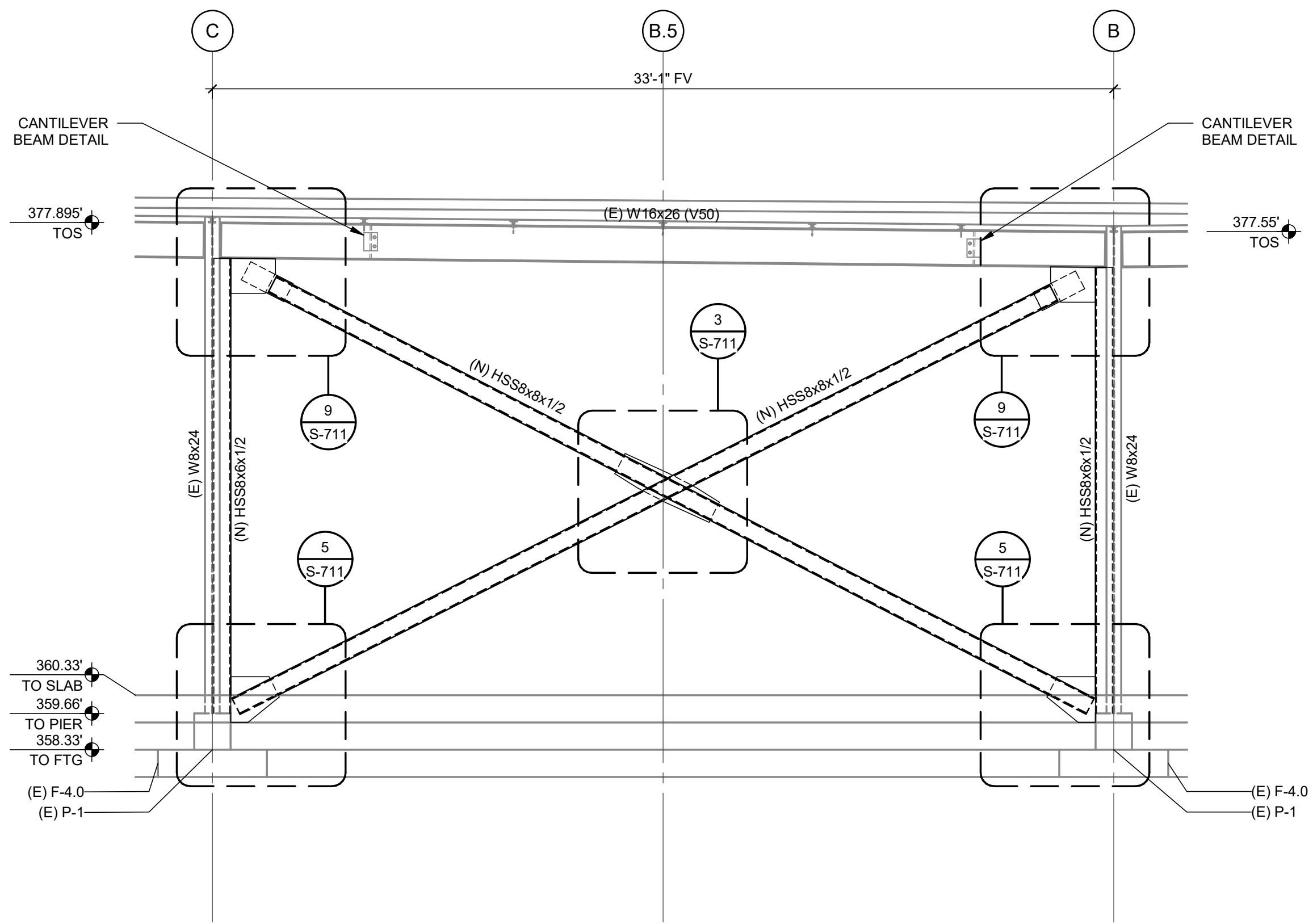
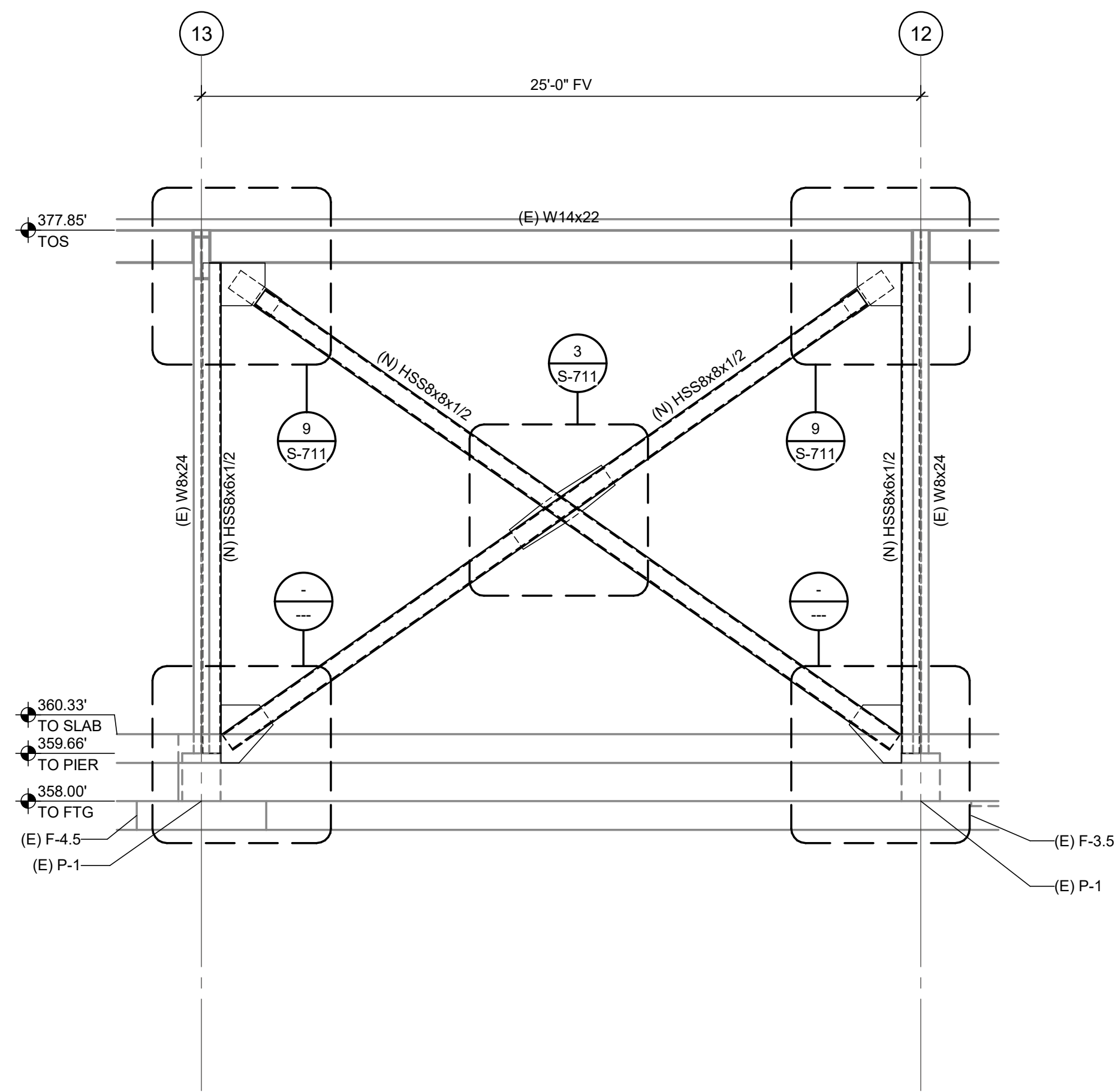

$$1'' = 1' - 0''$$

$$1^* = 1^* - 0$$

$$1'' = 1' - 0''$$


1/4" = 1'-0"



**3** SECTION BETWEEN GRIDLINES 3 AND 4 - BF-3
1/4" = 1'-0"**2** SECTION BETWEEN GRIDLINES D AND E - BF-2
1/4" = 1'-0"**1** SECTION BETWEEN GRIDLINES 3 AND 4 - BF-1
1/4" = 1'-0"**6** SECTION BETWEEN GRIDLINES C AND D - BF-6
1/4" = 1'-0"**5** SECTION BETWEEN GRIDLINES 8 AND 9 - BF-5
1/4" = 1'-0"**4** SECTION BETWEEN GRIDLINES D AND E - BF-4
1/4" = 1'-0"



3 SECTION BETWEEN GRIDLINES 12 AND 13 - BF-9
1/4" = 1'-0"

2 SECTION BETWEEN GRIDLINES B AND C - BF-8
1/4" = 1'-0"

1 SECTION BETWEEN GRIDLINES 10 AND 11 - BF-7
1/4" = 1'-0"

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S-701 - STRUCTURAL BRACING ELEVATIONS

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1	04/06/26	DD SUBMISSION
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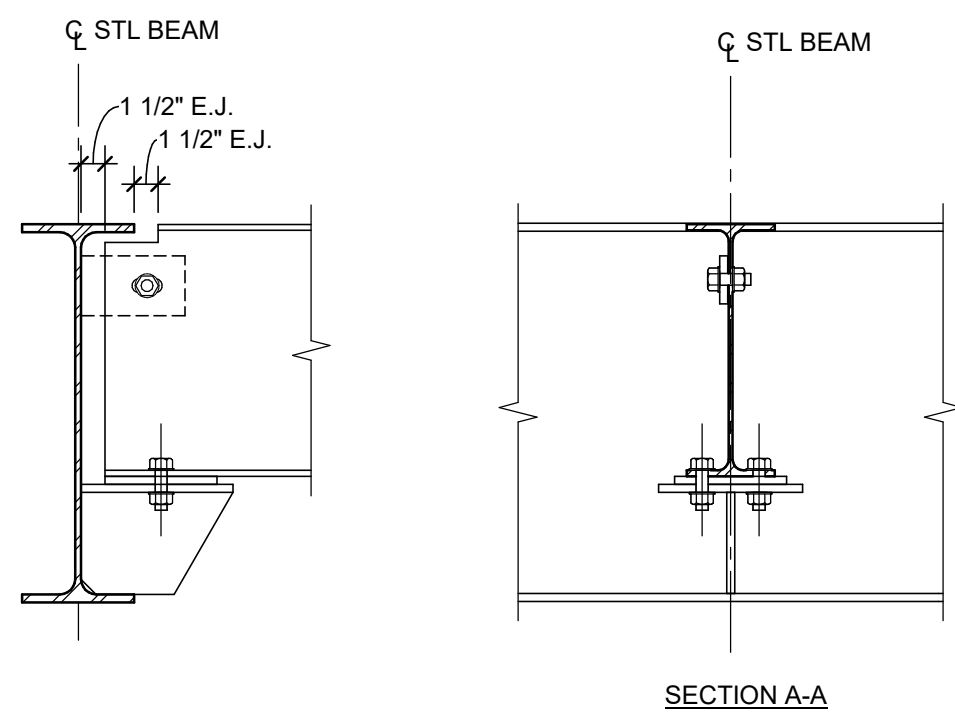
VSW JOB NO. 250202 SCALE 1/4" = 1'-0"

STRUCTURAL BRACING ELEVATIONS

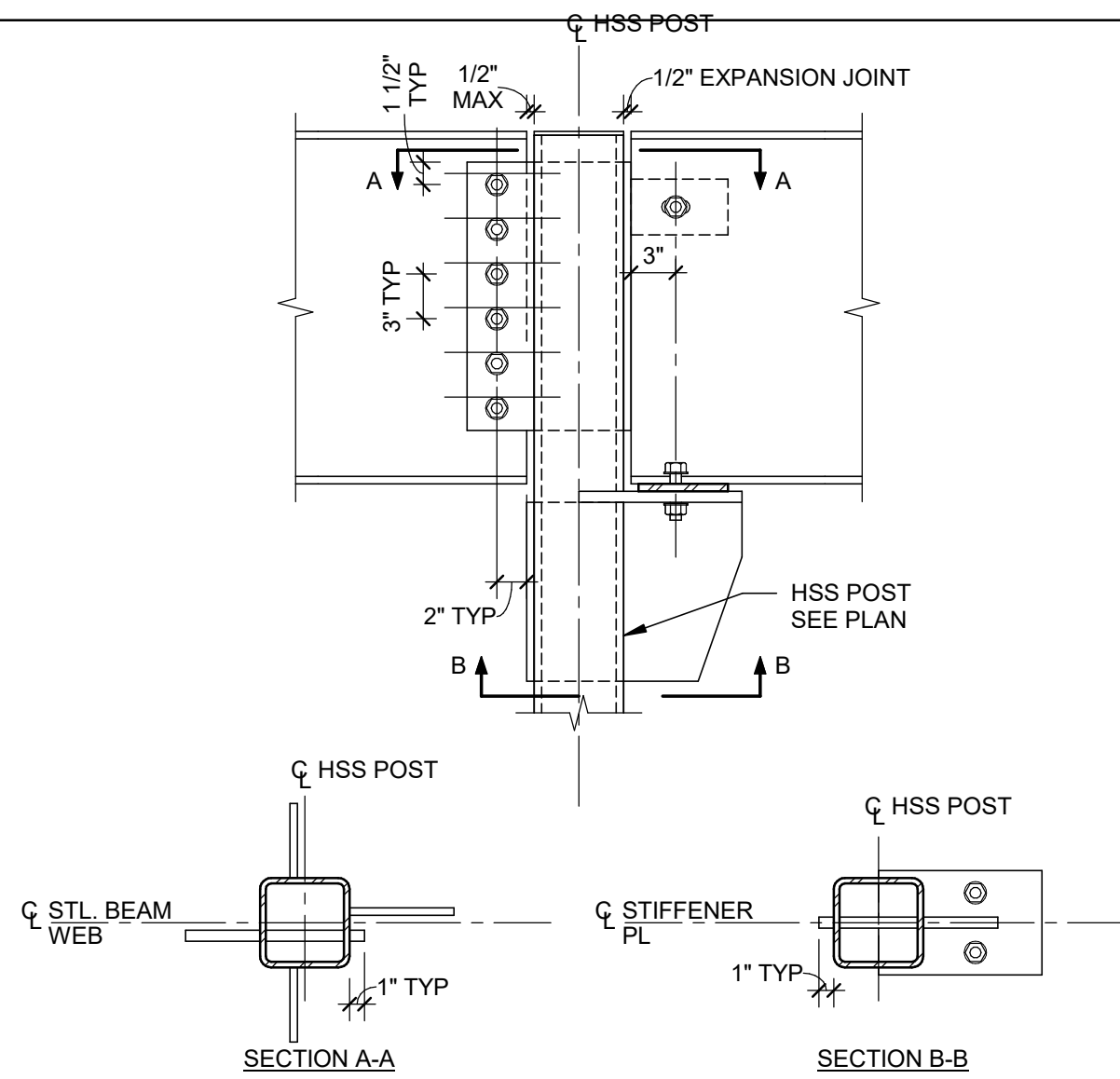
DRAWING

S-701

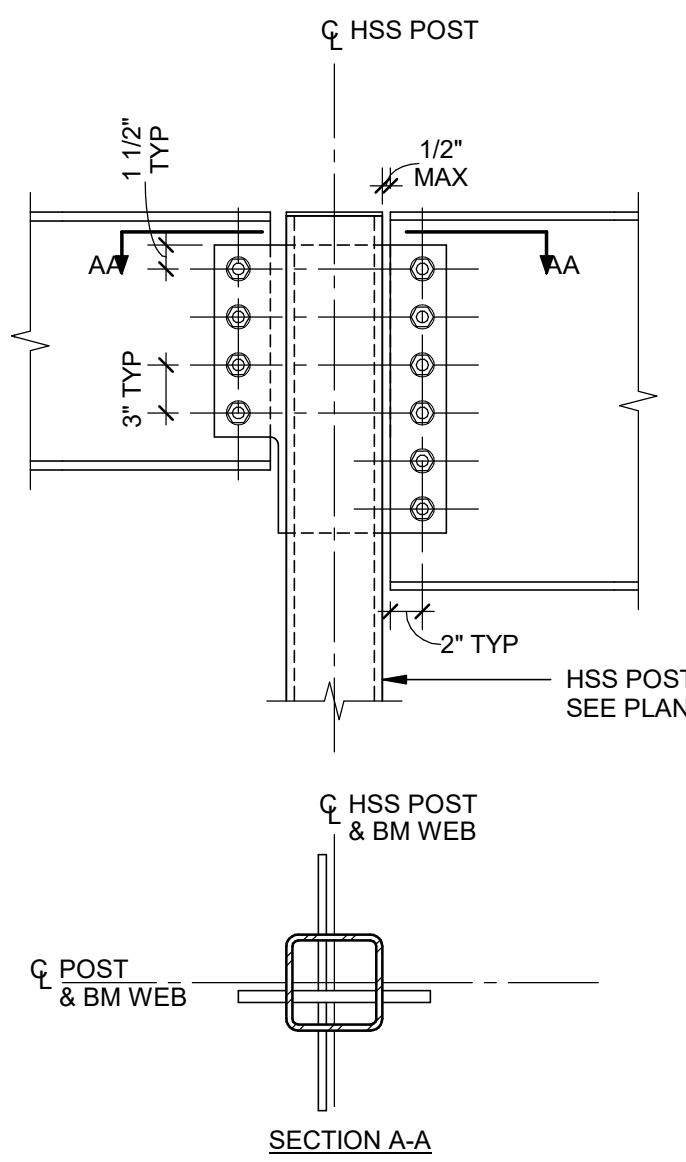
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5 TYPICAL BEAM TO BEAM CONNECTION AT EXPANSION JOINT
1" = 1'-0"



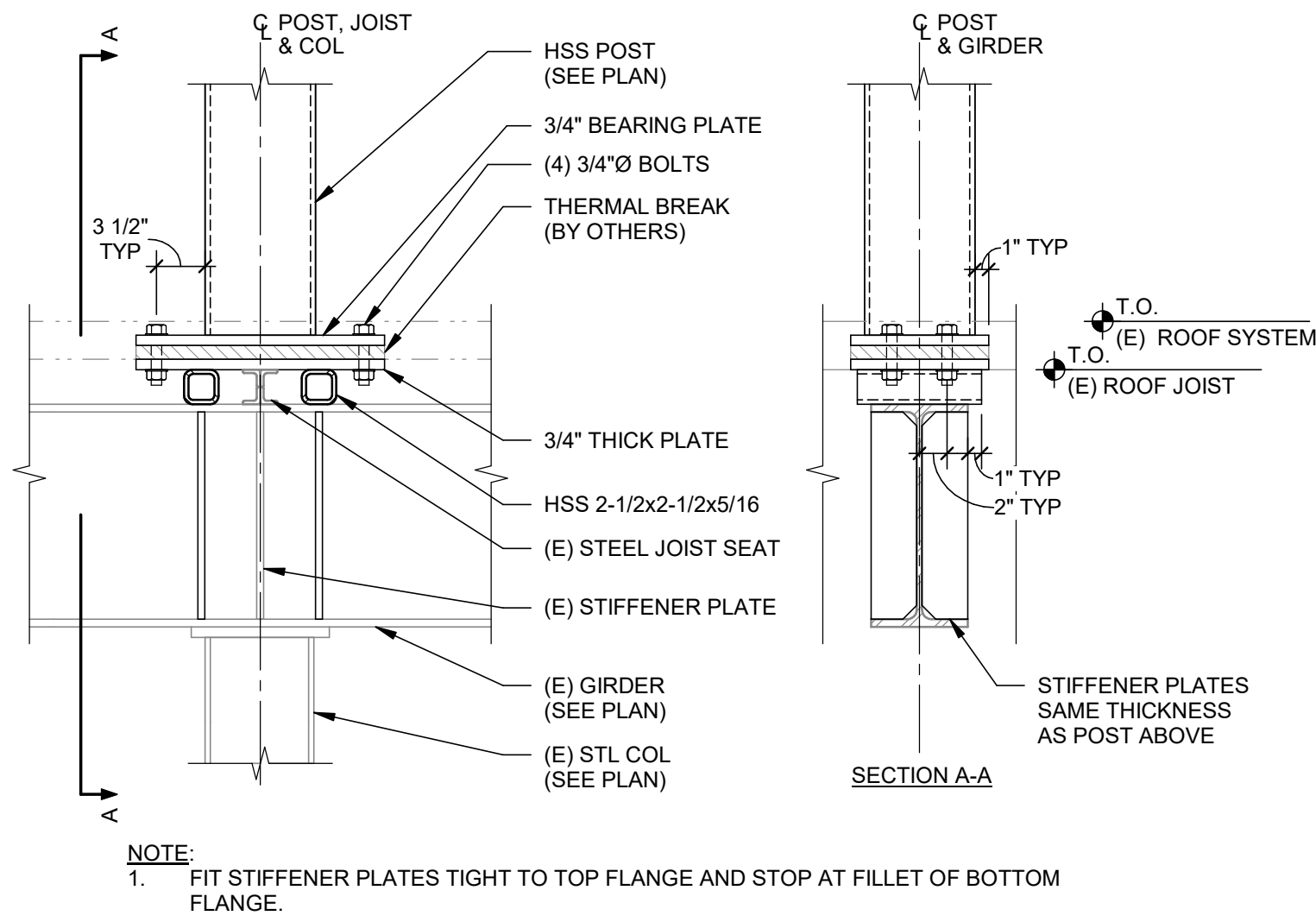
4 TYPICAL BEAM TO POST CONNECTION AT EXPANSION JOINT
1" = 1'-0"



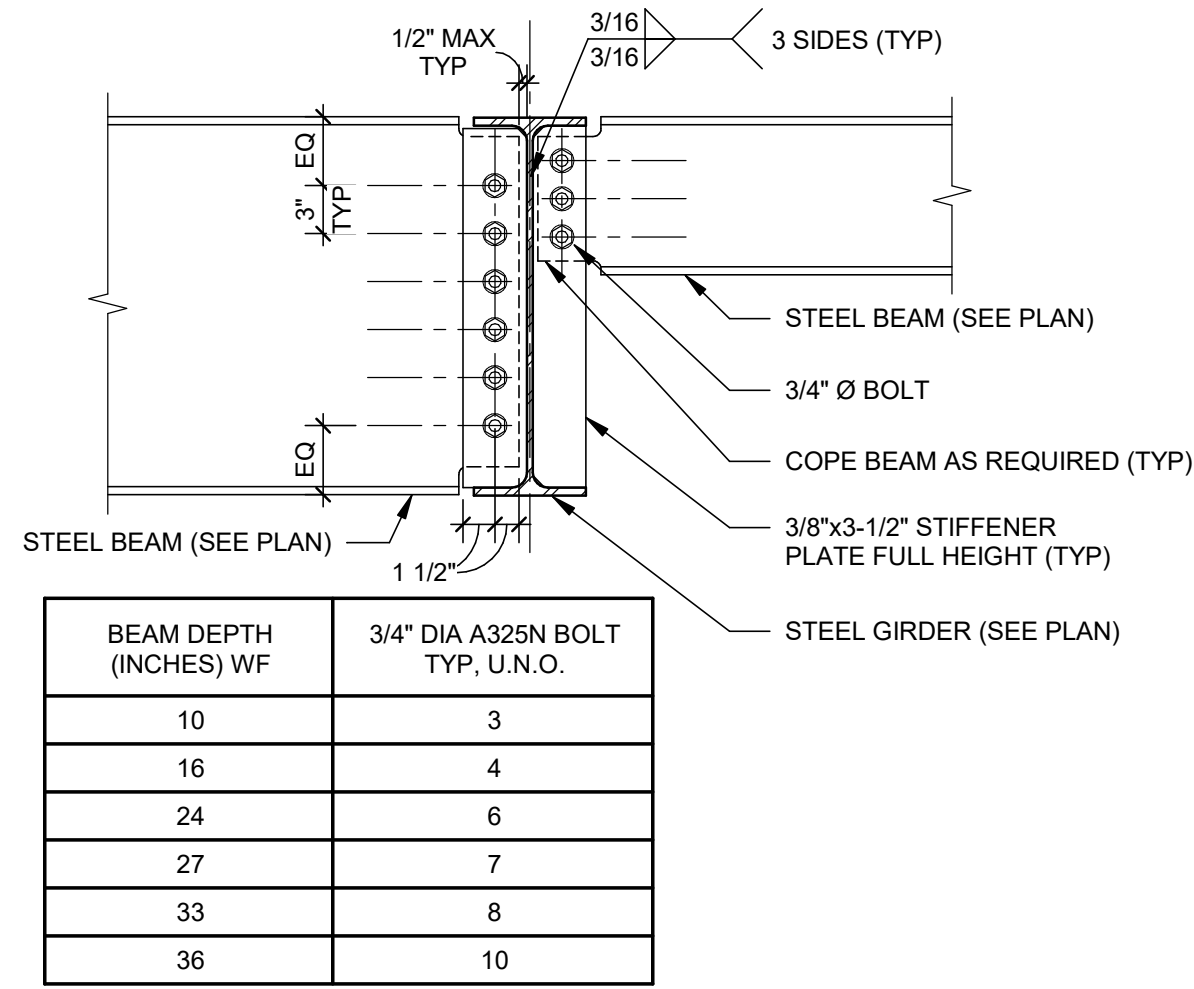
3 TYPICAL BEAM TO POST CONNECTION
1" = 1'-0"

NOTE:
1. SEE DETAIL S-710 FOR SHEAR PLATE THICKNESS AND BOLT REQUIREMENTS PER BEAM SIZE. FOR MOMENT CONNECTIONS, FULL-HEIGHT SHEAR PLATES ARE NOT REQUIRED.

2 TYPICAL BEAM TO BEAM MOMENT CONNECTION
1" = 1'-0"

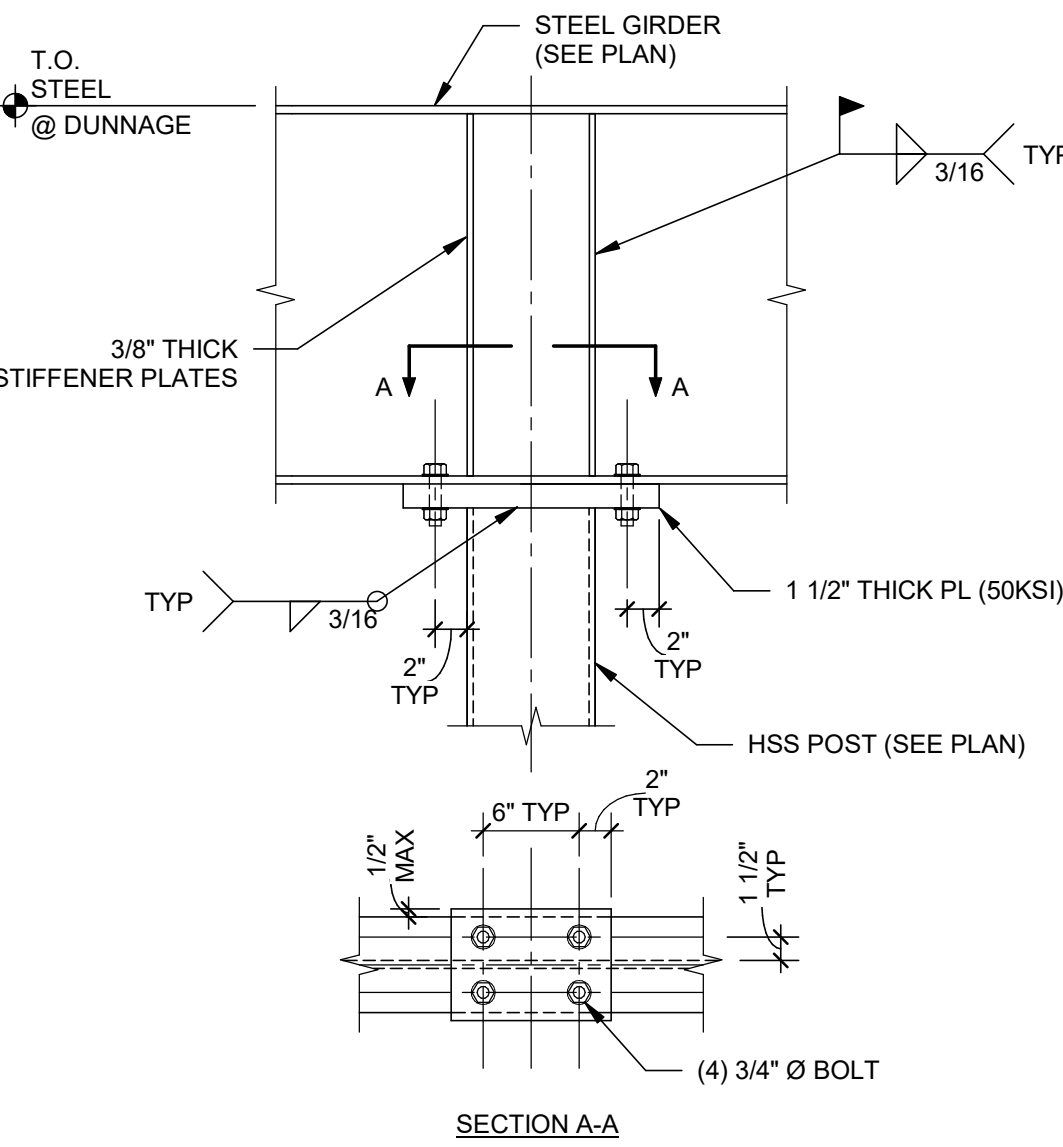


7 STEEL HSS POST TO TRANSFER GIRDER
NO SCALE

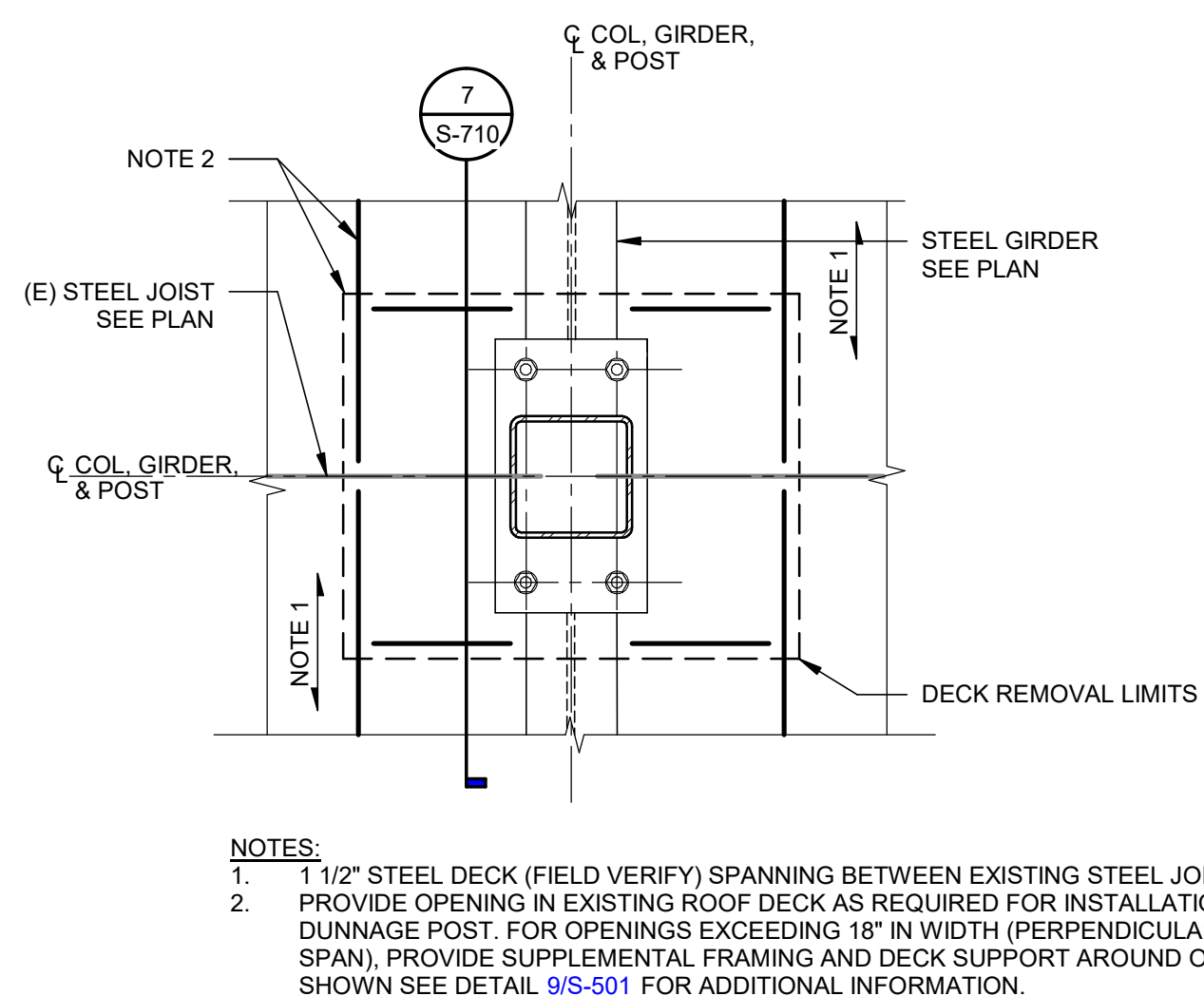


NOTE:
1. CONNECTION REFERS TO THE SHALLOWER BEAM IN A BEAM-TO-BEAM CONNECTION.

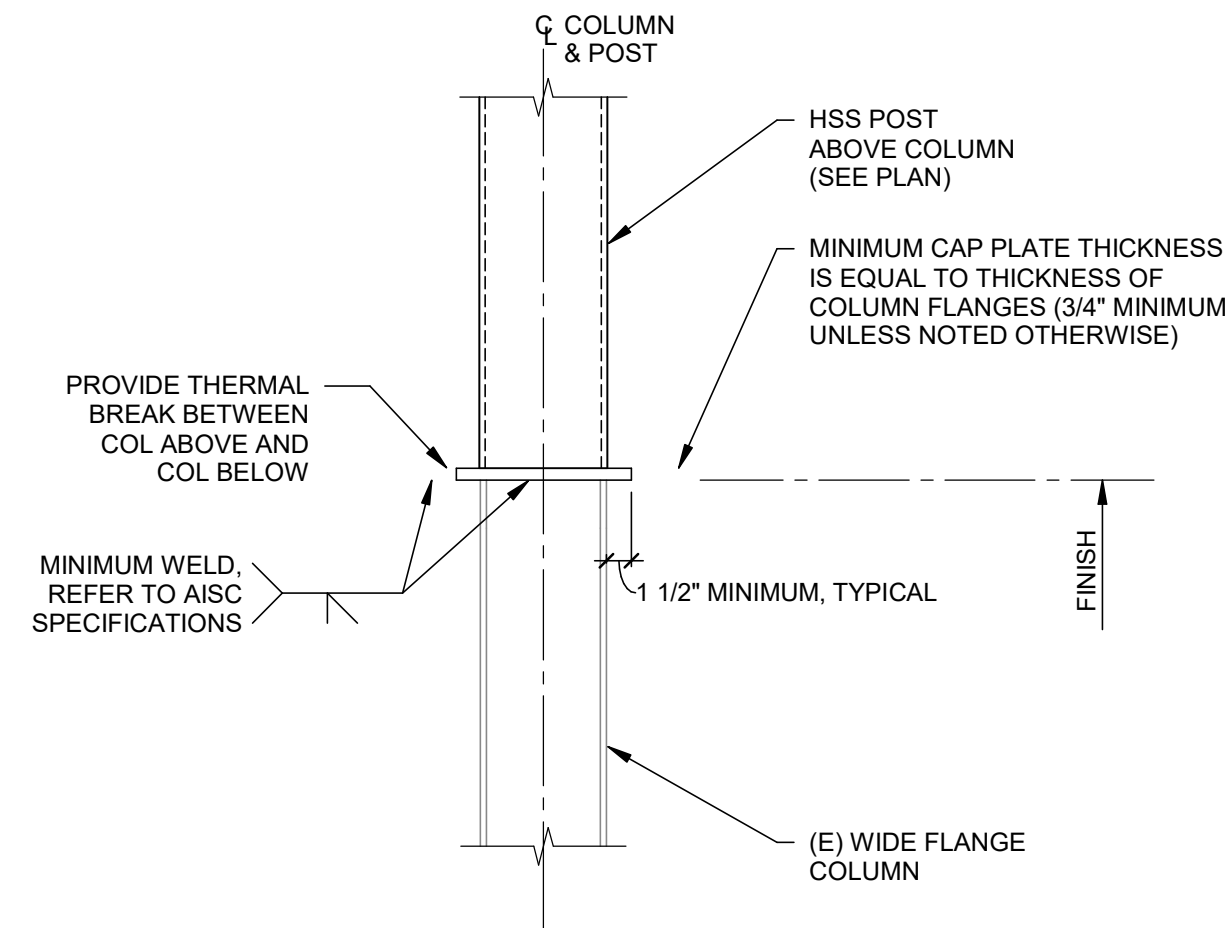
1 TYPICAL BEAM-TO-BEAM SHEAR CONNECTION
NO SCALE



6 BEAM OVER COLUMN CONNECTION
1" = 1'-0"



9 STEEL HSS POST TO TRANSFER GIRDER
1" = 1'-0"



8 WIDE FLANGE COLUMN TENSION SPLICE
NO SCALE

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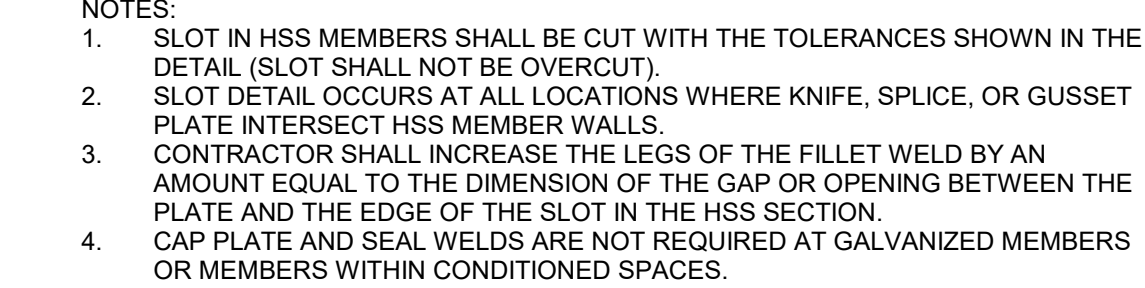
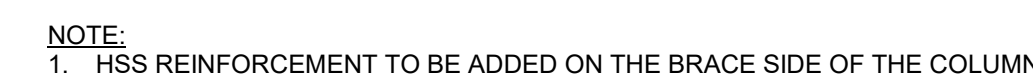
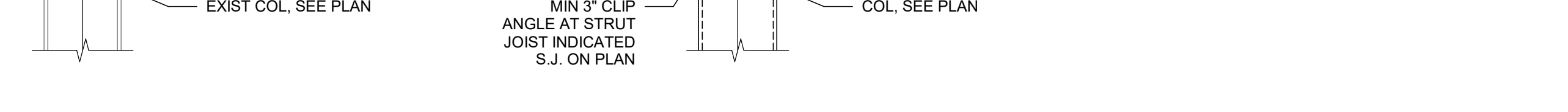
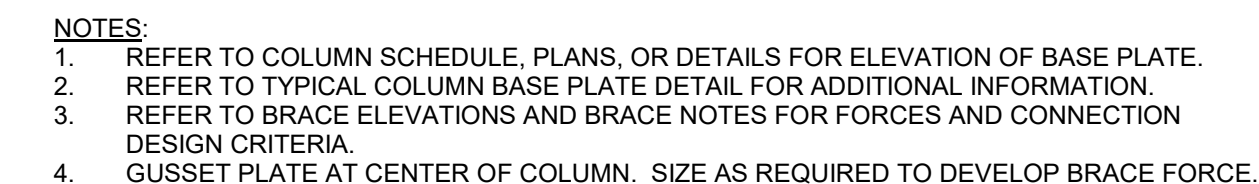
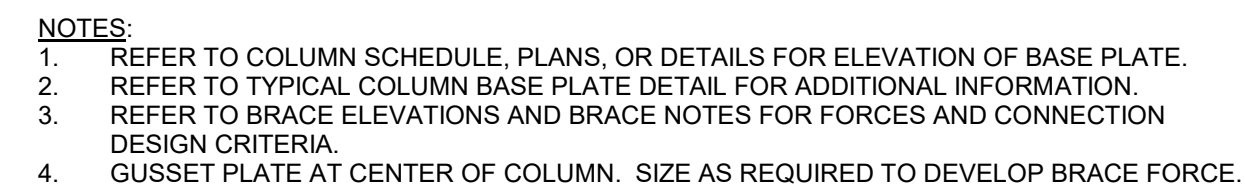
VSW JOB NO. SCALE
250202 1" = 1'-0"

STRUCTURAL DETAILS

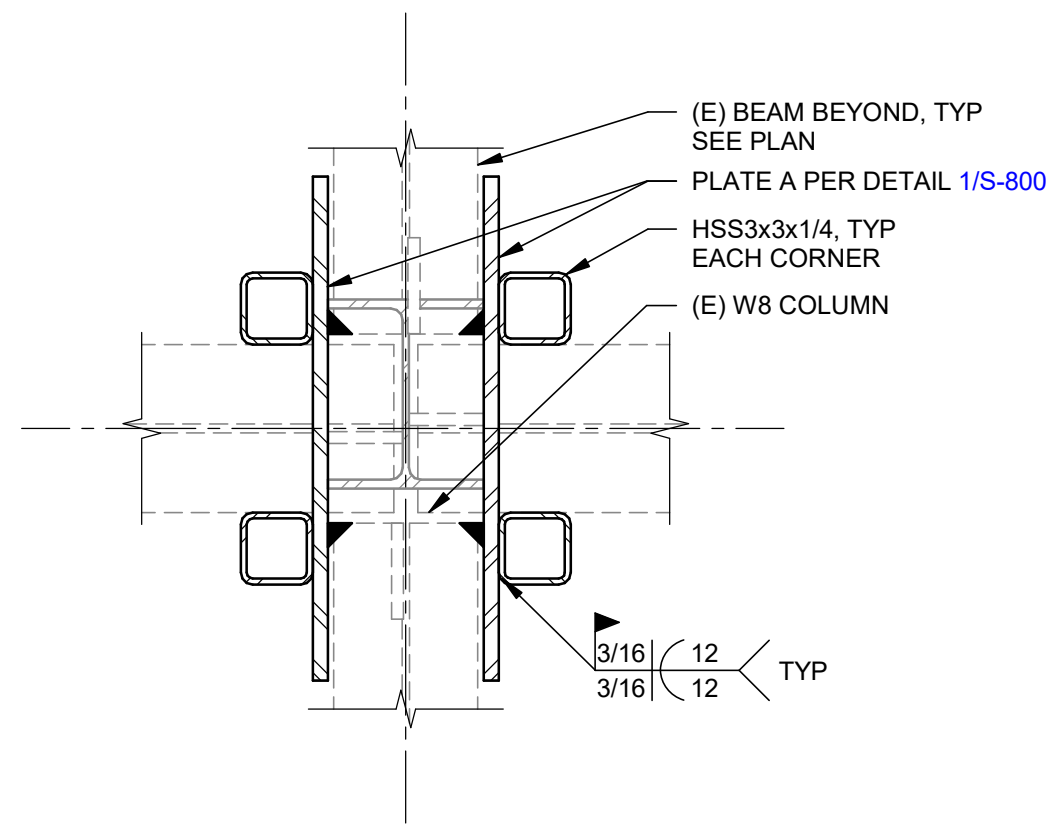
DRAWING

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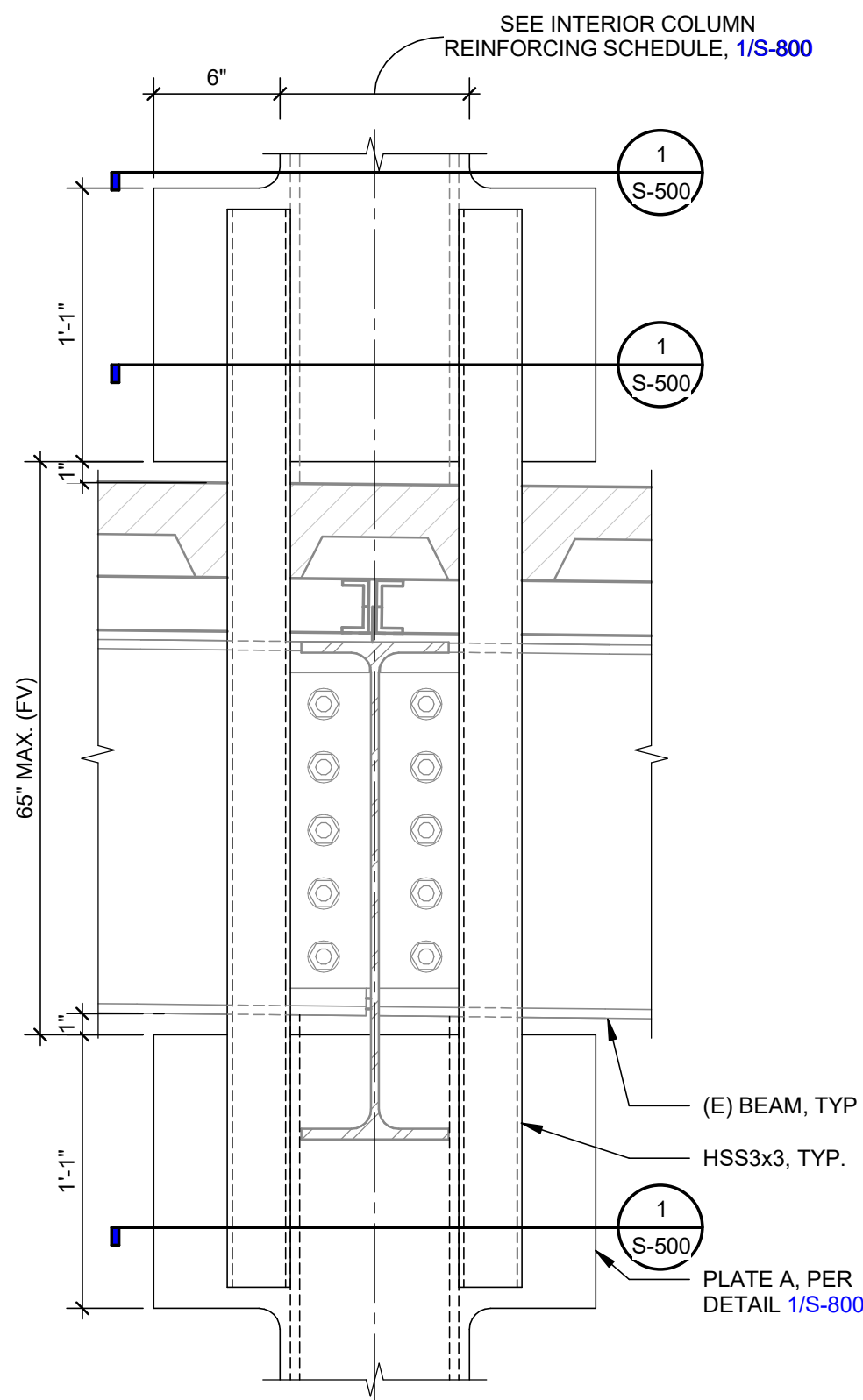
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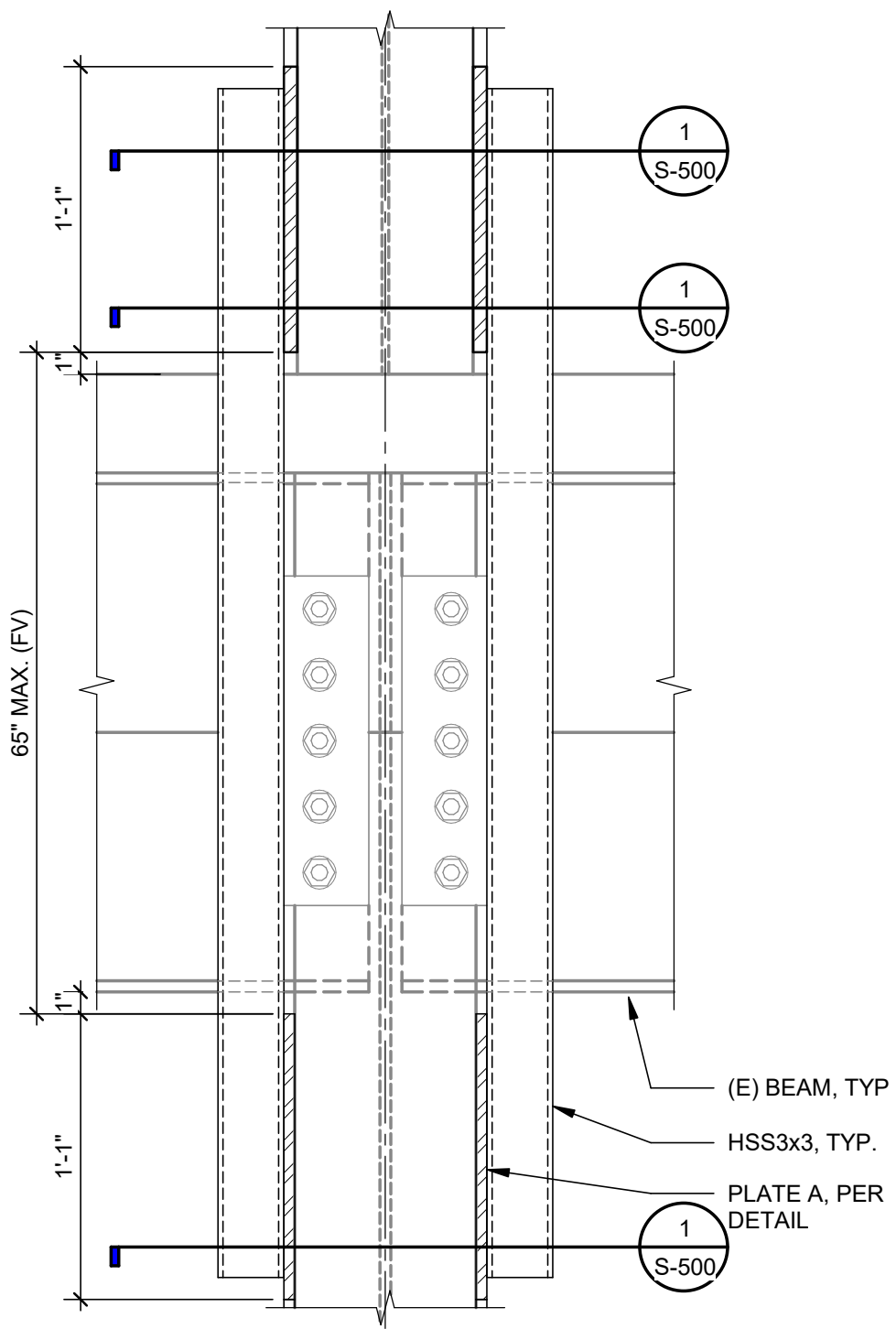
- TYPICAL BRACED FRAME NOTES



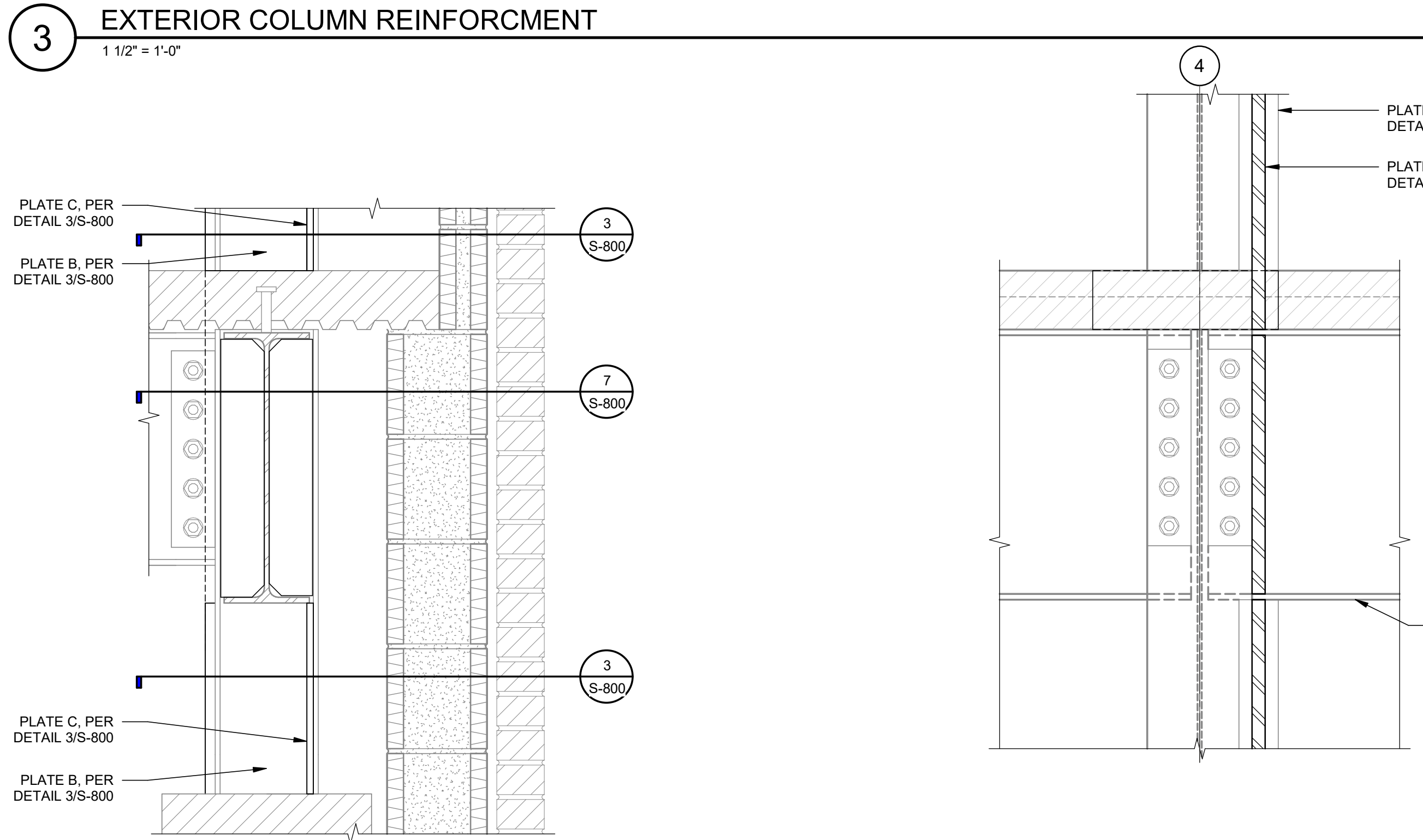
10 TYPICAL INTERIOR COLUMN REINFORCEMENT SPLICE
1 1/2" = 1'-0"



9 TYPICAL INTERIOR COLUMN REINFORCEMENT SPLICE
1 1/2" = 1'-0"



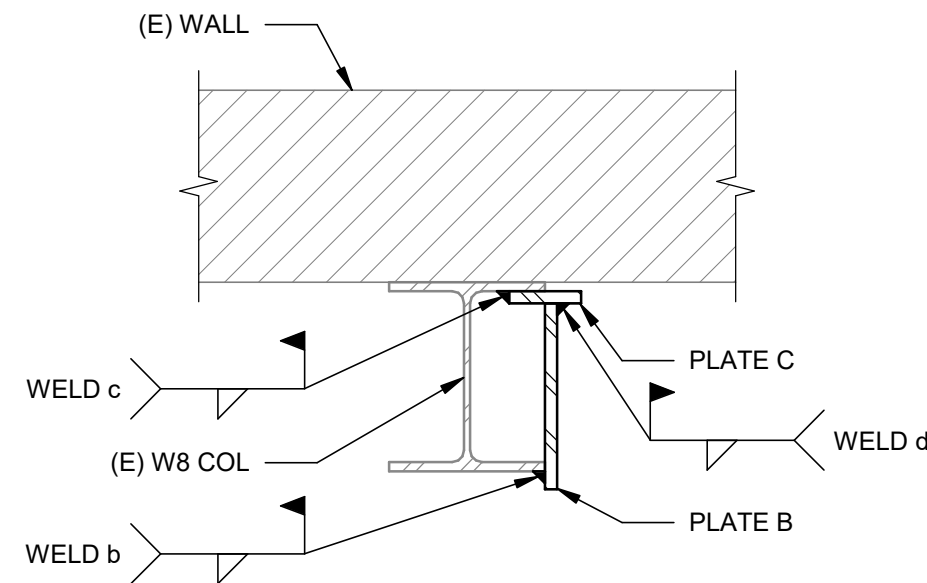
8 TYPICAL EXTERIOR COLUMN REINFORCEMENT SPLICE
1 1/2" = 1'-0"



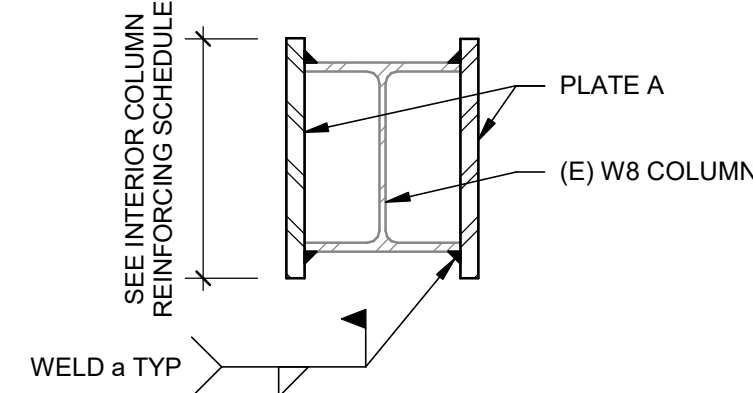
7 TYPICAL EXTERIOR COLUMN REINFORCEMENT SPLICE
1 1/2" = 1'-0"

MARK	PLATE B	PLATE C	WELD			COLUMN SIZE	NOTES
			b	c	d		
RE-1	7.75" x 0.5"	3" x 0.5"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	W8x24	E7018
RE-2	8" x 1"	3" x 0.75"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	W8x24	E7018
RE-3	7.75" x 0.5"	3" x 0.5"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	W8x28	E7018
RE-4	7.75" x 0.5"	3" x 0.5"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	W8x31	E7018
RE-5	7.75" x 0.75"	3" x 0.75"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	W8x31	E7018
RE-6	7.75" x 0.5"	3" x 0.5"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	5/16" FILLET 3"-9"	W8x35	E7018

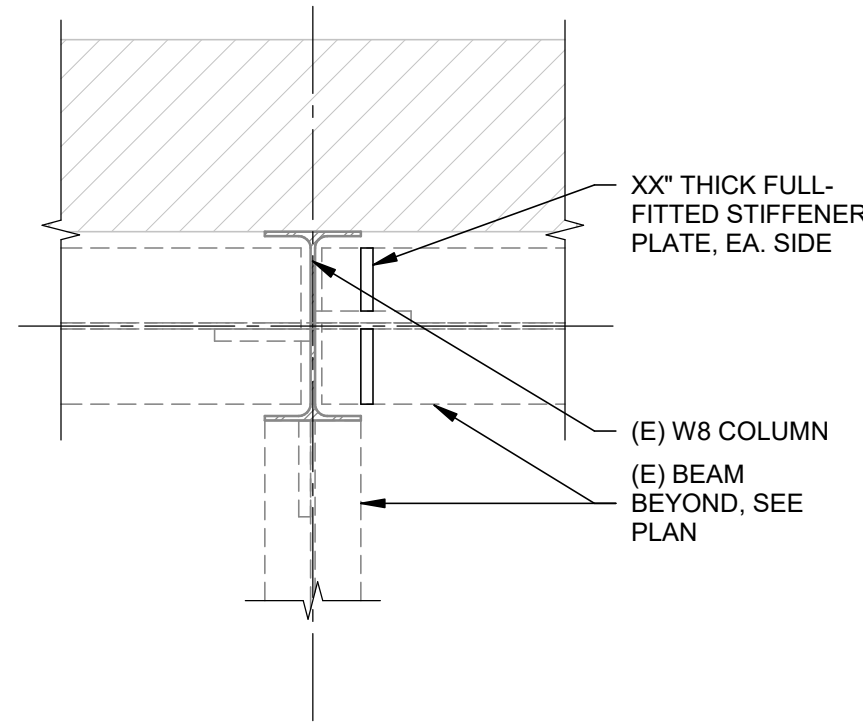
NOTE:
1. SEQUENCING: FIRST FULLY INSTALL PLATE C, THEN INSTALL PLATE B.



INTERIOR COLUMN REINFORCING SCHEDULE			
MARK	PLATE A	WELD a	COLUMN SIZE
RI-1	9" x 0.625"	5/16" FILLET 3"-9"	W8x31
RI-2	9" x 0.375"	5/16" FILLET 3"-9"	W8x31
RI-3	9.25" x 1"	5/16" FILLET 3"-9"	W8x40
RI-4	9.75" x 0.625"	5/16" FILLET 3"-9"	W8x58



1 TYPICAL INTERIOR COLUMN REINFORCEMENT
1 1/2" = 1'-0"



VSW

V Squared Workshop
7900 Westpark Dr, Suite A205
McLean, VA 22102
301.590.8800
www.vsw.co

MEP

AHA

3700 Mansell Road, Suite 200
Alpharetta, GA 30022
www.aha-engineers.com

STRUCTURAL

walter p moore

1700 K Street NW, Suite 1050,
Washington, DC, 20006
202.481.8881
www.walterpmoore.com

CIVIL

VIA

8180 Greensboro Dr.
Suite 200
Tysons, VA 22102

H5 DATA CENTERS

H5 Herndon
480 Springpark Pl, Herndon, VA 20170
H5 Data Centers

S-800 - COLUMN DETAILS AND REINFORCING SCHEDULES

FOR REVIEW ONLY,
NOT FOR
CONSTRUCTION

Professional Certification:
I hereby certify that these
documents were prepared by
me, and that I am a duly
licensed architect under the
laws of the State of Maryland.

Revisions	
02/03/26	DESIGN DEVELOPMENT /PRICING
1 04/06/26	DD SUBMISSION

ISSUED:
DD SUBMISSION
04/06/2026

VSW JOB NO. SCALE

250202 1 1/2" = 1'-0"

COLUMN DETAILS AND REINFORCING

SCHEDULE
DRAWING

S-800

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